



Deutsche Bank
Research

Who is buying Treasuries, Mortgages, Credit, and Munis?

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IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. UNTIL 19th MARCH 2021 INCOMPLETE DISCLOSURE INFORMATION MAY HAVE BEEN DISPLAYED, PLEASE SEE APPENDIX 1 FOR FURTHER DETAILS.



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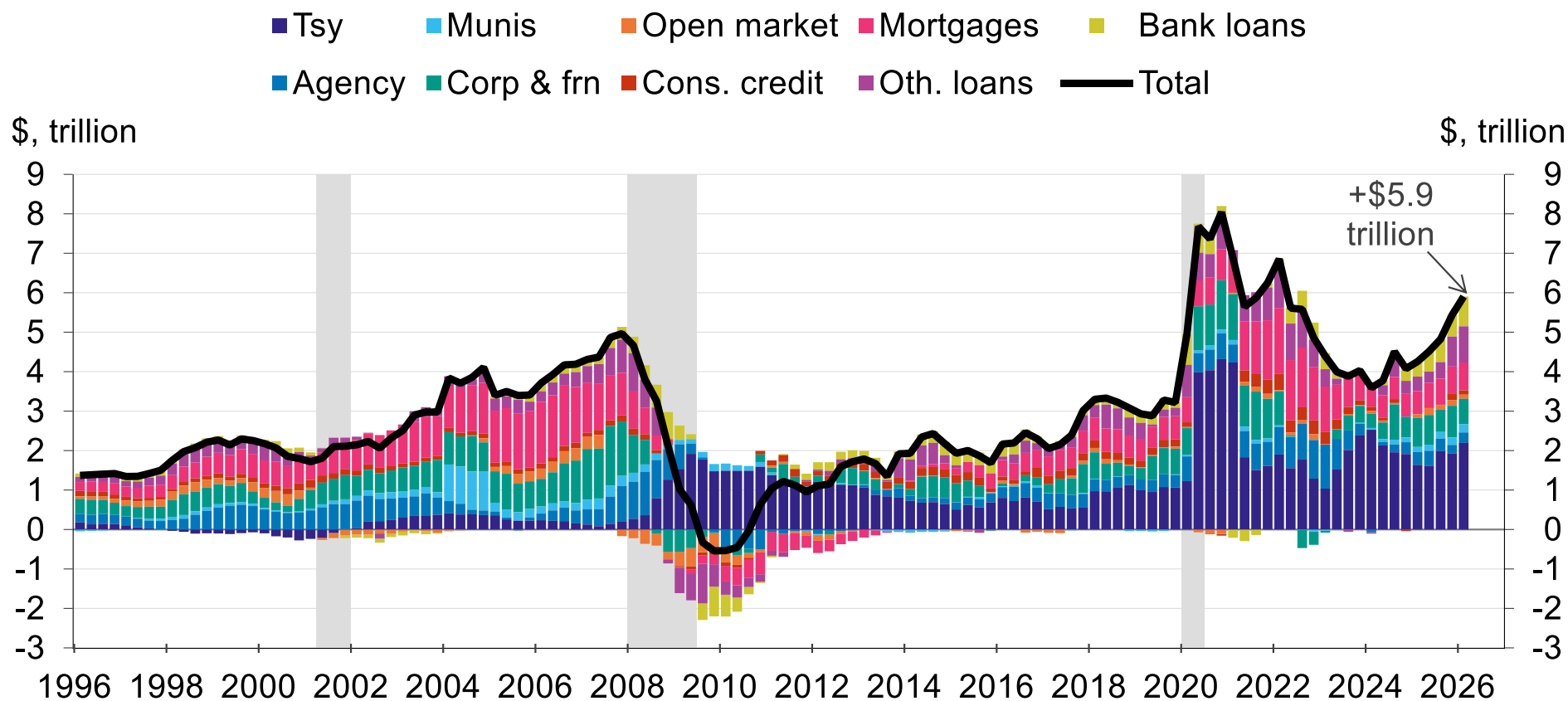


Featured charts

US fixed income markets grew by \$5.9trn in last 4 quarters



Change in debt securities outstanding

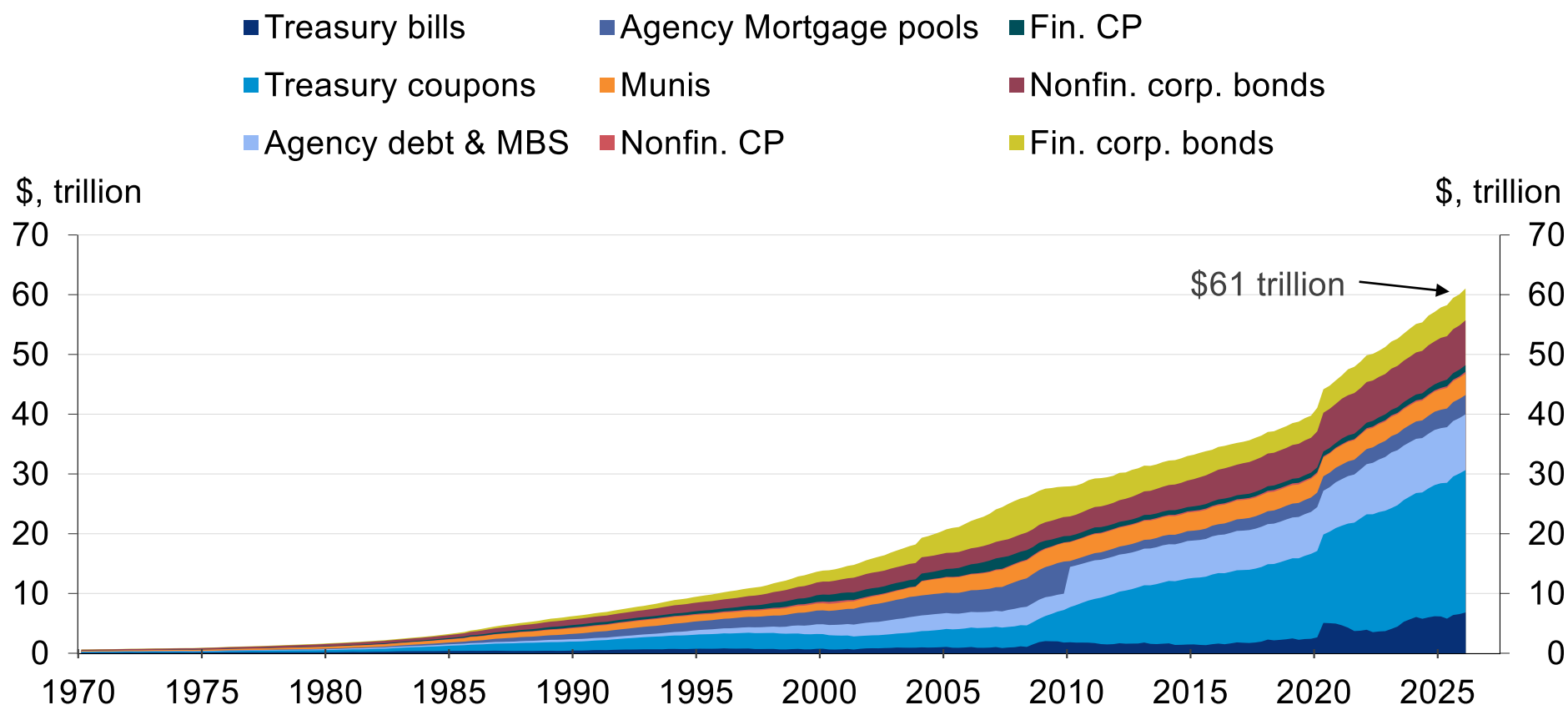


Source: Fed, Macrobond, Deutsche Bank

Total US fixed income securities outstanding: \$61trn as of Q1 2026



Outstanding debt securities

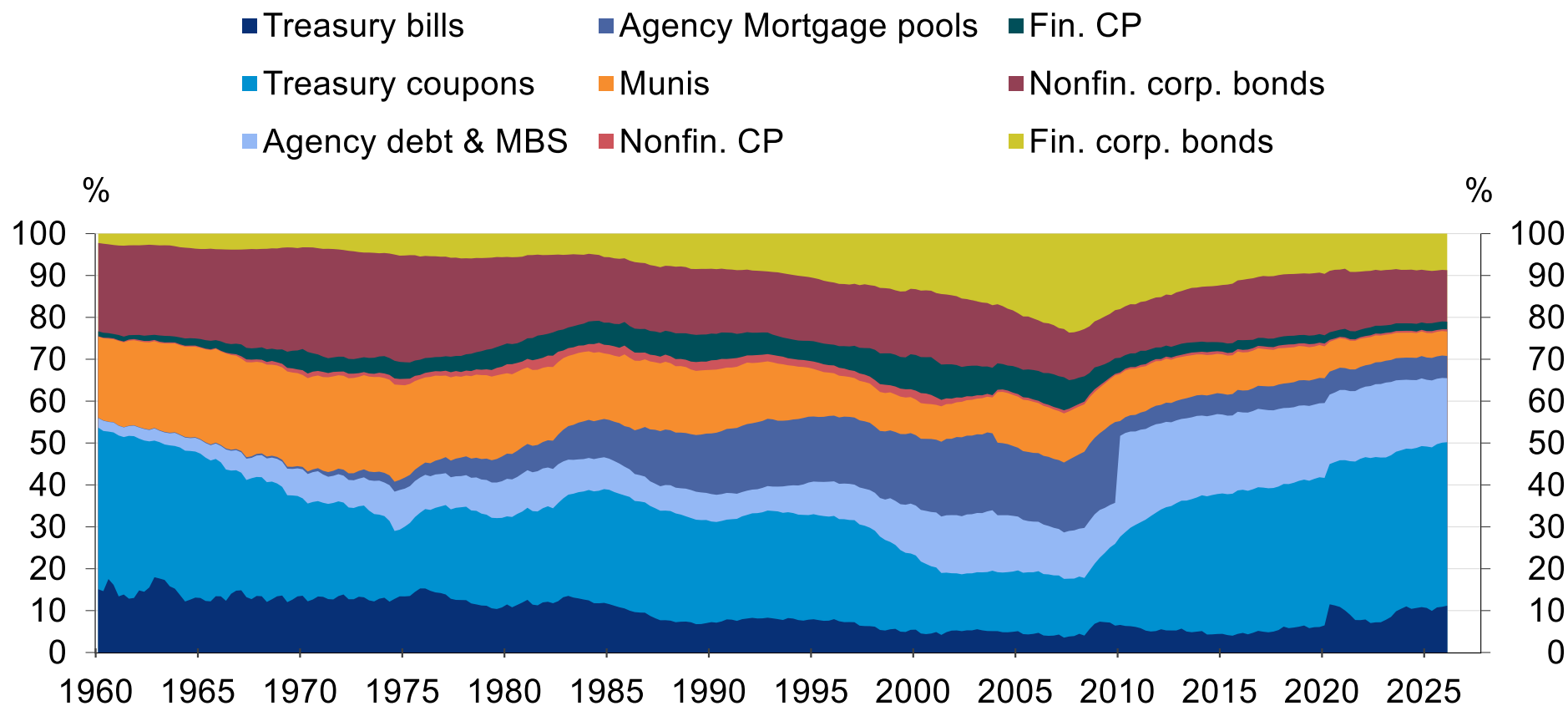


Source: Fed, Macrobond, Deutsche Bank

Treasury and agency securities make up about 70% of US fixed income



Outstanding debt securities

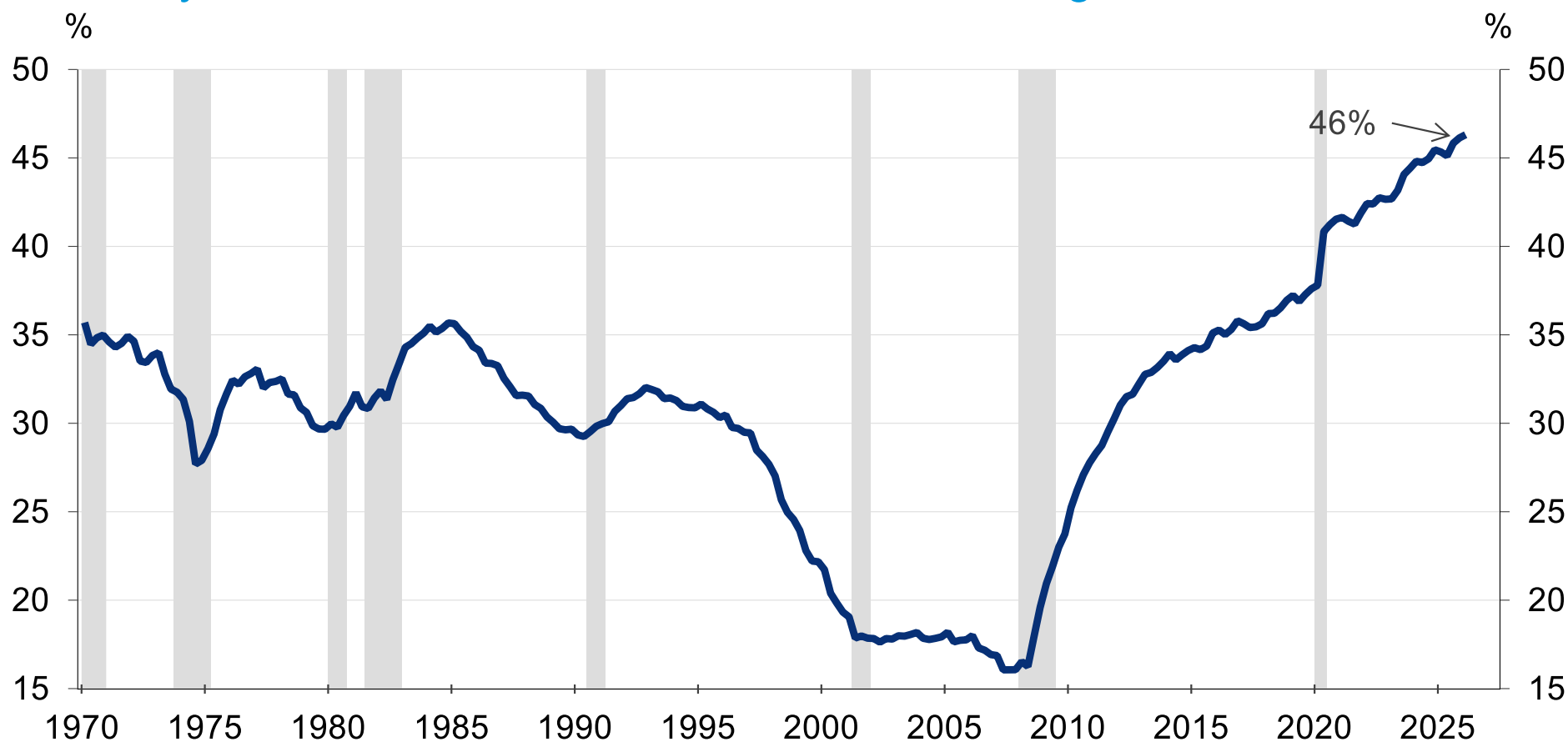


Source: Fed, Macrobond, Deutsche Bank

Treasuries' share of US fixed income has expanded significantly since the Global Financial Crisis



Treasury securities as a share of debt outstanding

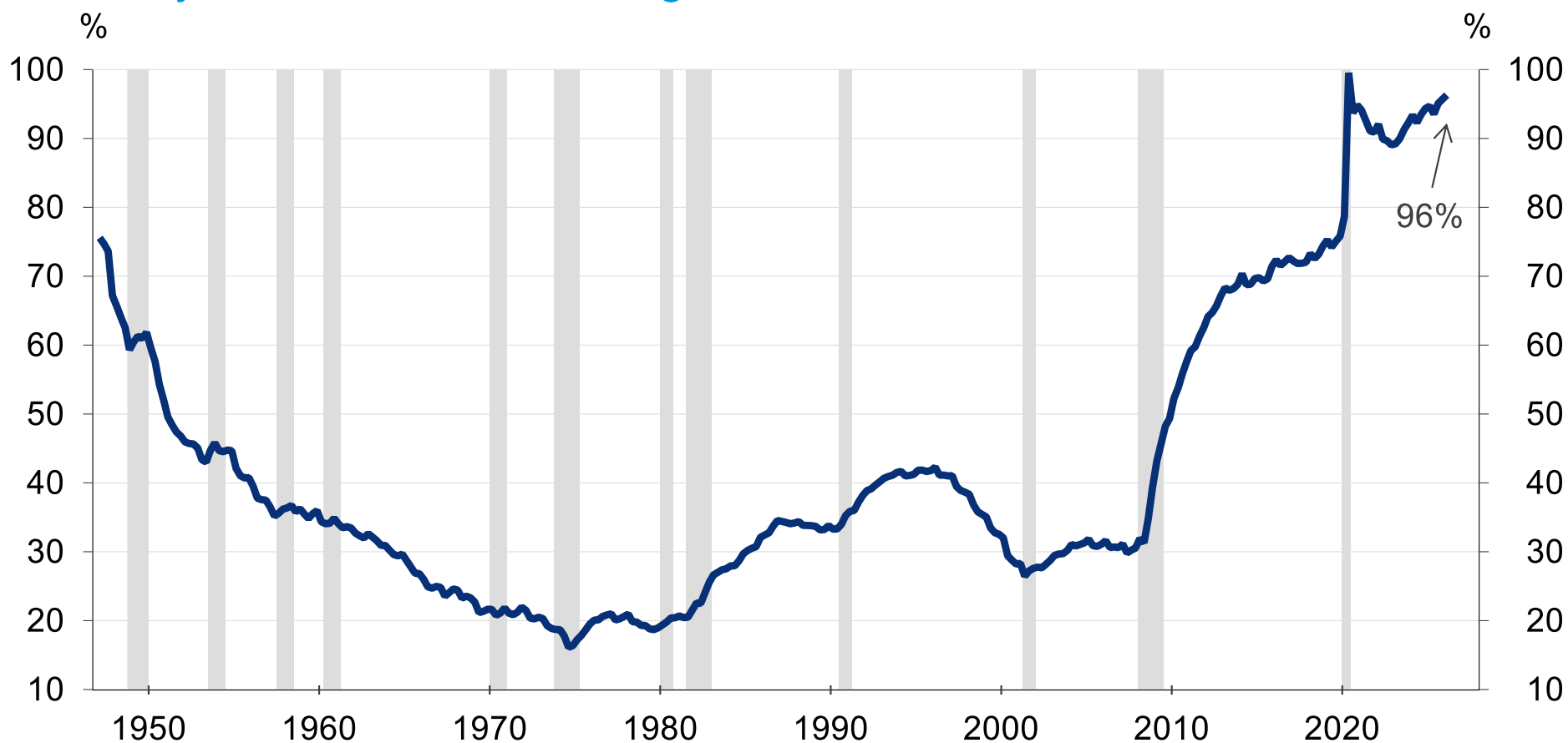


Source: Fed, Macrobond, Deutsche Bank

Treasury debt as a percentage of GDP is currently 96%, up from 75% before Covid



Treasury securities outstanding as a share of GDP

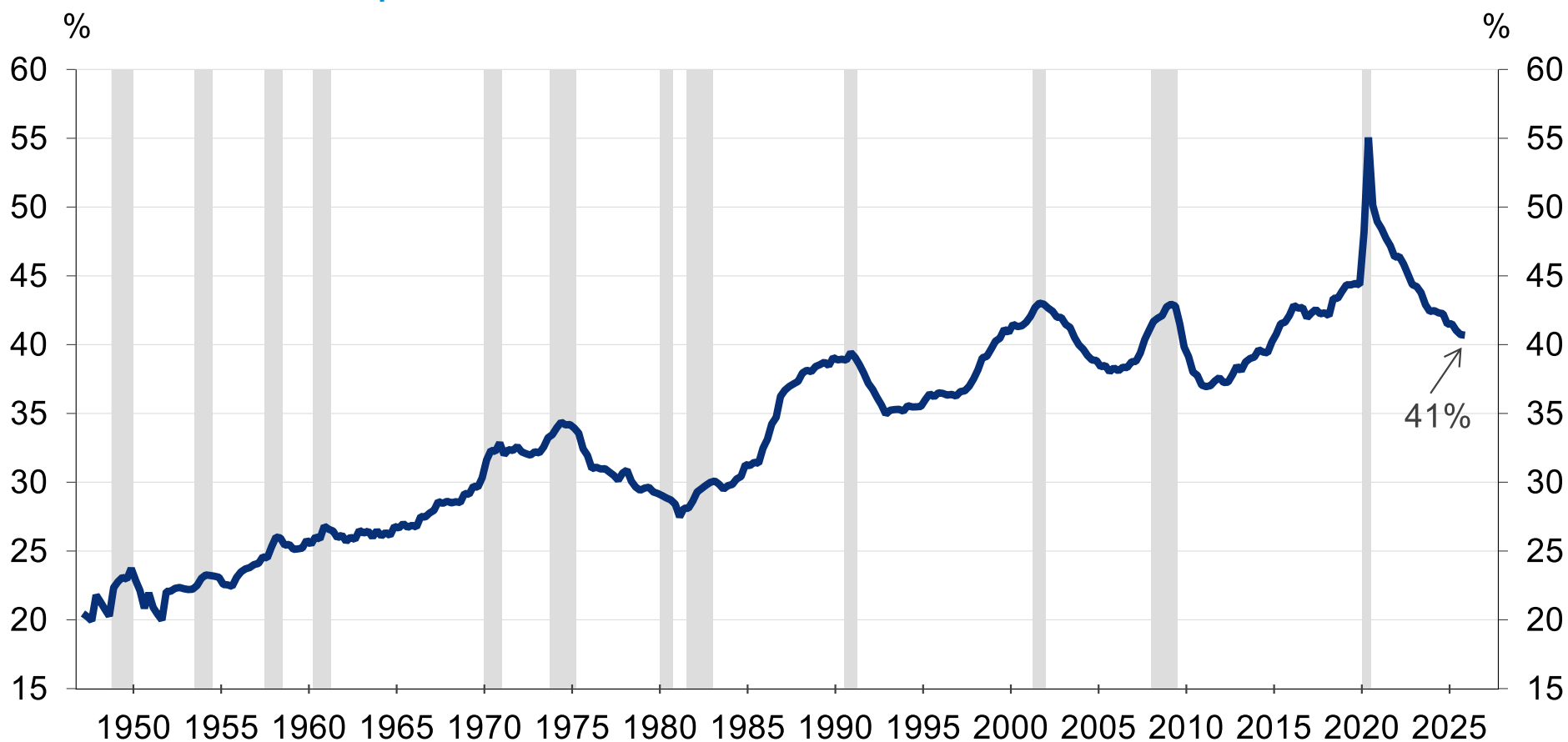


Source: Fed, BEA, Macrobond, Deutsche Bank

Non-financial corporate debt as a percentage of GDP declined from a peak of 55% during Covid to 41% currently



Non-financial corporate debt securities and loans as a share of GDP

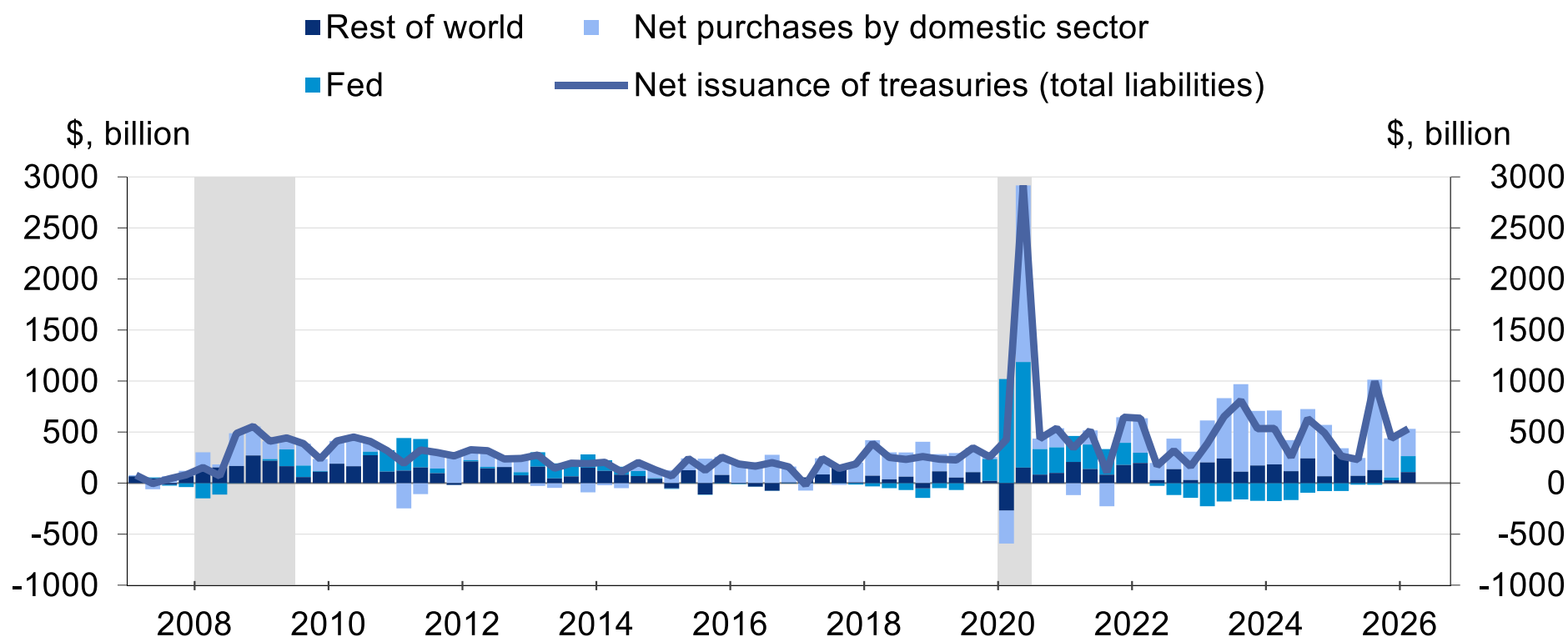


Source: Fed, BEA, Macrobond, Deutsche Bank

Domestic sector funds the bulk of the US budget deficit



Transactions in US Treasury securities



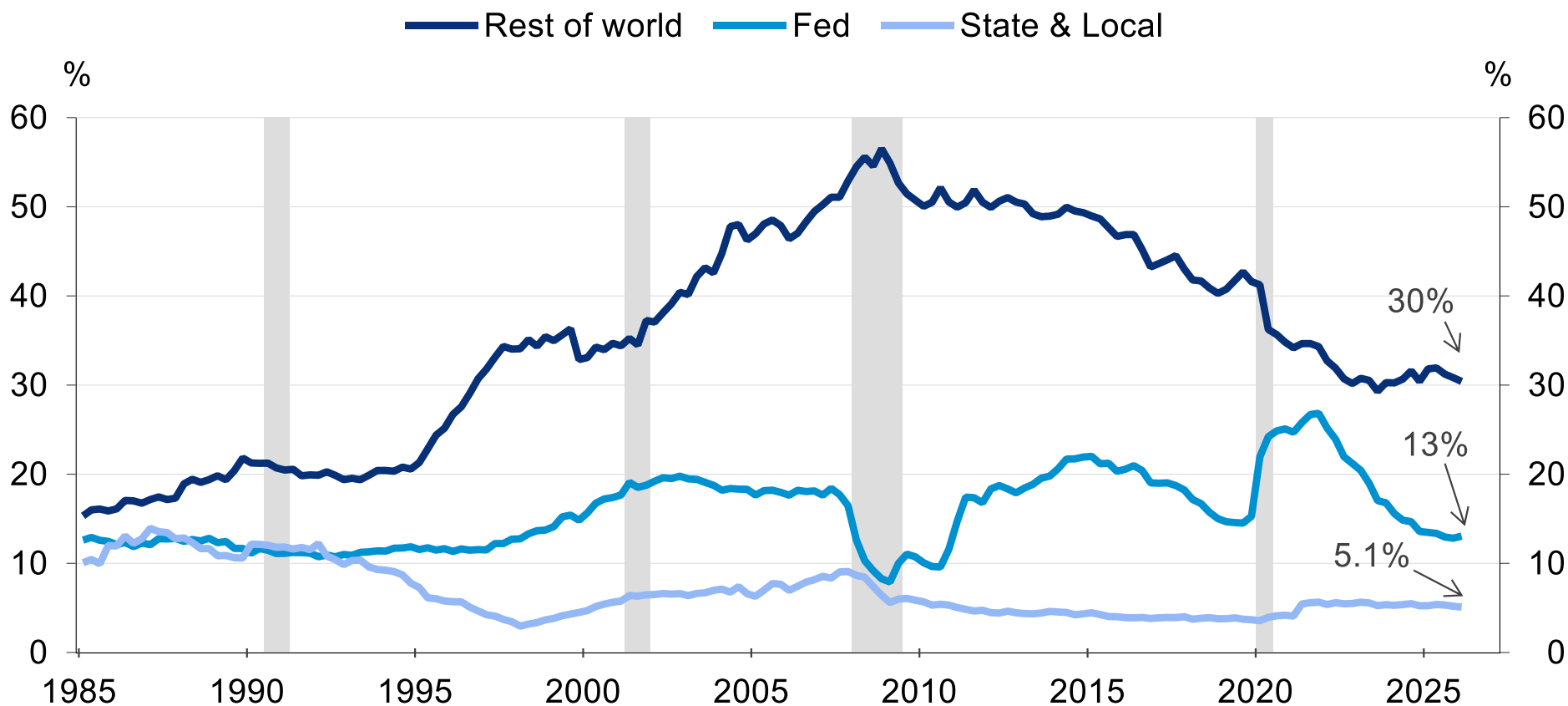
Source: Fed, Macrobond, Deutsche Bank

Note: Domestic sector includes households, banks, pension funds, insurance, govt. retirement funds, mutual funds, GSEs and others.

Foreign ownership of US Treasuries fell slightly to 30% in Q1 2026, not far from the multi-decade low of 29% seen in 2023 Q3



Holdings of Treasuries as % of treasury debt outstanding

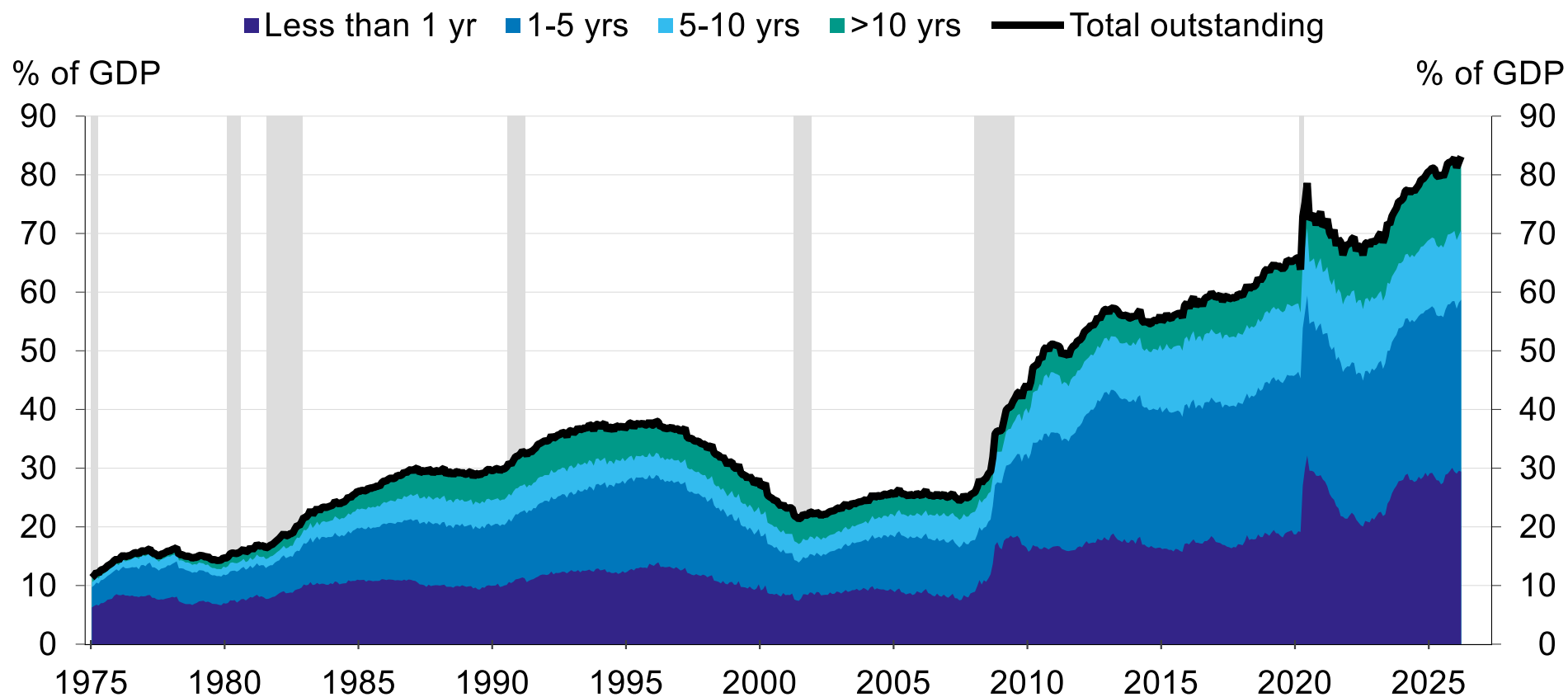


Source: Fed, Macrobond, Deutsche Bank

Private holdings of US Treasuries concentrated in maturity of 5 years or less



Marketable debt held outside the Federal Reserve: Maturity breakdown

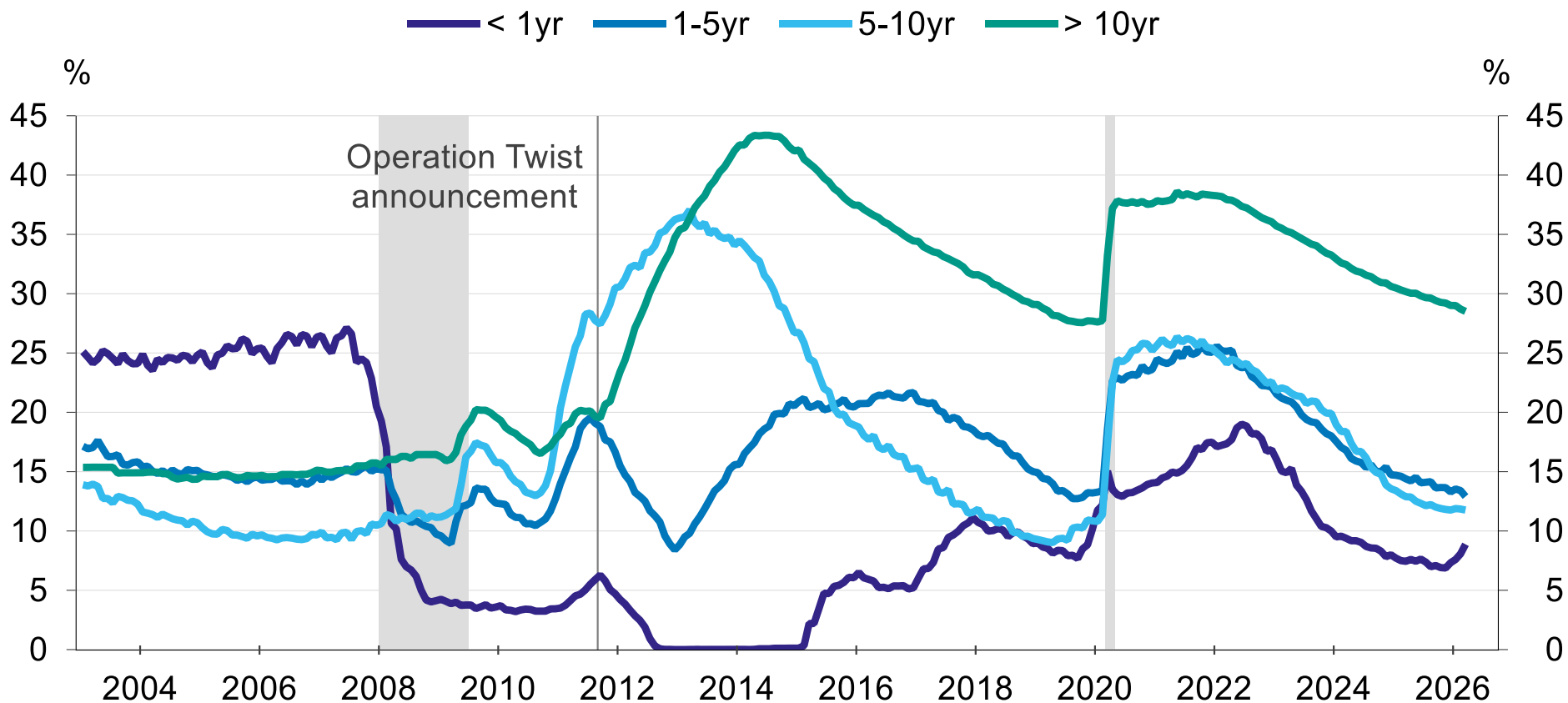


Source: U.S. Treasury, BEA, Macrobond, Deutsche Bank

Federal Reserve still owns close to 30% of Treasuries with 10-year or longer maturity



Fed holdings of Treasury securities outstanding (% by maturity)

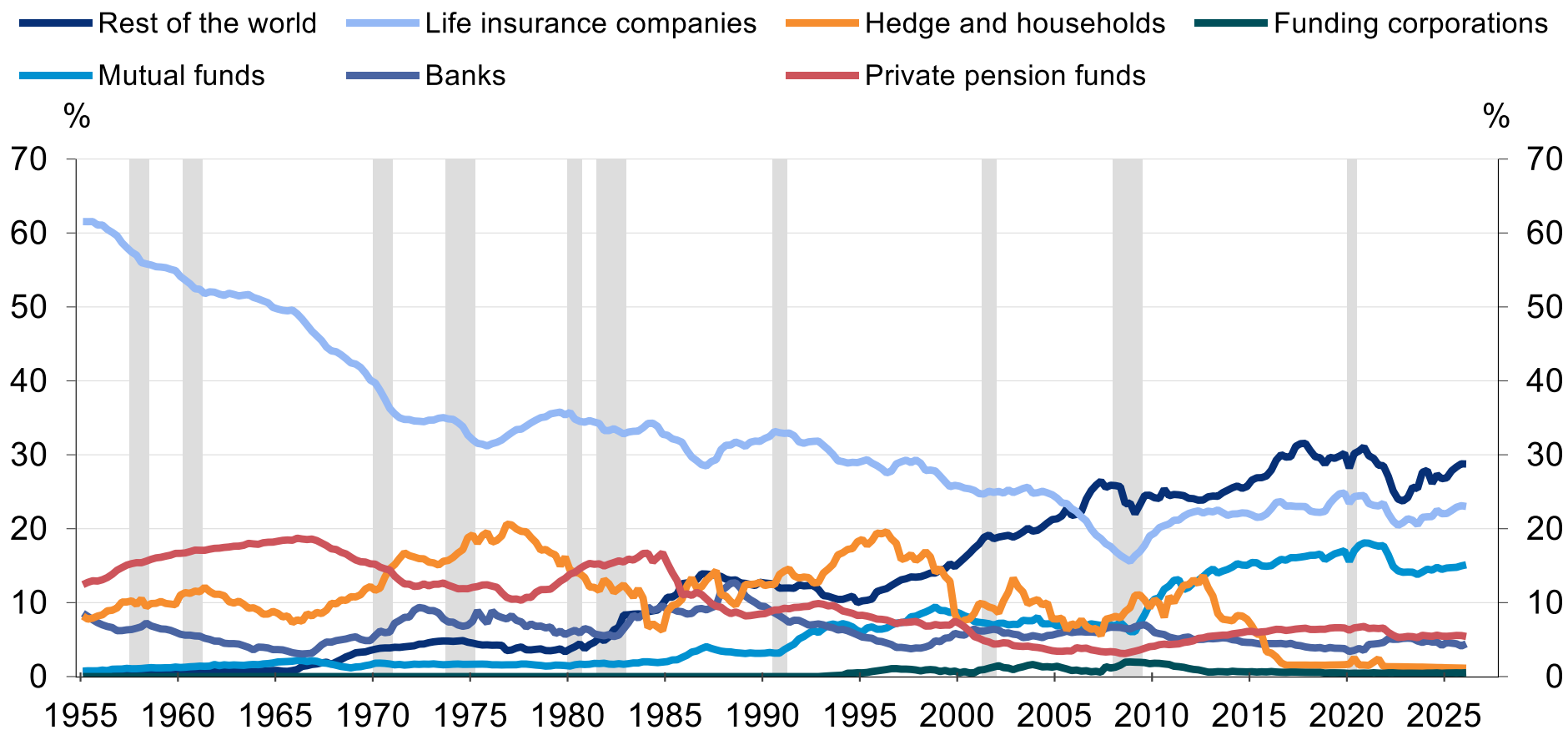


Source: Fed, U.S. Treasury, Macrobond, Deutsche Bank

Foreigners, insurance, and mutual funds make up majority of the investor in credit markets (IG, HY, and loans)



Holdings of corporate and foreign bonds as % outstanding

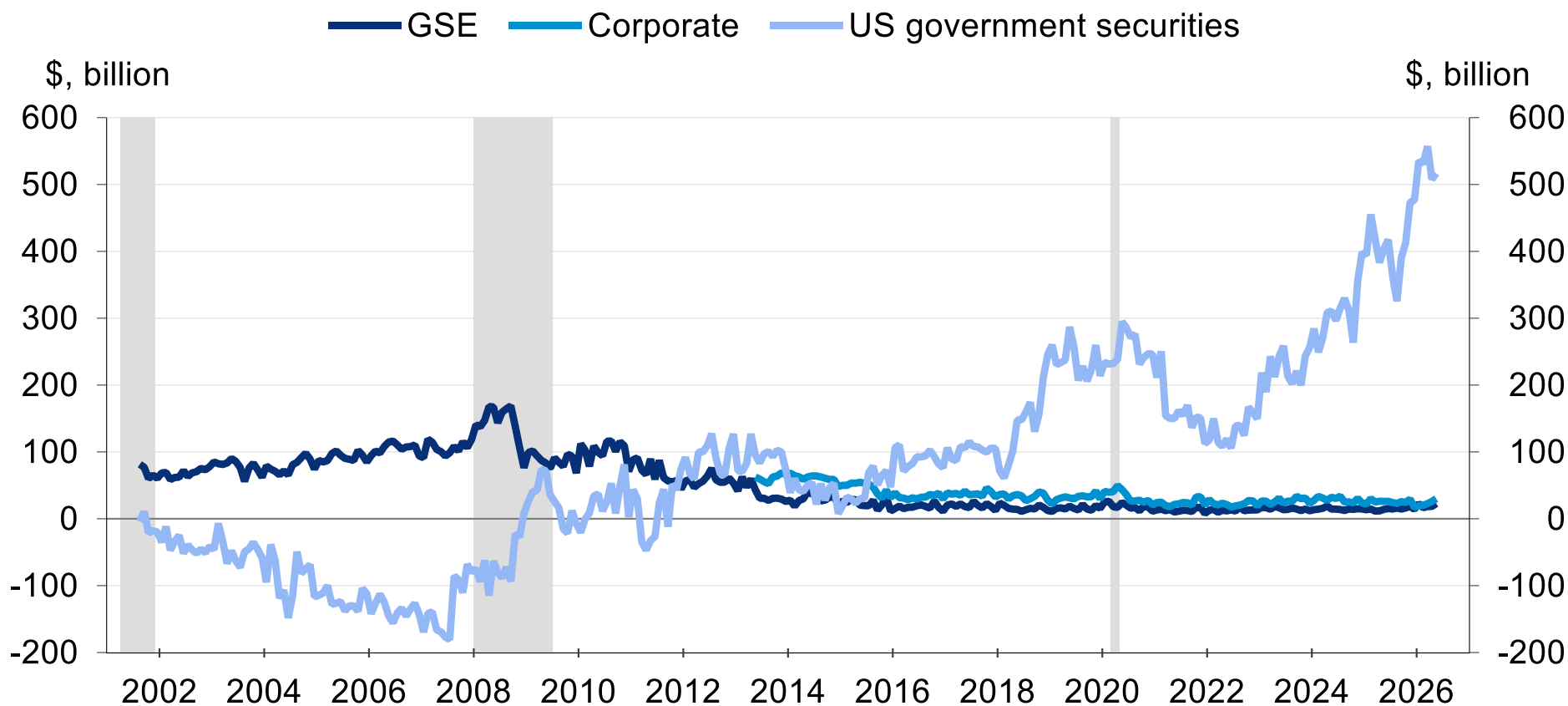


Source: Fed, Macrobond, Deutsche Bank

Primary dealer net holdings of Treasuries have risen sharply since 2022



Primary dealer positions by asset class

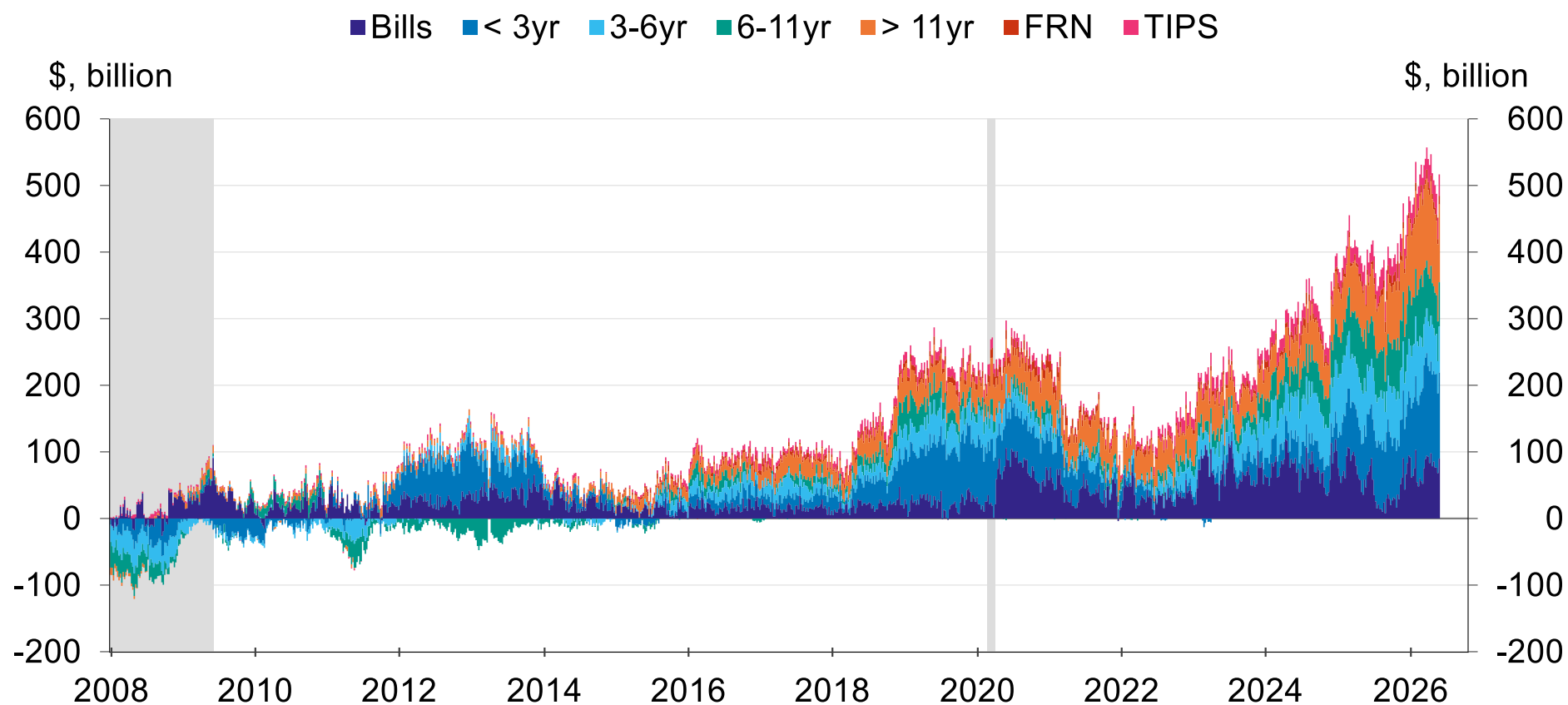


Source: New York Fed, Macrobond, Deutsche Bank

Maturity composition of primary dealers' inventory of Treasuries



Primary Dealer Net Outright Position in US Govt. securities by category

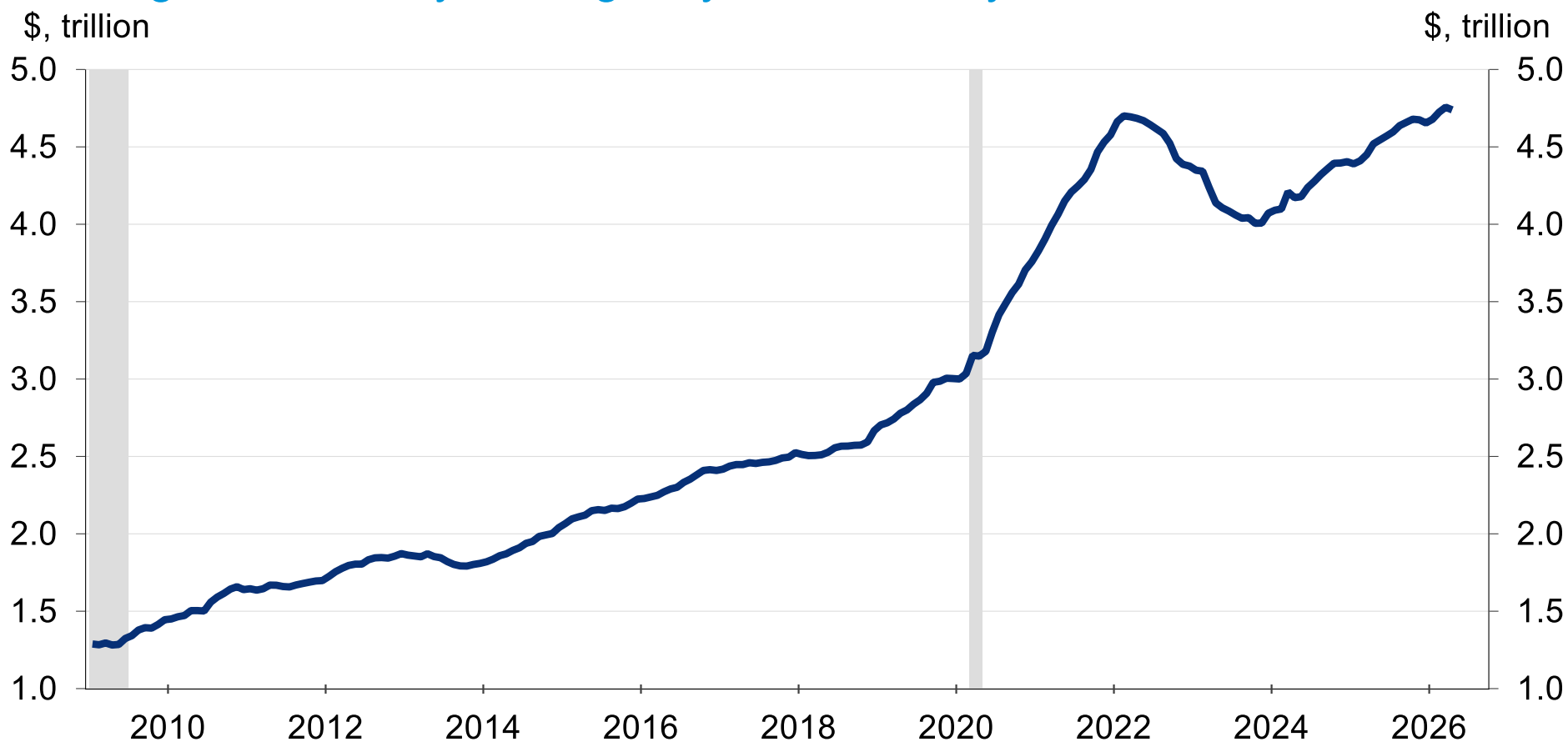


Source: New York Fed, Macrobond, Deutsche Bank

Bank holdings of Treasury and agency debt expanding again after contracting during the Fed tightening cycle



Holdings of Treasury and agency securities by commercial banks



Source: Fed, Macrobond, Deutsche Bank

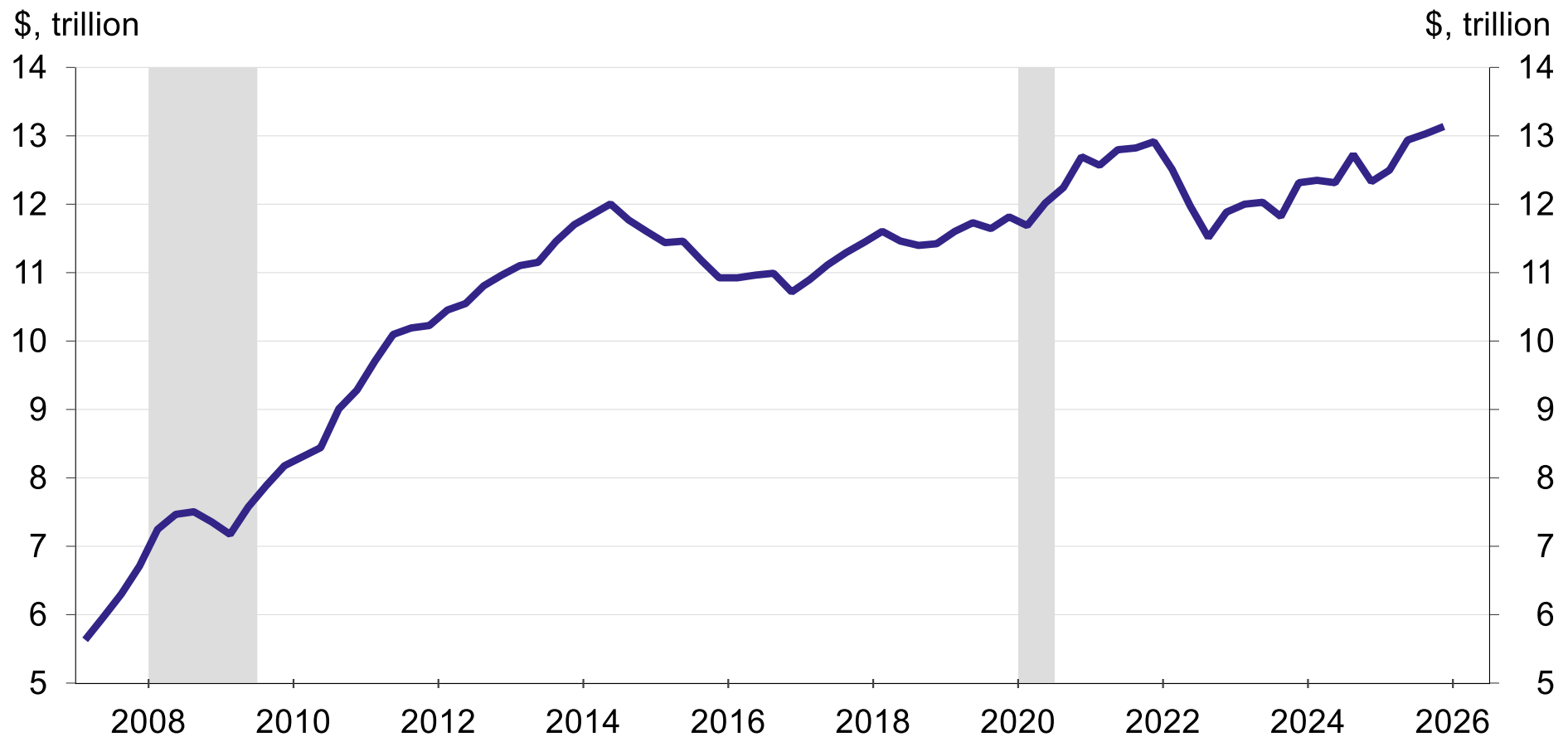


What are foreign central banks doing?

Global FX reserves: \$13trn as of 2025 Q3



Global FX Reserves

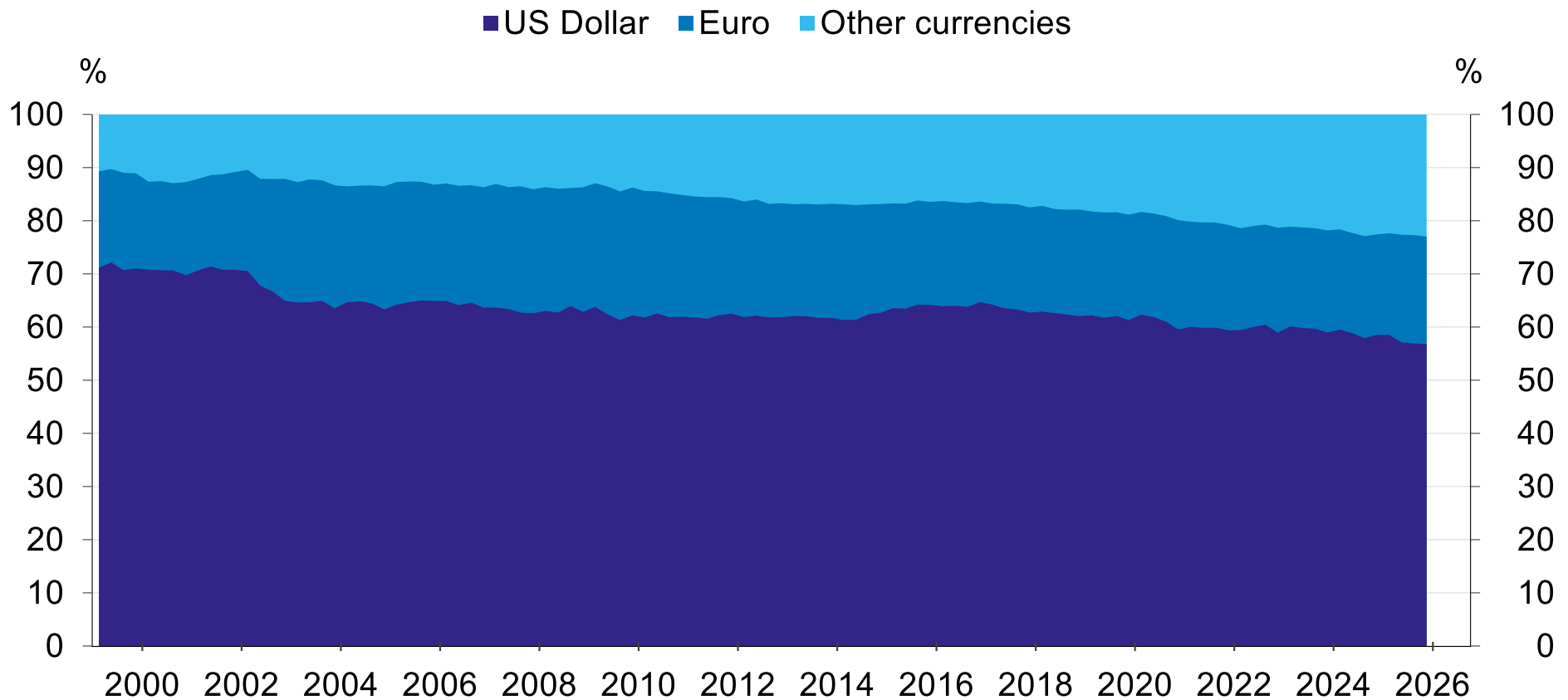


Source: IMF, Macrobond, Deutsche Bank

Global FX reserves: USD share trending lower but remains above 50%



Allocated reserves in different currencies

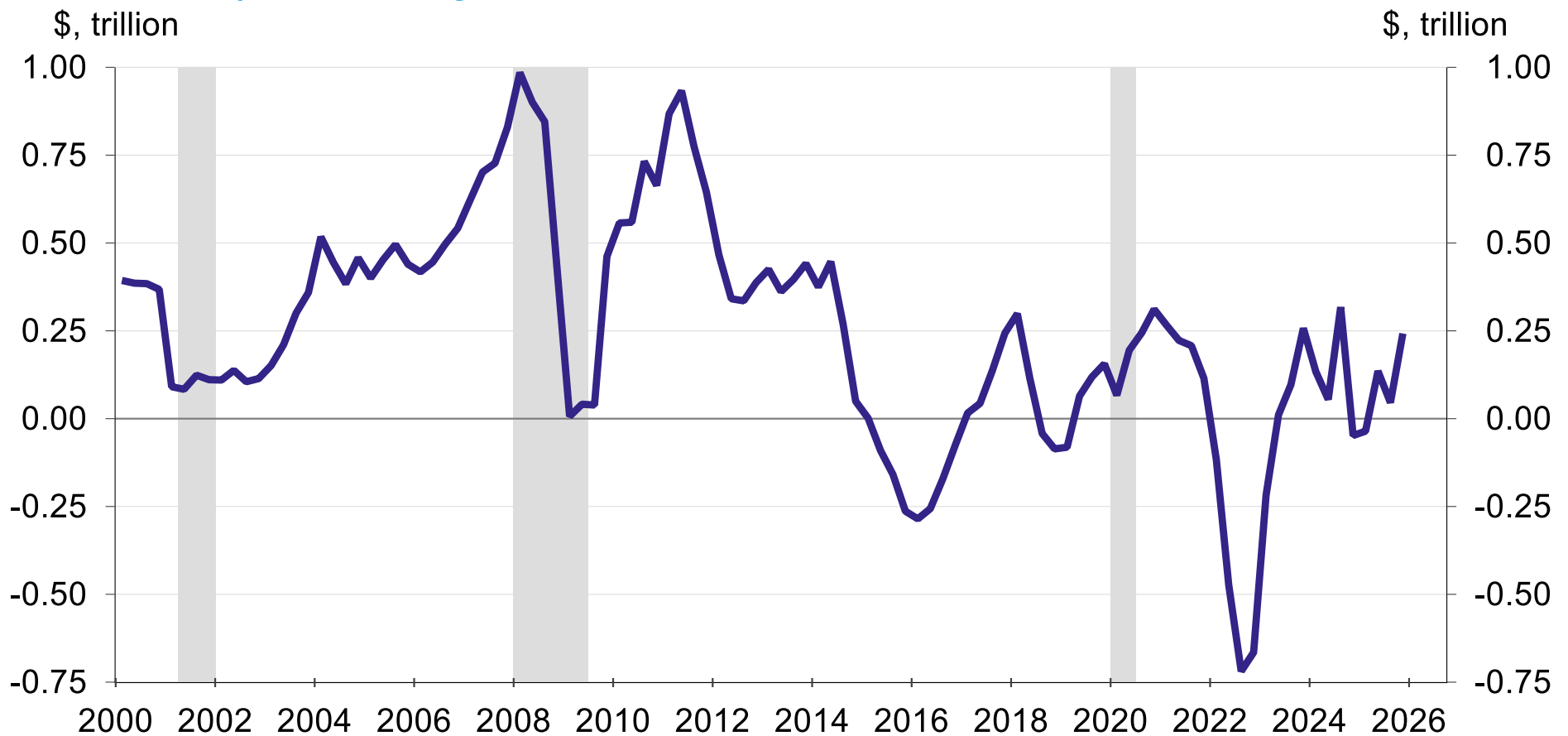


Source: IMF, Macrobond, Deutsche Bank

Value of USD reserves recovering after falling sharply in the last Fed tightening cycle



Year-over-year change in world USD reserves

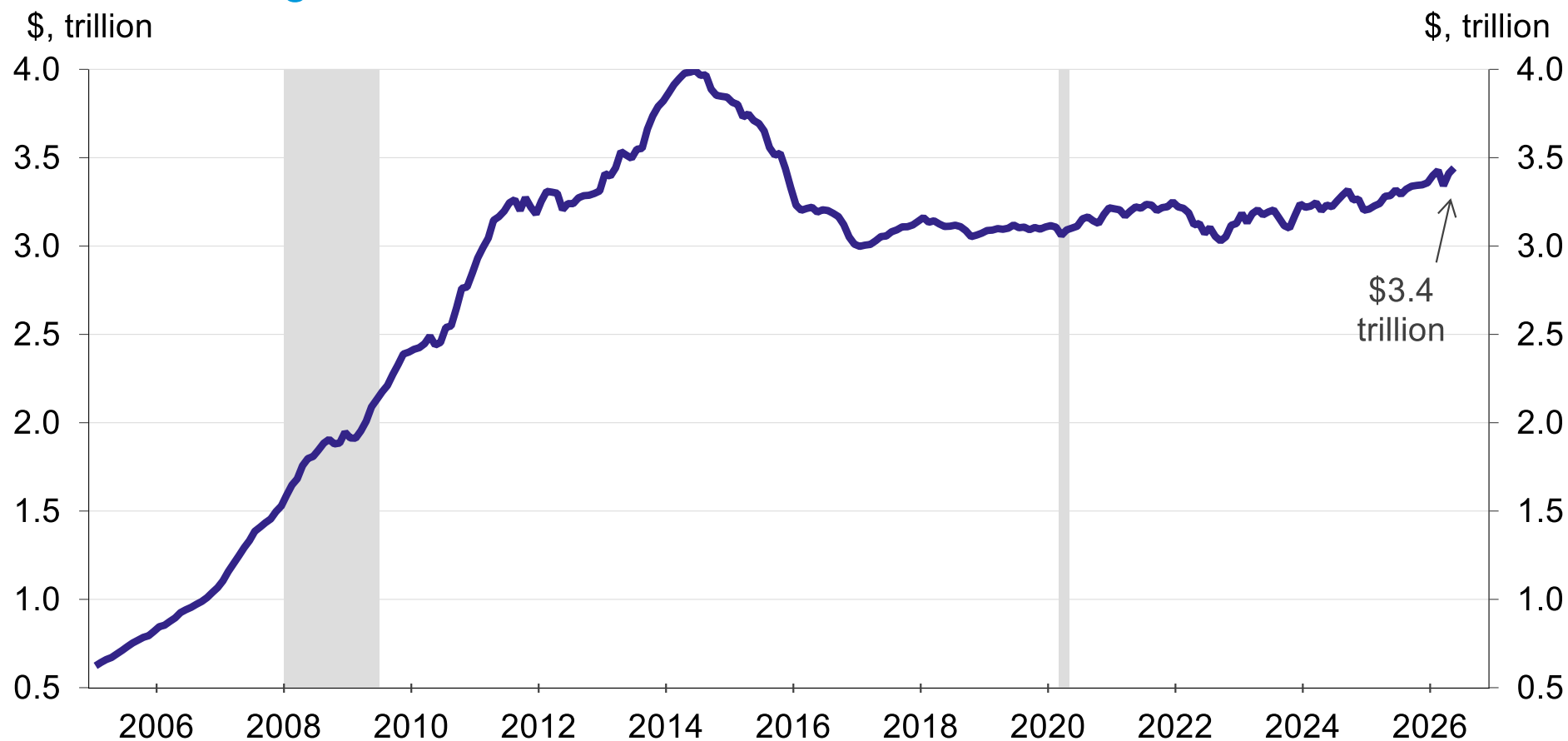


Source: IMF, Macrobond, Deutsche Bank

China's FX reserves rose to \$3.4 trillion as of May 2026



China's foreign reserves

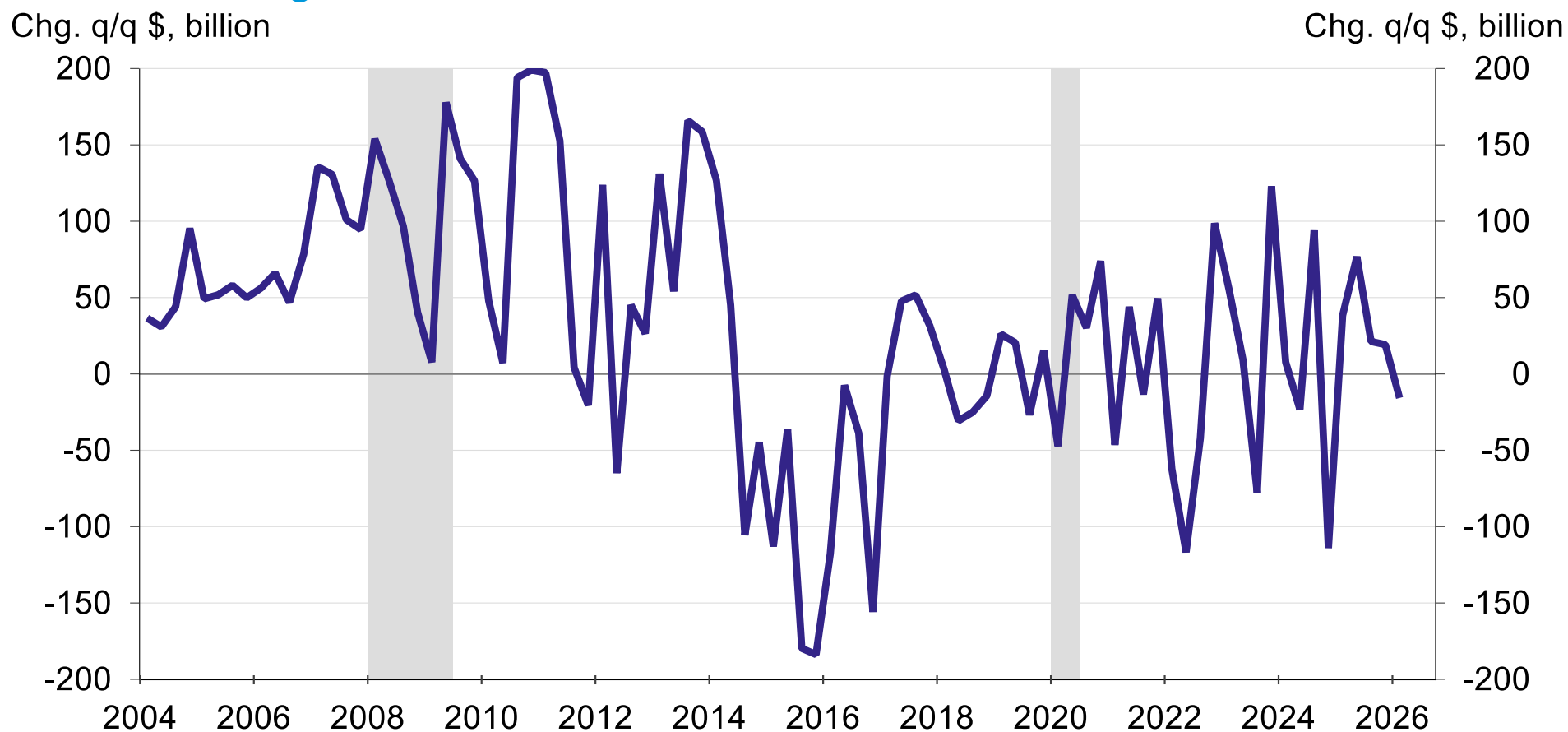


Source: SAFE, Macrobond, Deutsche Bank

Increased volatility in China's FX reserves in recent years



China's foreign reserves

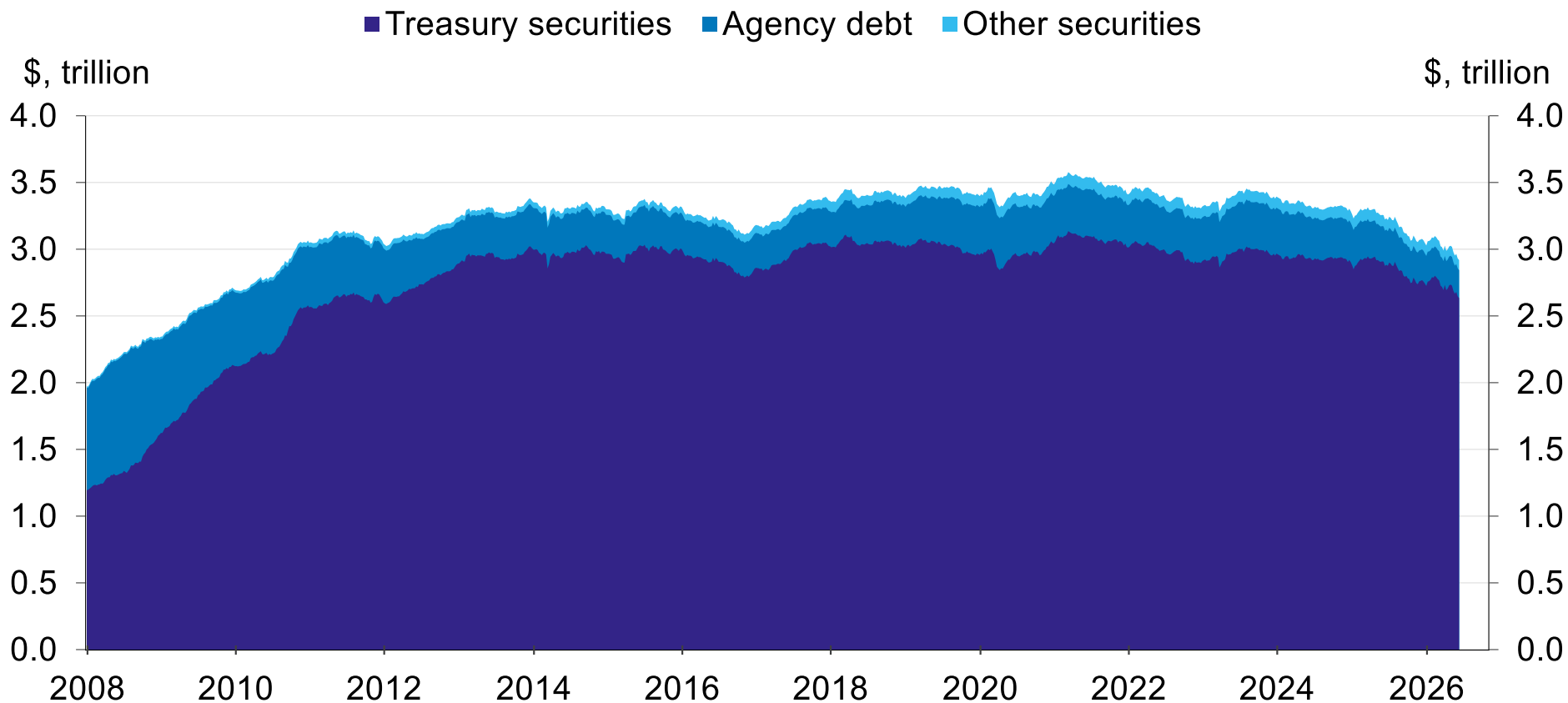


Source: SAFE, Macrobond, Deutsche Bank

Federal Reserve's custodial holdings for foreign official accounts have declined



Fed custody holdings of securities for foreign central banks

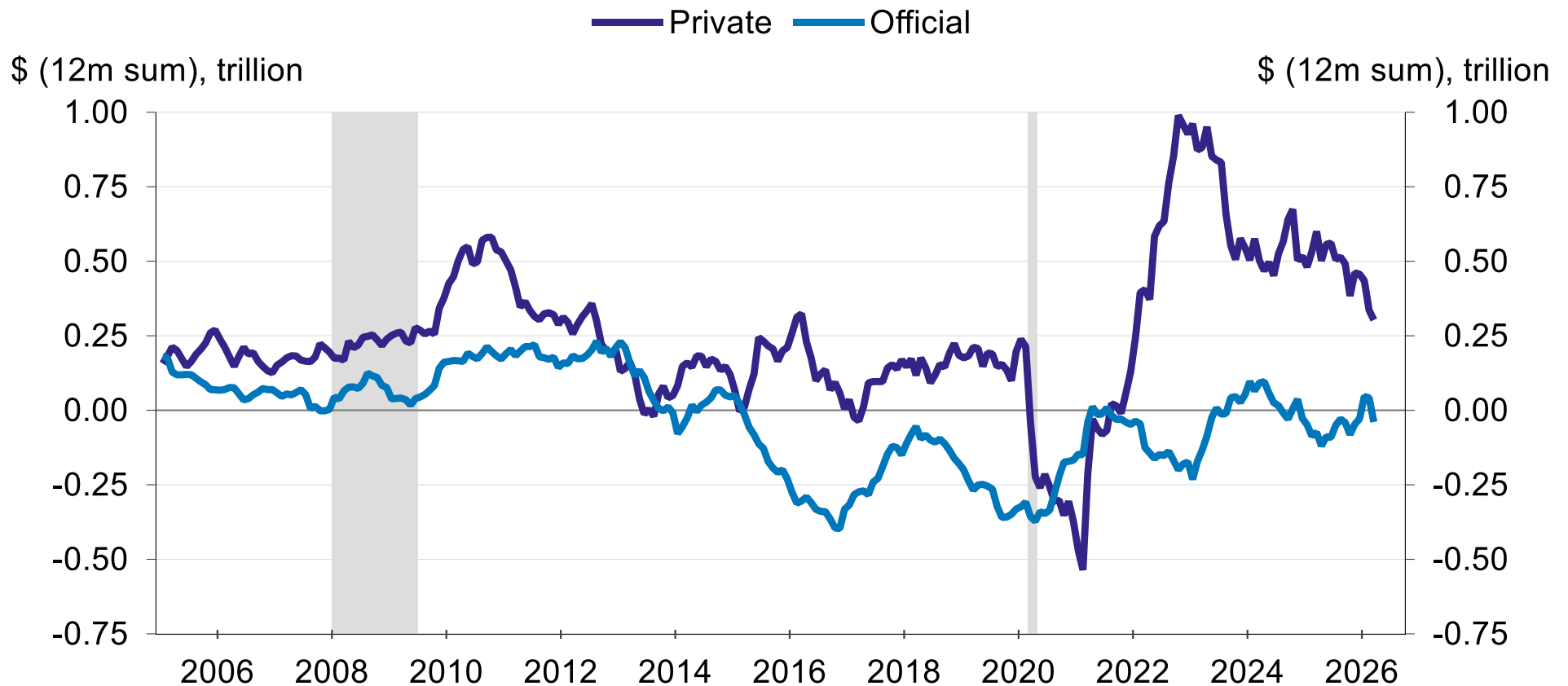


Source: Fed, Macrobond, Deutsche Bank

Foreign private purchases of Treasuries remain positive while foreign official purchases are flat



Net foreign purchases: Treasury bonds and notes

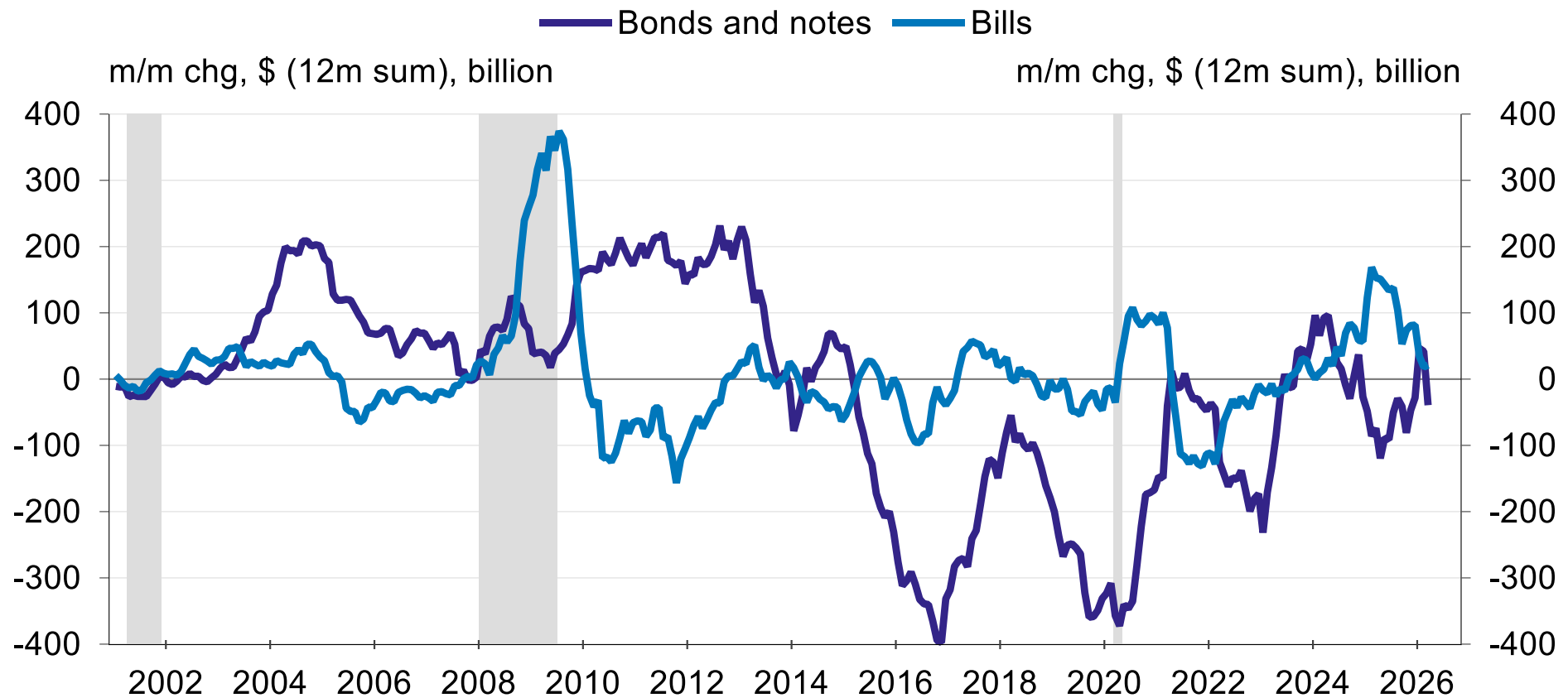


Source: U.S. Treasury, Macrobond, Deutsche Bank

Foreign official holdings shifted from Treasury notes and bonds into T-bills over last year



Holdings of US Treasury: Foreign official institutions

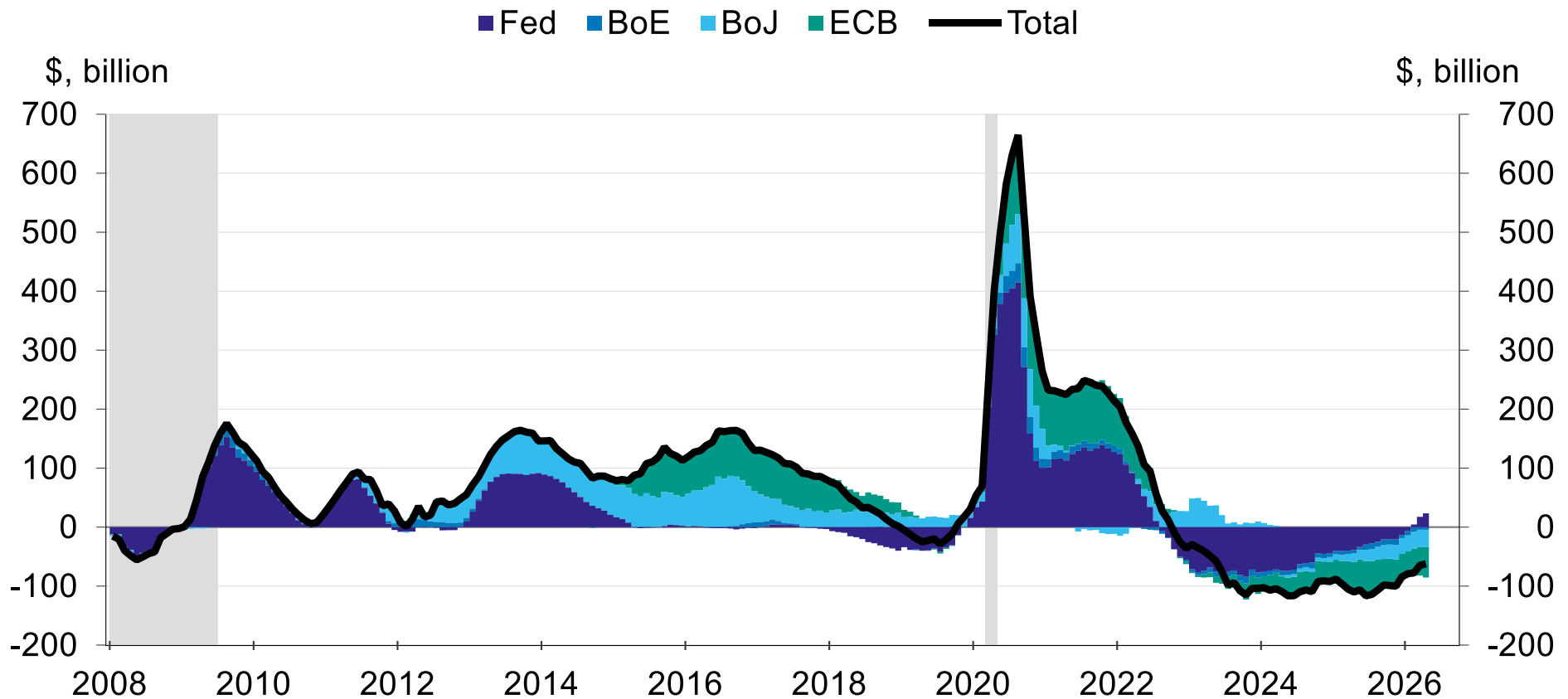


Source: U.S. Treasury, Macrobond, Deutsche Bank

In aggregate, global central banks continued to reduce assets over the past six months



Monthly flow of G4 central bank asset purchases (6m moving average)



Source: Fed, BOJ, ECB, ONS, Macrobond, Deutsche Bank

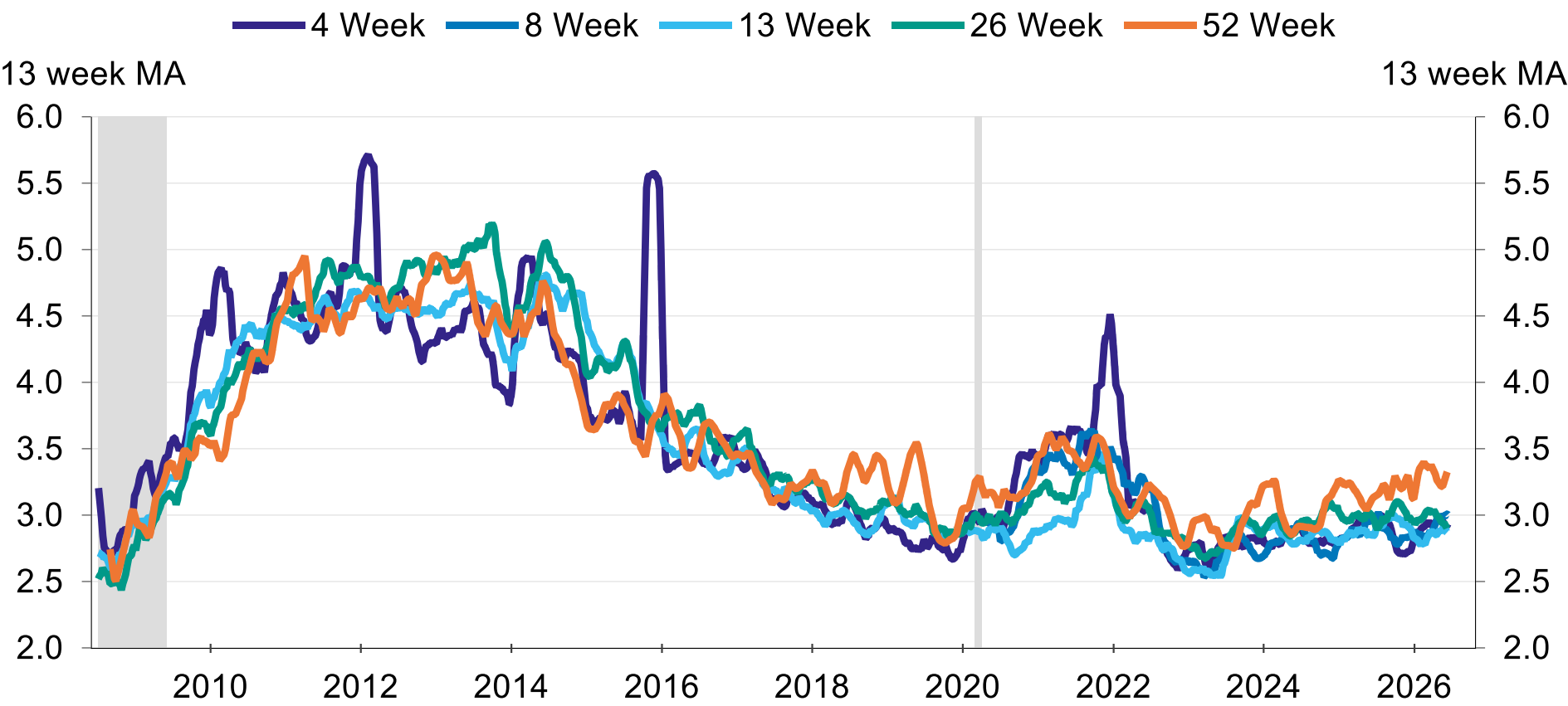


Auction demand

Demand for 52-week bills has increased



Bid-to-cover ratios for T-bill auctions

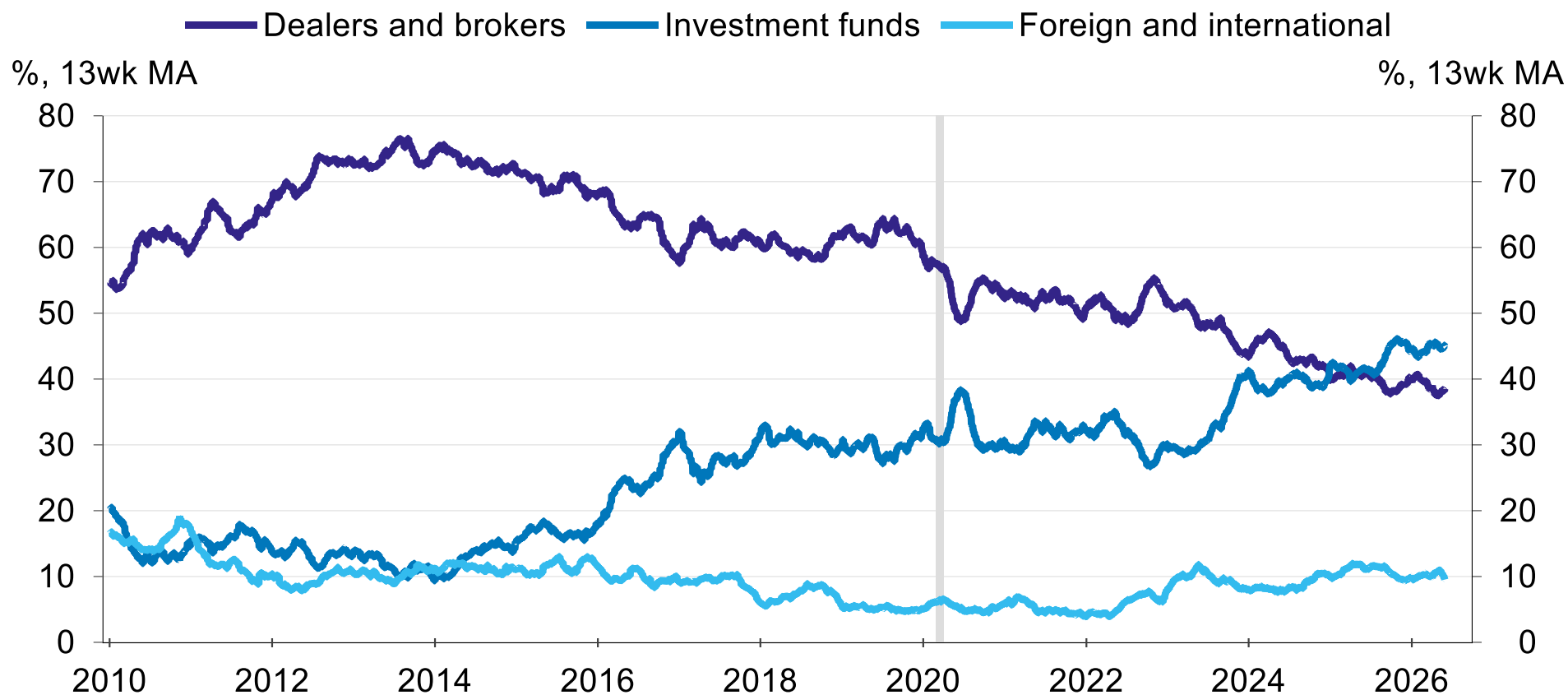


Source: U.S. Treasury, Macrobond, Deutsche Bank

Investment funds have overtaken primary dealers in T-bill auction allotments



Percent awarded by investor class for T-bills

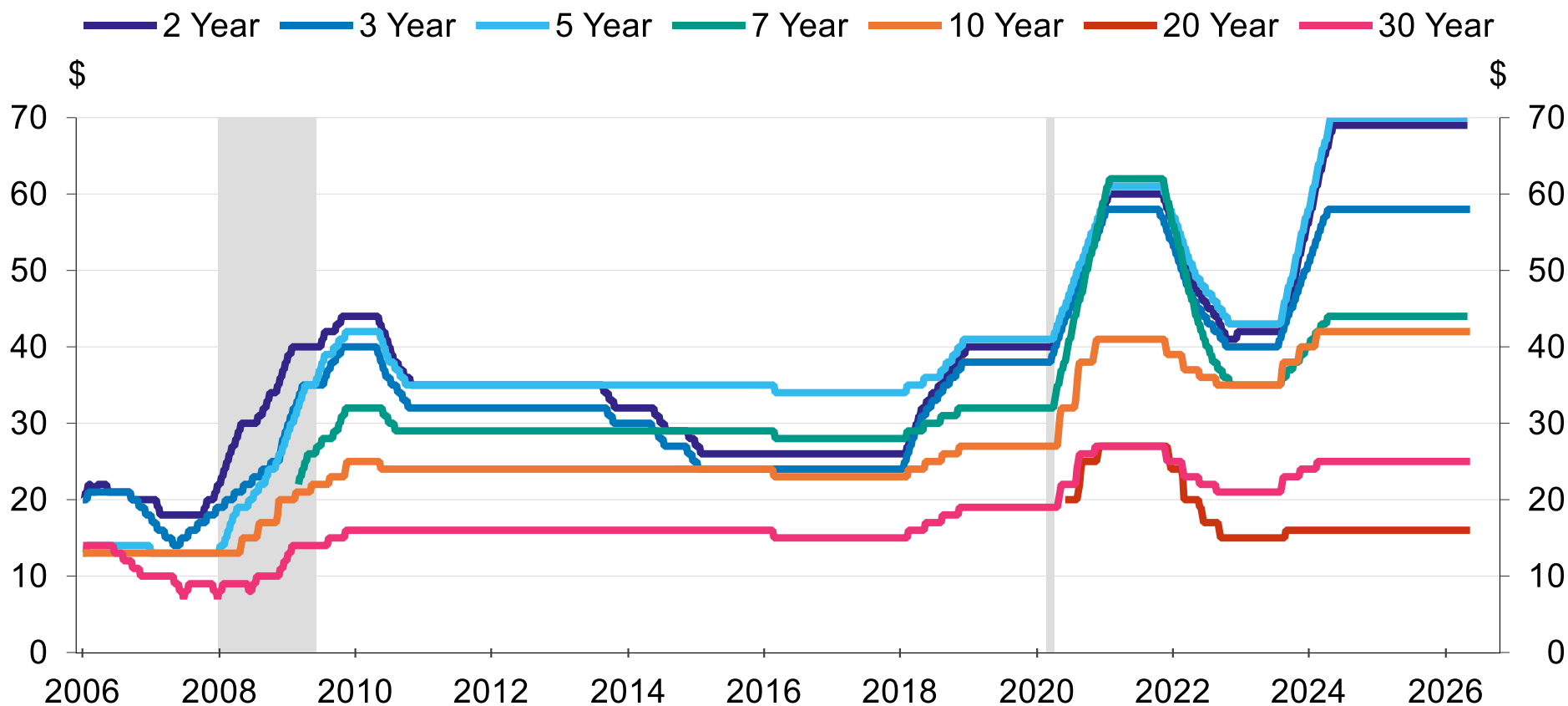


Source: U.S. Treasury, Macrobond, Deutsche Bank

Steady auction sizes for nominal Treasury coupon securities since 2024



Nominal security auction sizes

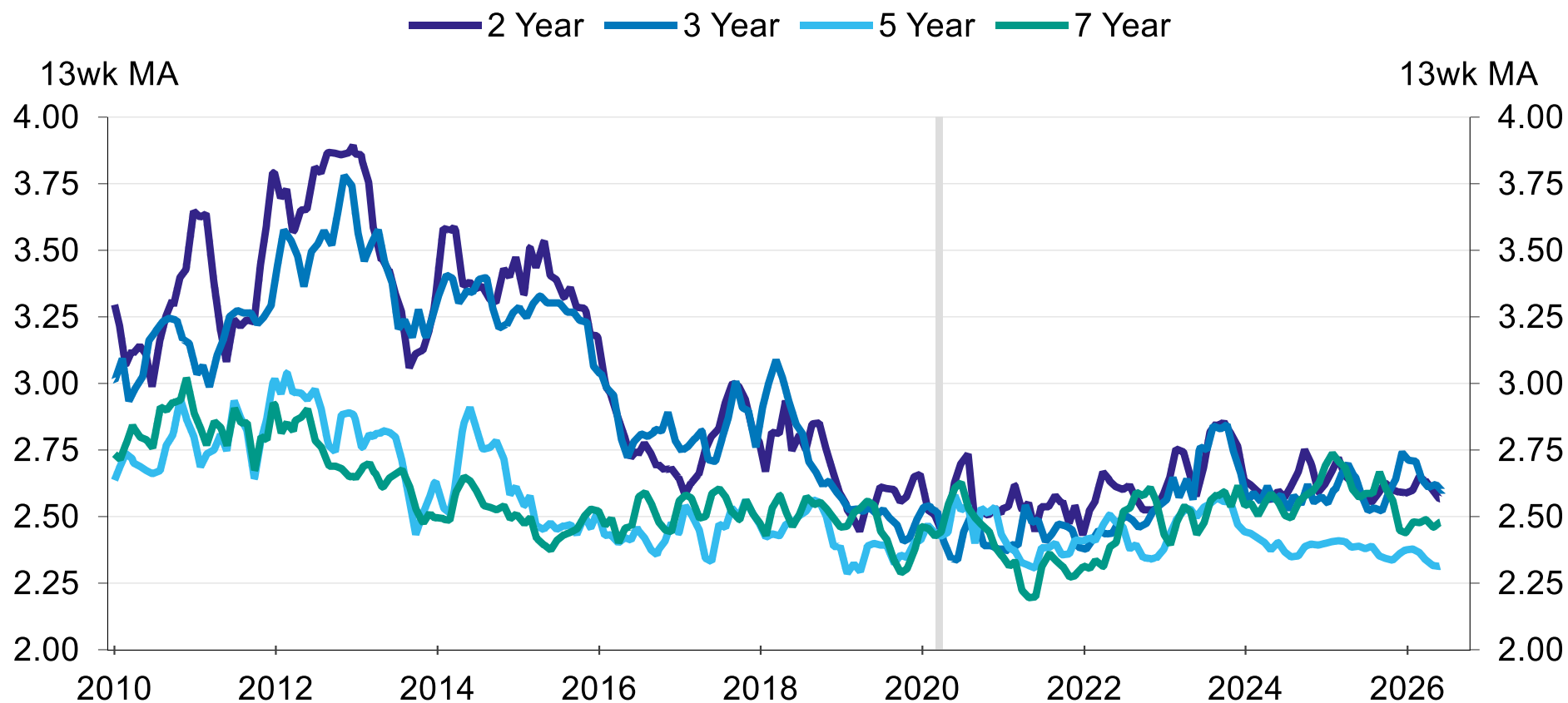


Source: U.S. Treasury, Macrobond, Deutsche Bank

Generally stable demand for 2- to 7-year auctions



Bid-to-cover ratios for 2yr, 3yr, 5yr & 7yr auctions

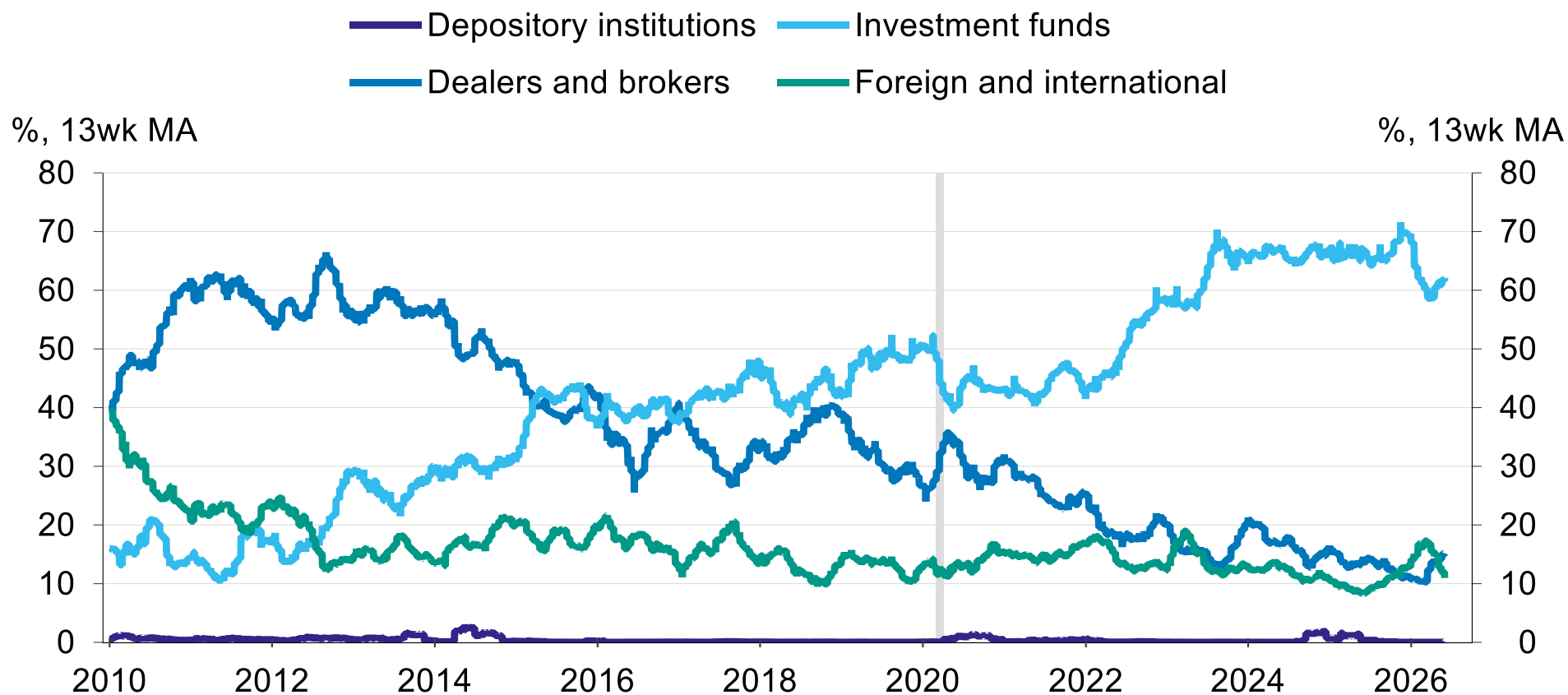


Source: U.S. Treasury, Macrobond, Deutsche Bank

Investment fund allotments in 2- to 7-year auctions fell from recent highs



Percent awarded by investor class: 2yr, 3yr, 5yr & 7yr auctions

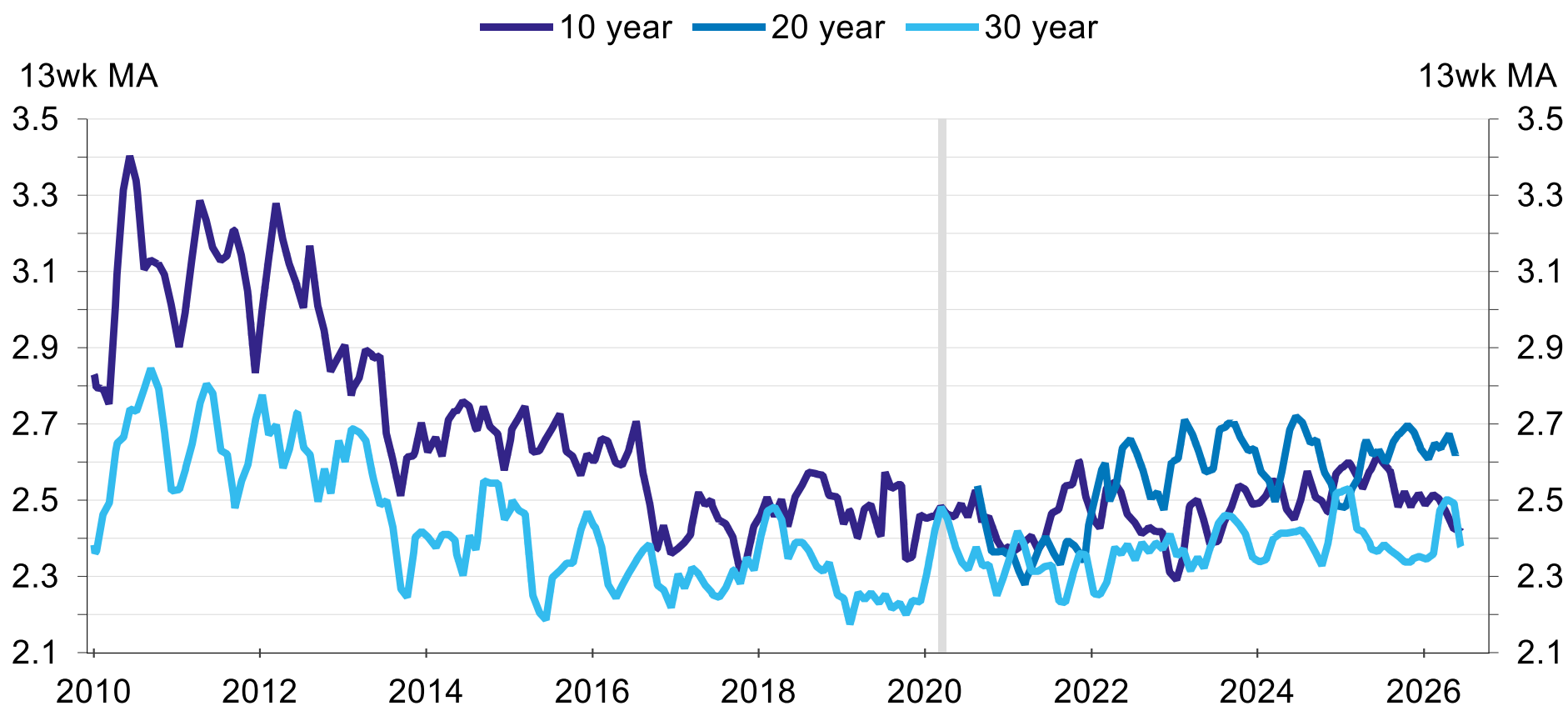


Source: U.S. Treasury, Macrobond, Deutsche Bank

Stable demand for 10- to 30-year auctions



Bid-to-cover ratios for 10yr, 20yr & 30yr auctions

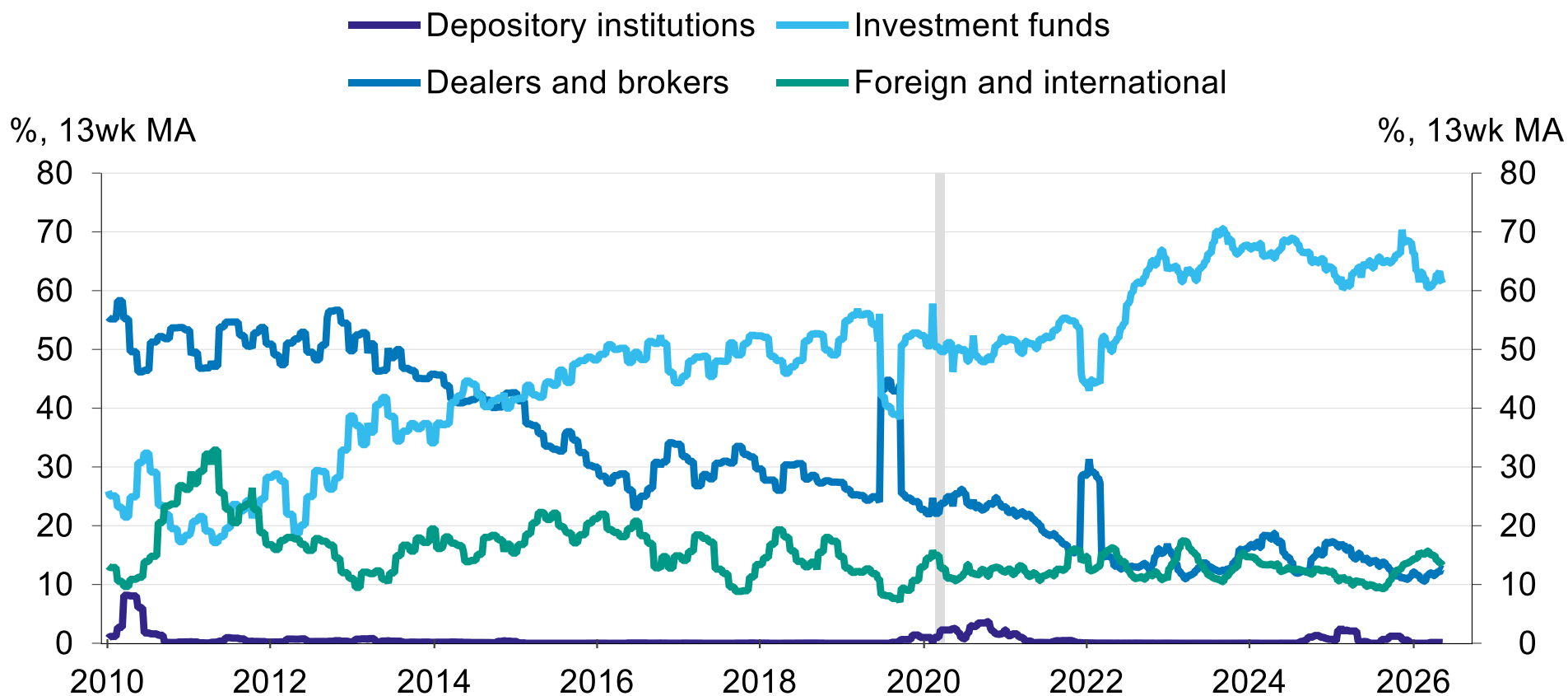


Source: U.S. Treasury, Macrobond, Deutsche Bank

Investment fund allotments in 10- to 30-year auctions fell from recent highs



Percent awarded by investor class: 10yr, 20yr, & 30yr auctions



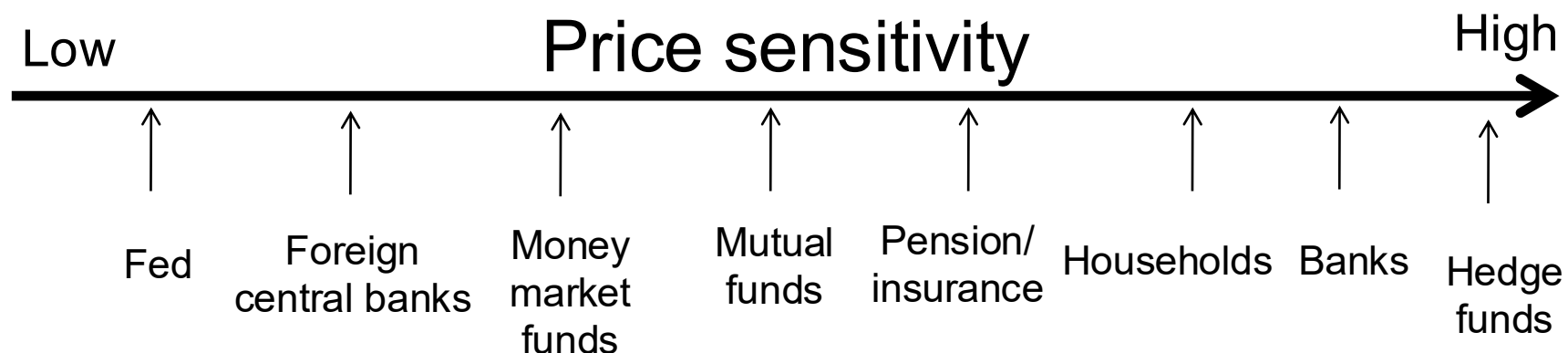
Source: U.S. Treasury, Macrobond, Deutsche Bank



Who is buying Treasuries?



Different buyers of fixed income have different price sensitivities:



Source: Deutsche Bank

Foreigners remain the largest investors in Treasury debt



Holdings of Treasury securities (%)

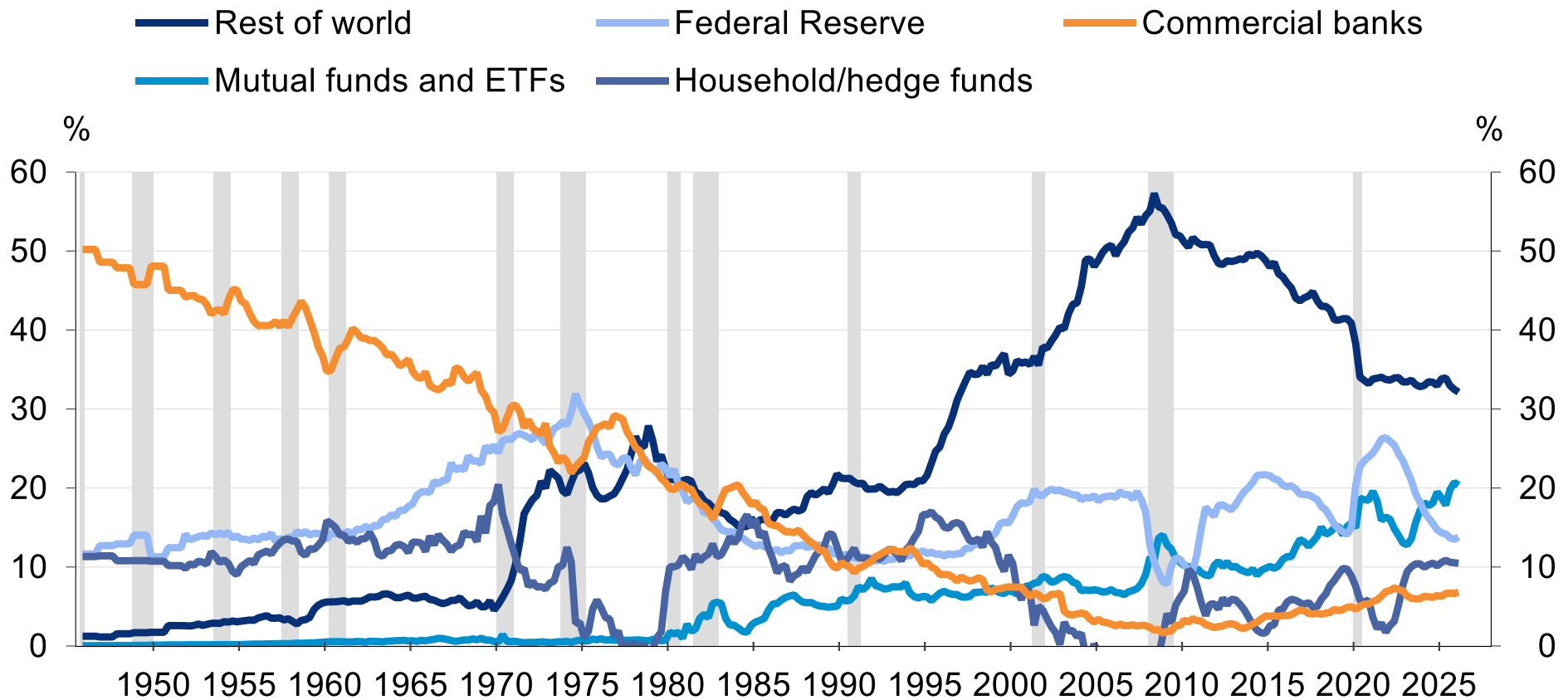
	2026 Q1	2019 Q2	2007 Q2
Rest of world	32.1	41.4	54.3
Mutual funds and ETFs	20.4	14.1	7.3
Federal Reserve	13.8	14.5	19.6
Household/hedge funds	10.5	9.8	-6.8
Others	7.0	0.6	9.1
Commercial banks	6.8	4.6	2.5
State and local governments	5.4	3.9	8.9
Investment banks	2.6	2.0	-2.7
Insurance companies	2.1	2.3	4.4
Private pensions	2.1	2.6	3.2
Government retirement funds	2.1	2.6	4.5
GSEs	0.8	0.9	0.4
ABS issuers	0.1	0.2	1.7

Source: Fed, Macrobond, Deutsche Bank. Note “Others” category includes i. Other Financial Business, ii. Non-financial Corporate and Noncorporate Business, iii. Credit Unions and iv. Instrument Discrepancies. Total may not sum to total 100% due to rounding.

Top 5 investor categories in US Treasuries



Holdings of Treasuries as a % of Treasury debt outstanding - Top 5

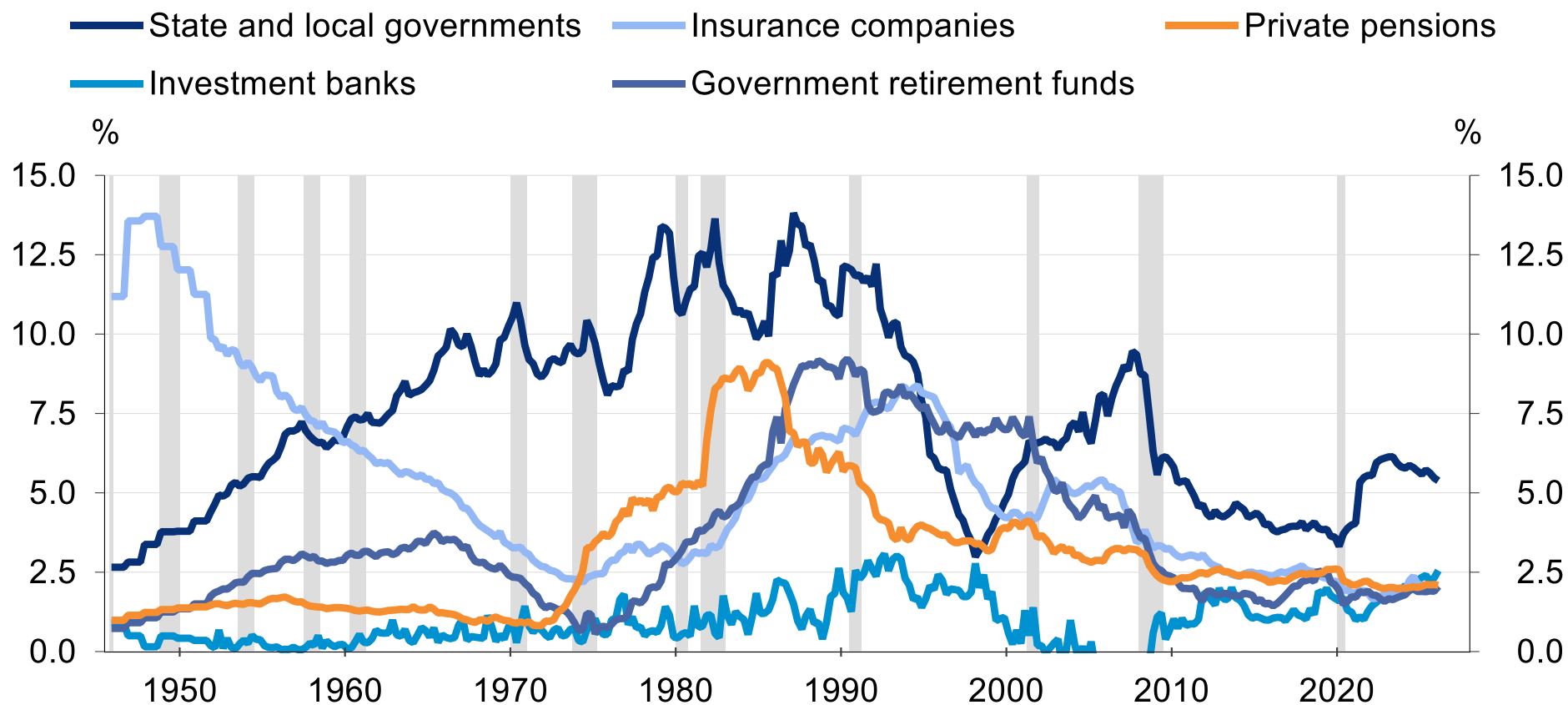


Source: Fed, Macrobond, Deutsche Bank

Other investor categories (6th-10th) in US Treasuries



Holdings of Treasuries as a % of Treasury debt outstanding - Next 5

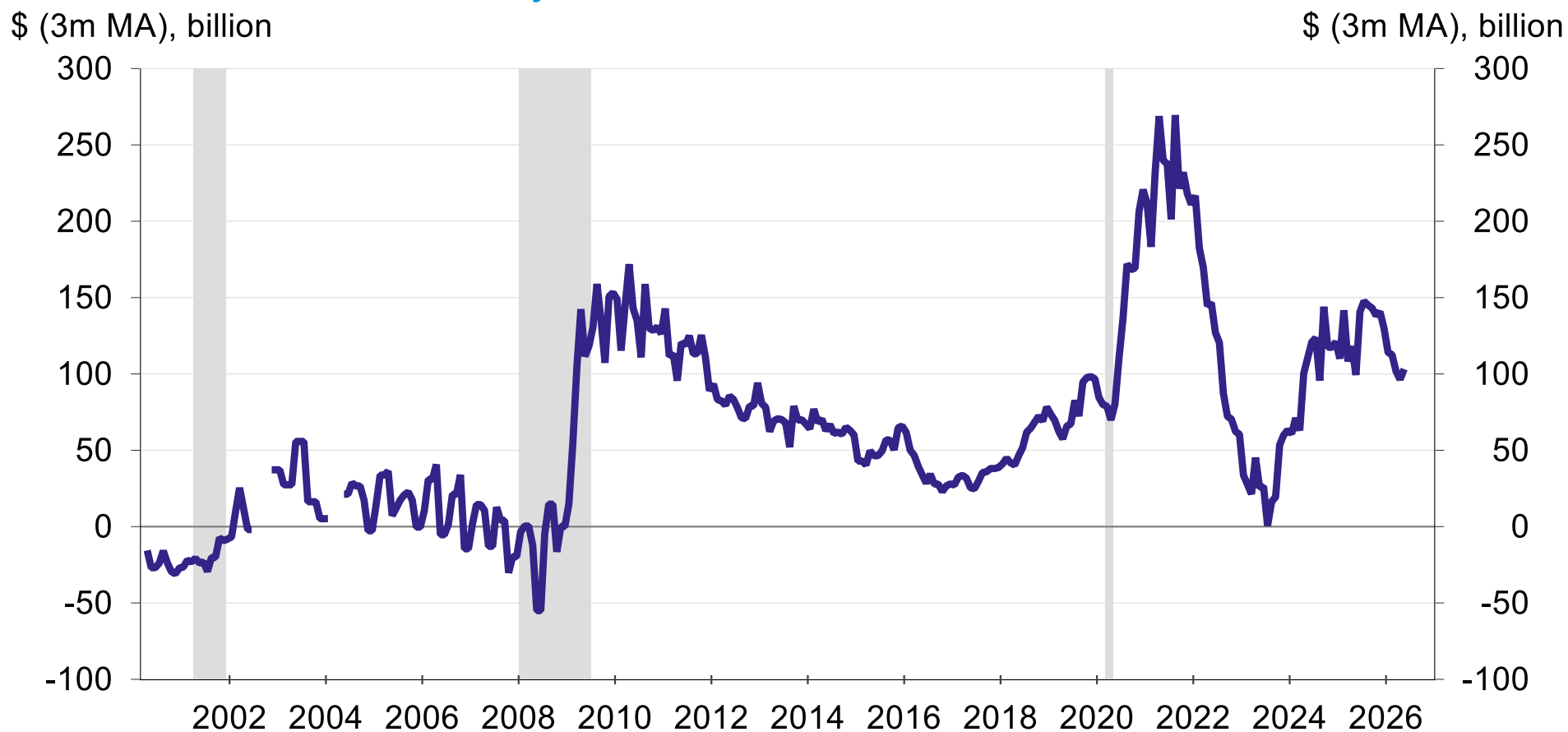


Source: Fed, Macrobond, Deutsche Bank

Net issuance of Treasury notes and bonds remains elevated relative to pre-Covid levels



Net issuance of Treasury notes and bonds

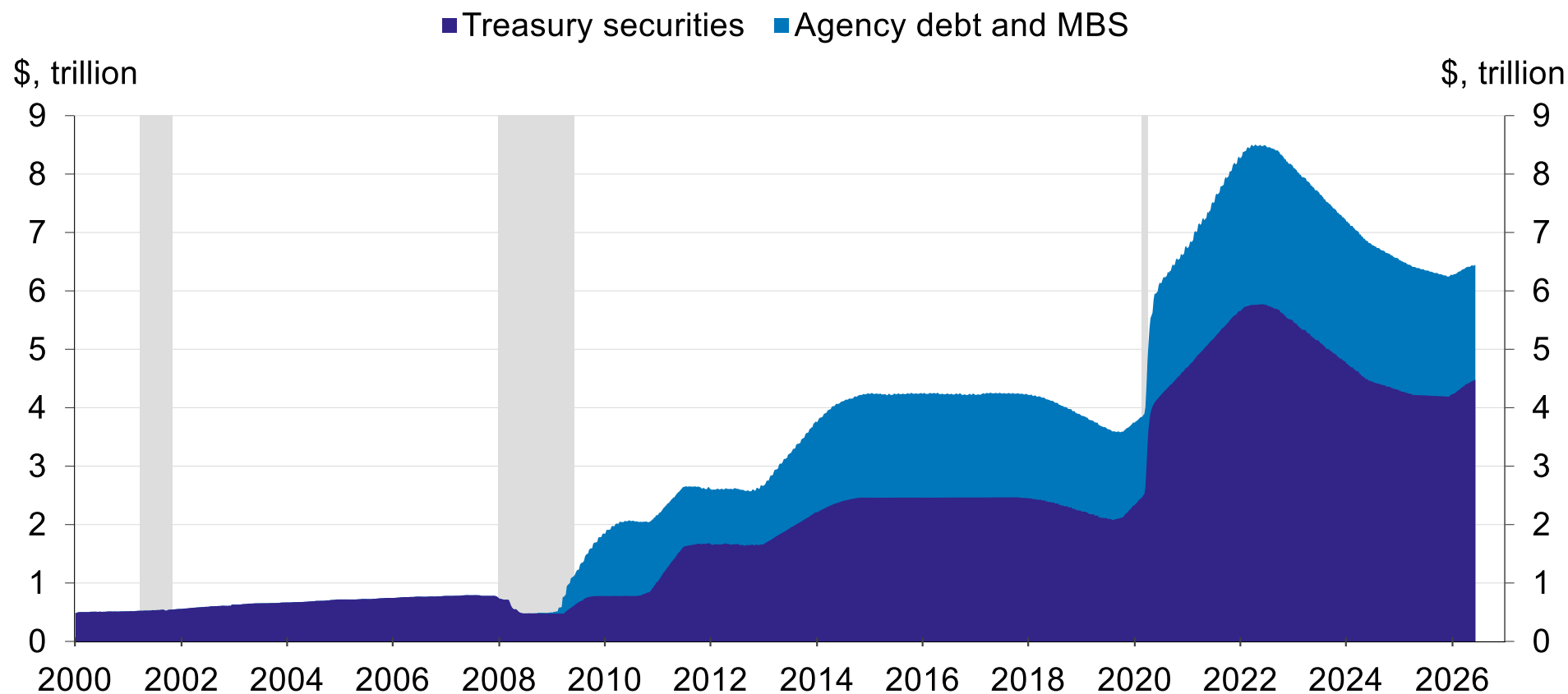


Source: SIFMA, Macrobond, Deutsche Bank

The Fed's bond portfolio fell by more than \$2trn during QT but is rising again with reserve management purchases (RMPs)



Treasuries and MBS outstanding held by the Fed

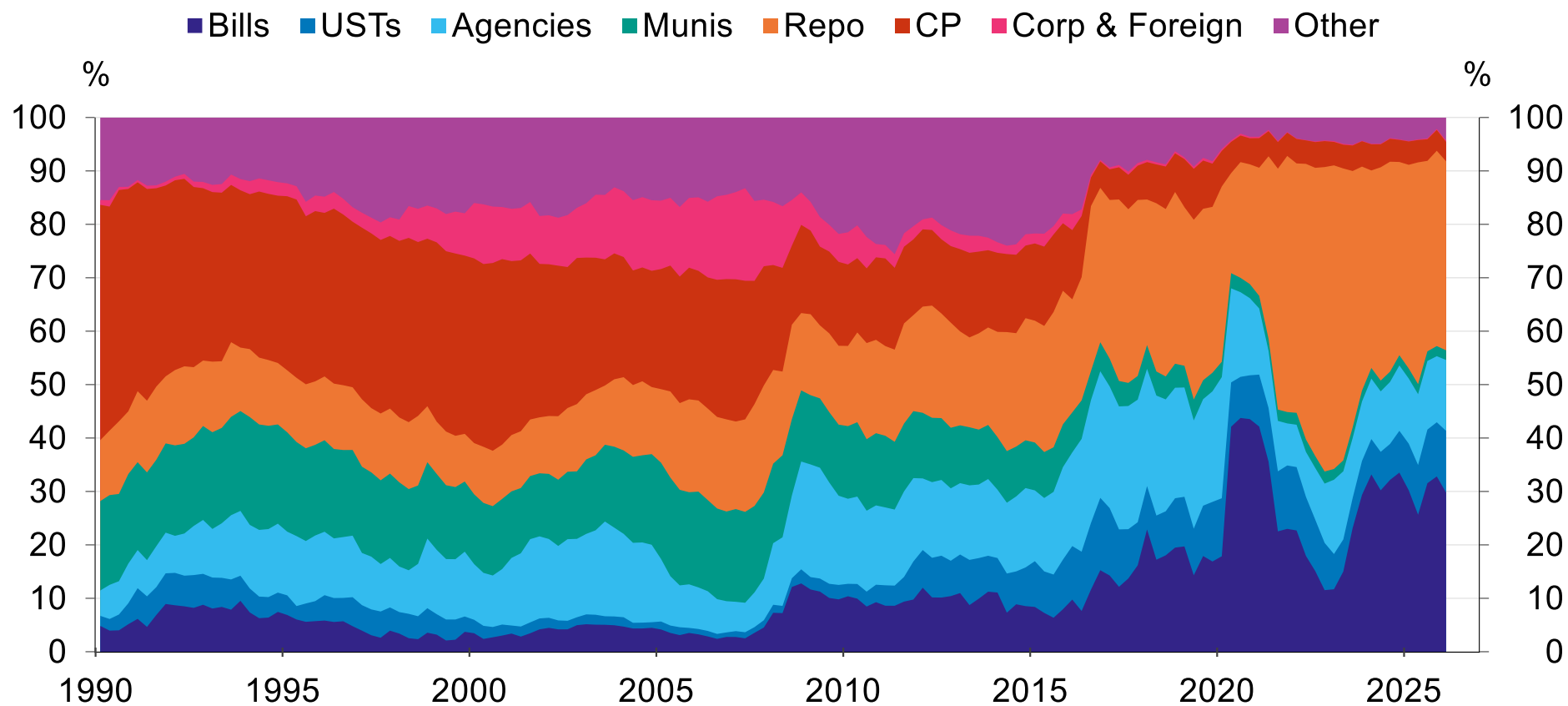


Source: Fed, Macrobond, Deutsche Bank

Money market funds hold primarily short-term Treasury securities and repo



Money market mutual funds holdings

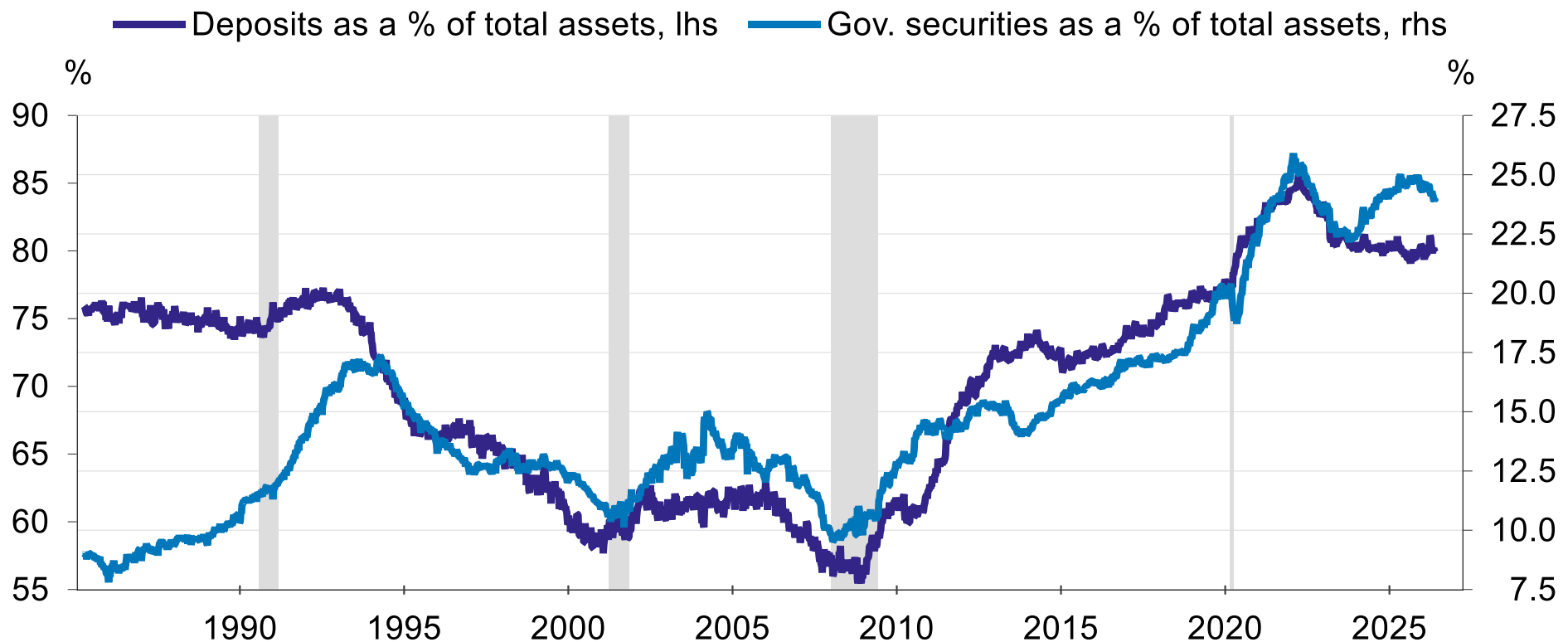


Source: Fed, Macrobond, Deutsche Bank

Bank holdings of Treasury and agency securities track their deposit liabilities



Large domestically chartered commercial banks holdings of US Treasuries & Deposits

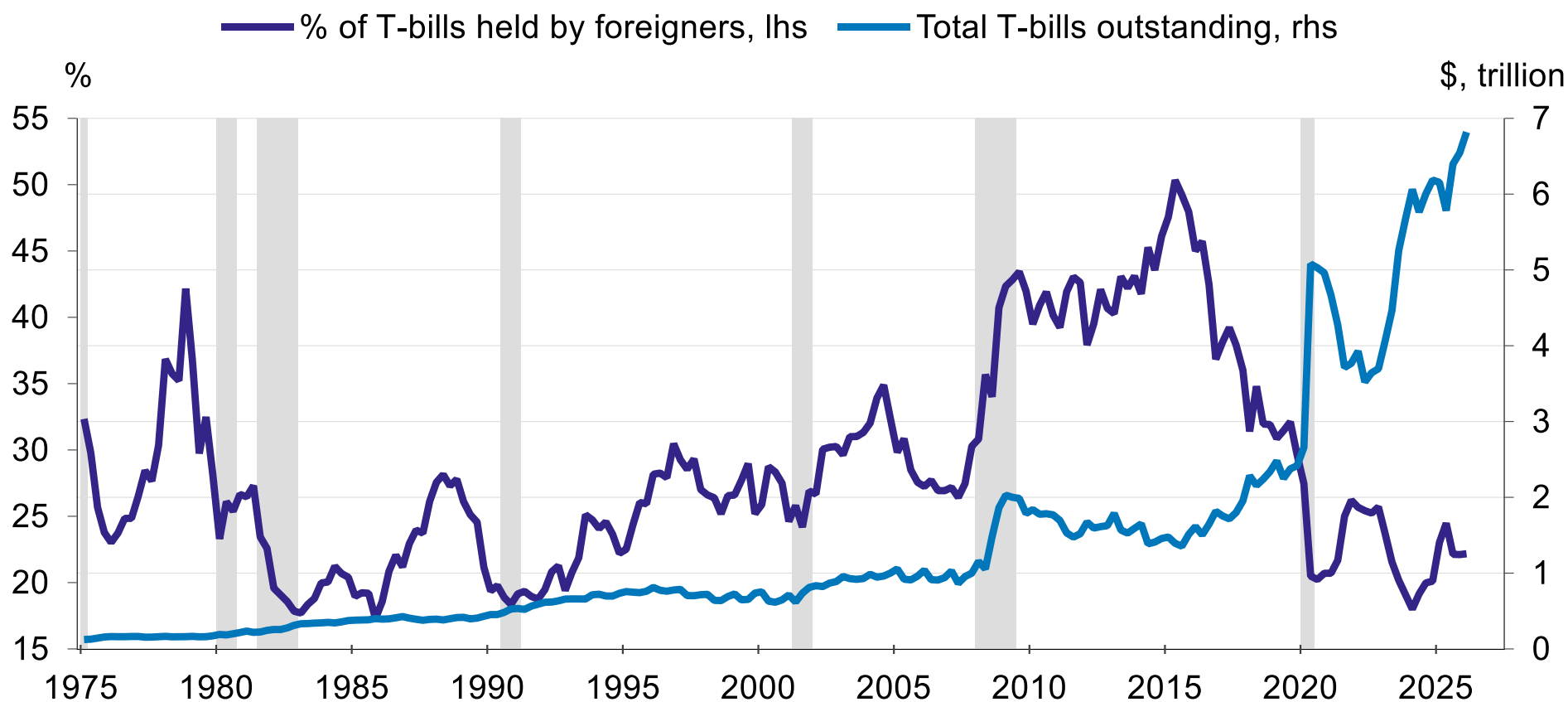


Source: Fed, Macrobond, Deutsche Bank

Foreign ownership of T-bills nears a multi-decade low



Treasury bills held by foreigners as % of outstanding

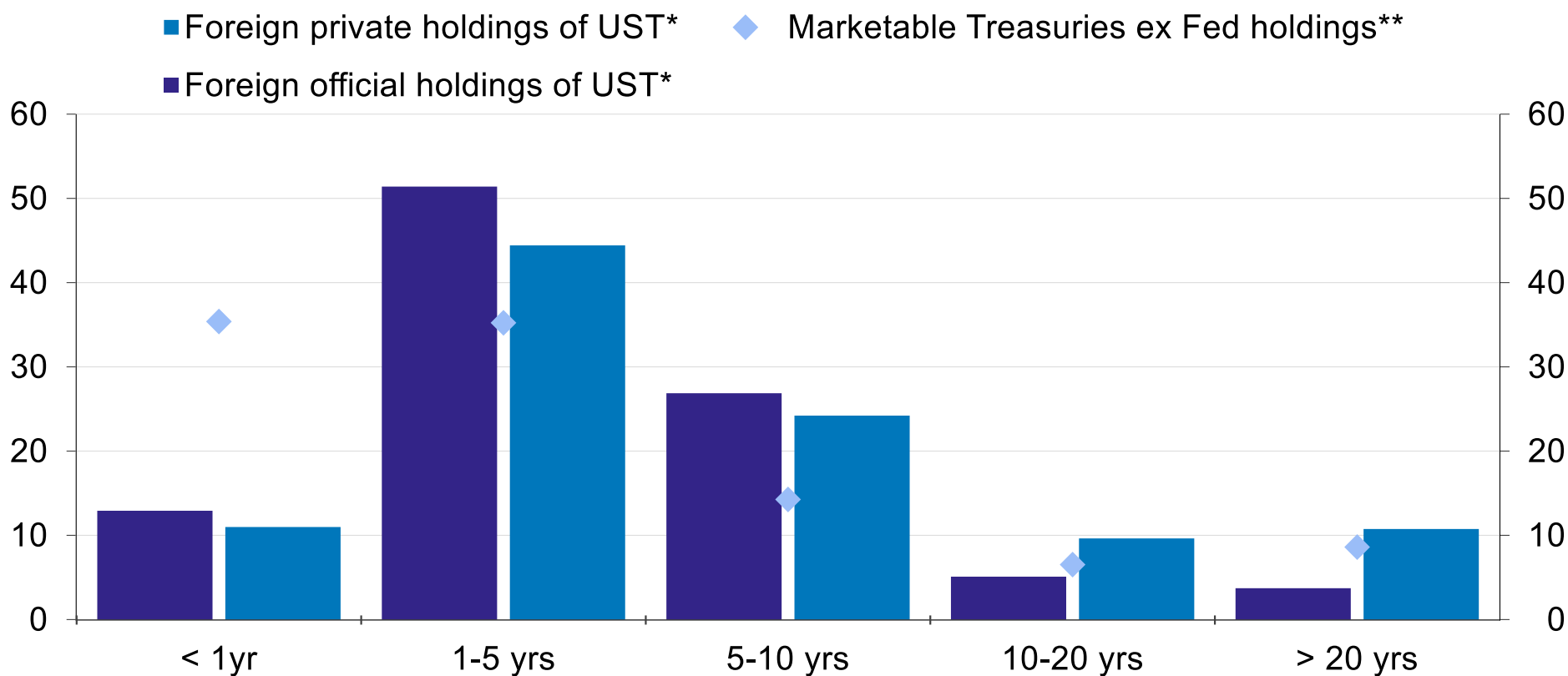


Source: Fed, Macrobond, Deutsche Bank

Foreign private holdings skew toward longer maturities compared to foreign official holdings



Maturity structure of Treasury holdings



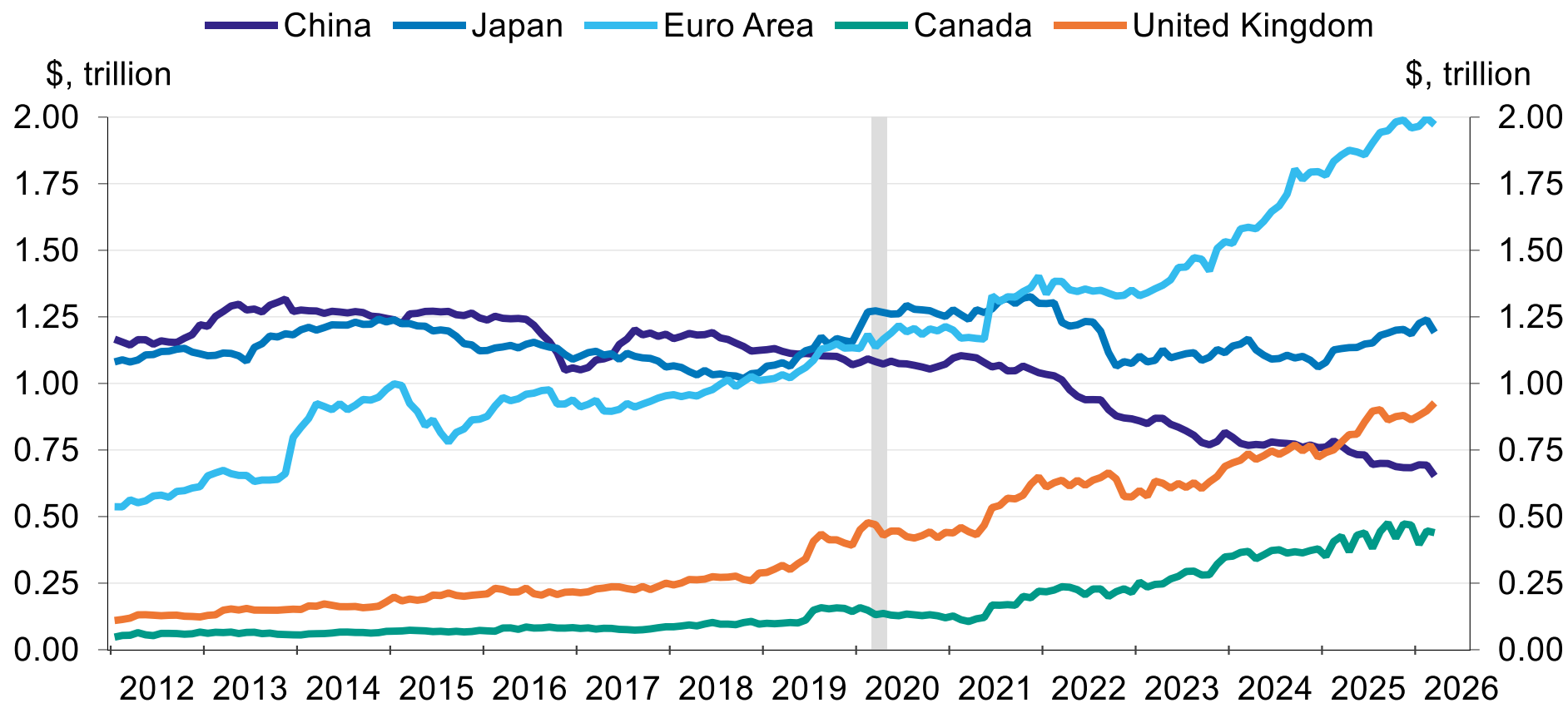
Source: U.S. Treasury, Macrobond, Deutsche Bank

* Latest data for 2025 released on 5/1/2026 ** Latest data for March 2026 released on 6/3/2026

China's reported holdings of Treasuries continues to fall, while Europe, UK, and Canadian holdings are rising



Holdings of US Treasury securities

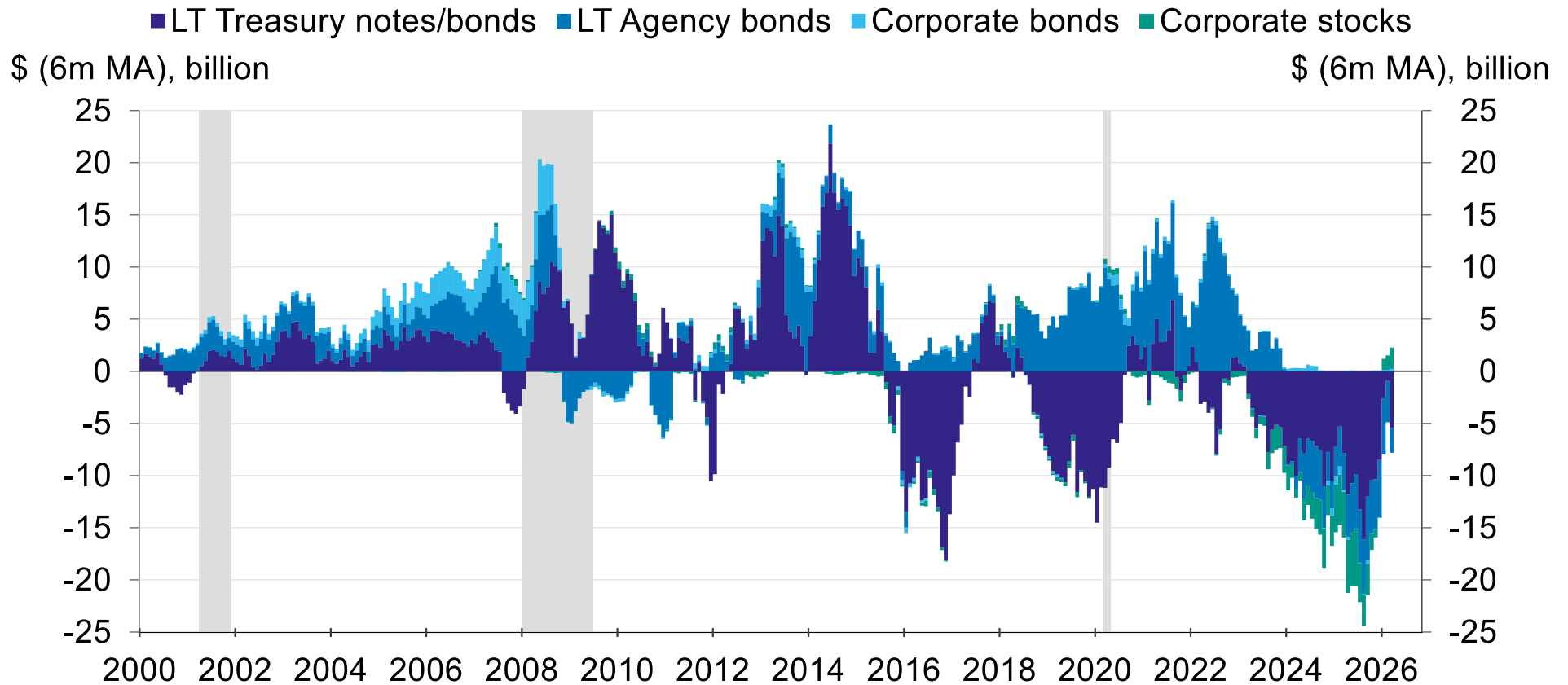


Source: U.S. Treasury, Macrobond, Deutsche Bank

China appears to have restarted buying USD equities in early 2026



China net purchases by asset class

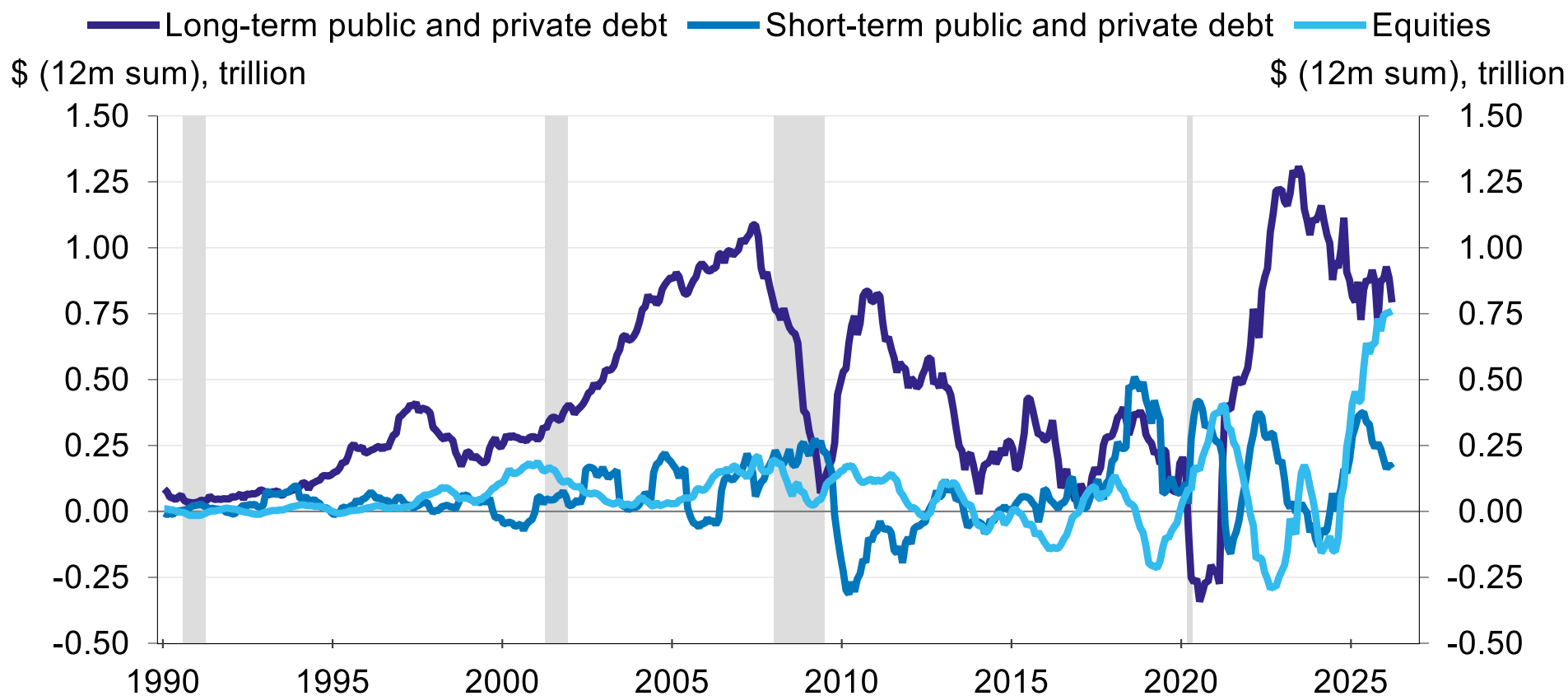


Source: U.S. Treasury, Macrobond, Deutsche Bank

Foreign buying of equities has accelerated in recent quarters on a 12m basis



Net foreign buying of all US securities

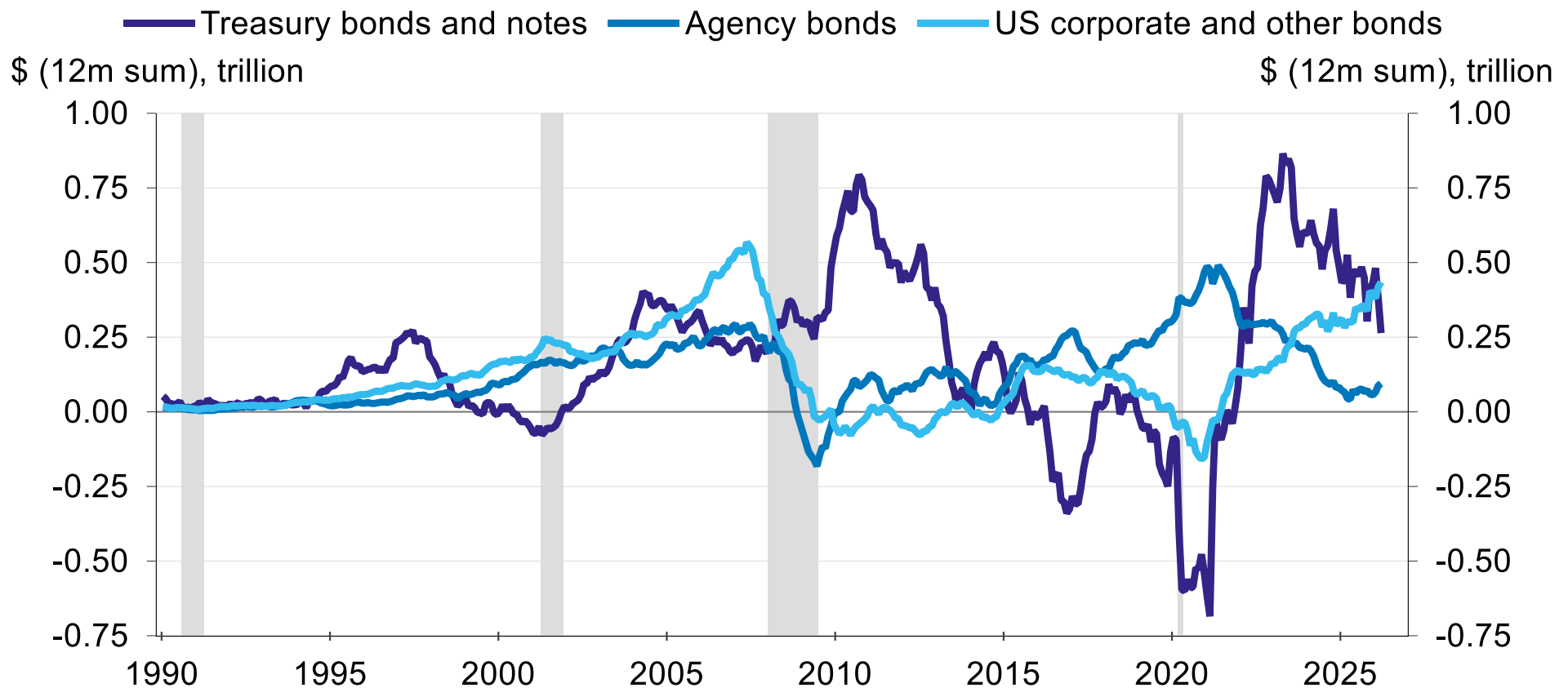


Source: U.S. Treasury, Macrobond, Deutsche Bank

Foreign demand is rising for corporate bonds and falling for Treasury and agency bonds



Net foreign buying of US debt securities



Source: U.S. Treasury, Macrobond, Deutsche Bank



Who is buying GSE MBS and agency debt?

Commercial banks are the largest investor with 25% of total agency securities outstanding, steady since 2019



Holdings of agency and GSE-backed debt securities (%)

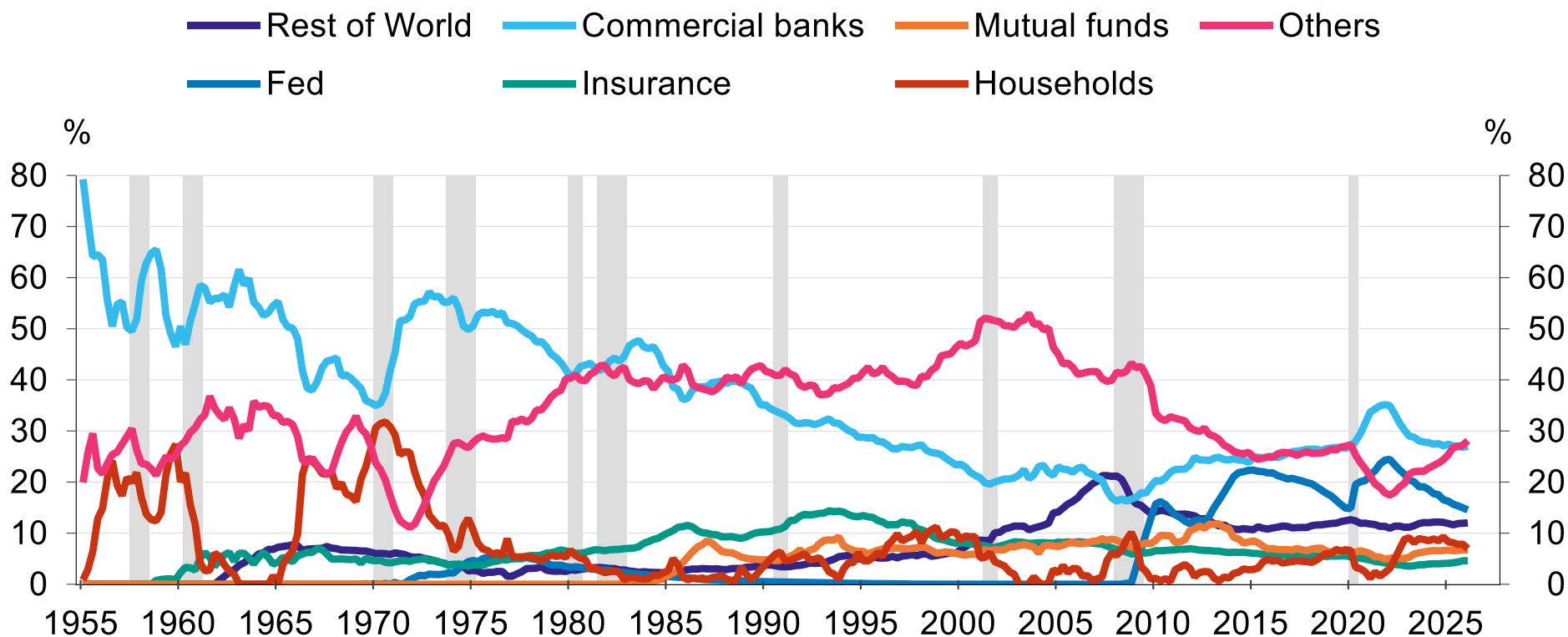
	2026 Q1	Q2 2019	Q2 2007
Commercial banks	24.8	25.1	19.2
Federal Reserve	14.6	16.4	0.0
Others	14.8	6.7	11.5
Mutual funds	16.0	13.9	11.0
Rest of world	12.0	11.9	21.2
Households/Hedge funds	6.9	6.8	1.4
Investment banks	5.5	3.1	7.4
Insurance companies	4.6	5.5	7.8
GSEs	2.6	2.7	10.3
Private pensions	2.4	2.3	2.7
Government retirement funds	1.7	1.6	3.8
ABS issuers	0.0	0.0	5.0

Source: Fed, Macrobond, Deutsche Bank. Note “Others” includes i) Non-financial Corporate ii) Government, iii. Credit Unions , iv) REITs, Holding companies and v) Instrument Discrepancies. Total may not sum to total 100% due to rounding.

Agency MBS ownership relatively spread out among investor categories



Holdings of agency securities as % of outstanding

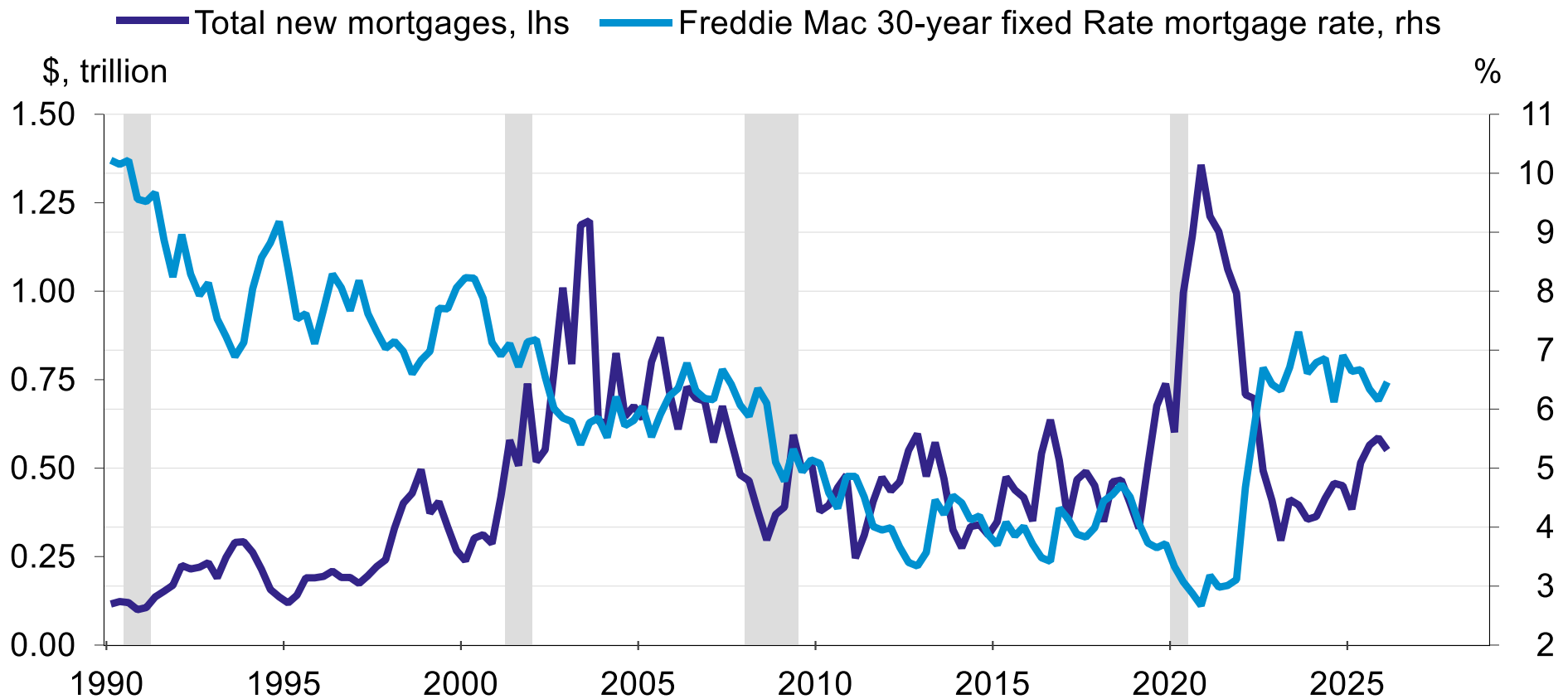


Source: Fed, Macrobond, Deutsche Bank. Note: Banks consists of Commercial banking , U.S. Chartered depository institutes, Foreign banking offices, Banks in U.S. possessions and Holding Companies

Mortgage activity perked up slightly as mortgage rates have come down slightly



Total new mortgages



Source: MBA, Freddie Mac, Macrobond, Deutsche Bank



Who is buying Corporate and Foreign Bonds?

Foreign investors and insurance companies own combined over half of corporate and foreign bonds



Holdings of corporate and foreign bonds (%)

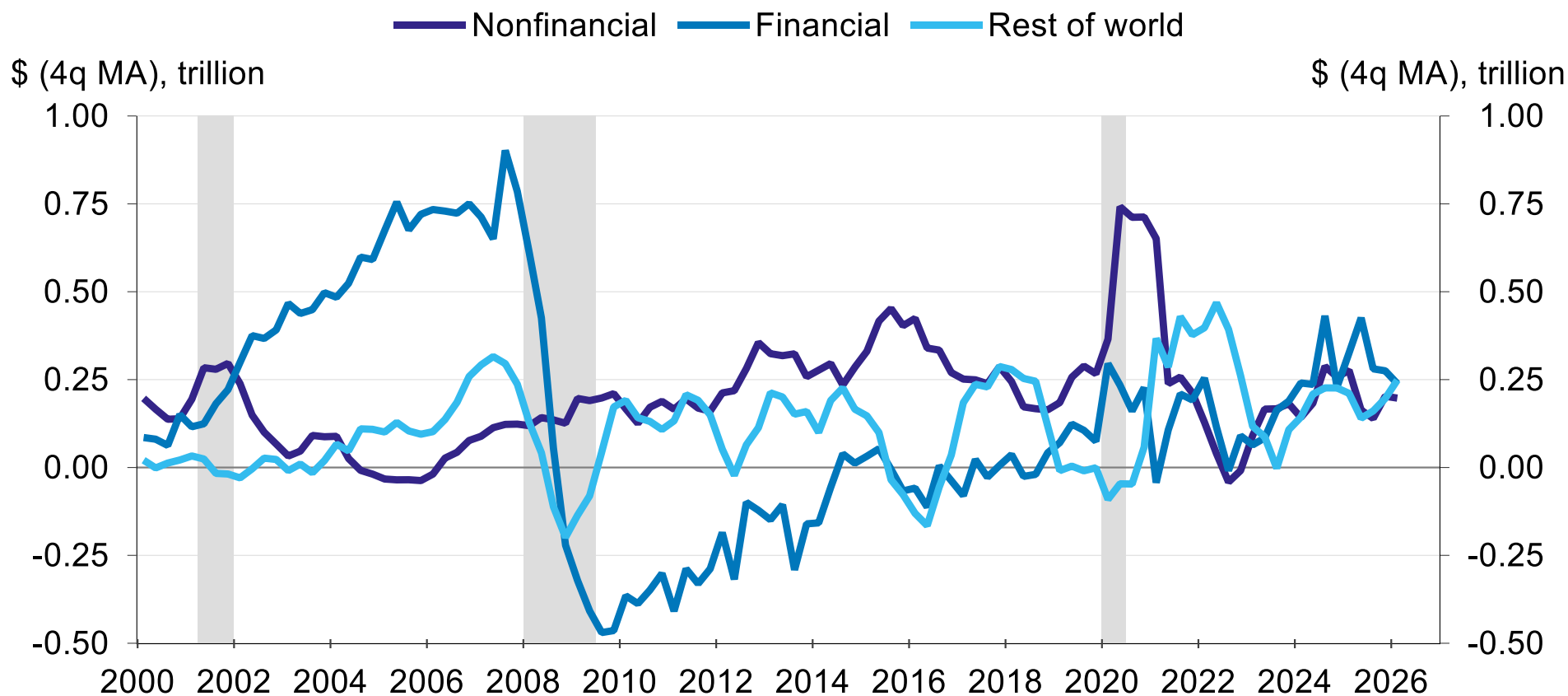
	2026 Q1	Q2 2019	Q2 2007
Insurance companies	28.0	28.3	22.3
Rest of World	25.9	26.6	18.1
Mutual funds and ETFs	23.8	20.8	12.5
Others	3.5	3.3	23.6
Commercial banks	5.4	5.0	9.9
Private pension funds	5.4	6.5	3.5
Government retirement funds	4.0	3.8	4.0
Government	1.8	2.5	1.7
Households/hedge funds	1.1	1.6	5.7
Finance companies	0.1	0.4	2.0

Source: Fed, Macrobond, Deutsche Bank. Note “Others” includes i) Instrument Discrepancies, ii) Holding Companies, iii) Security Brokers & Dealers, iv) REITs, v) Govt-Sponsored Enterprises and vi) Credit Unions. Total may not sum to total 100% due to rounding.

Growth in financial corporate and foreign bond supply remains positive



Net issuance of corporate & foreign bonds



Source: Fed, Macrobond, Deutsche Bank



Who is buying municipal securities and loans?

Household sector accounts for close to half of total municipal bonds outstanding



Holders of municipal securities and loans (%)

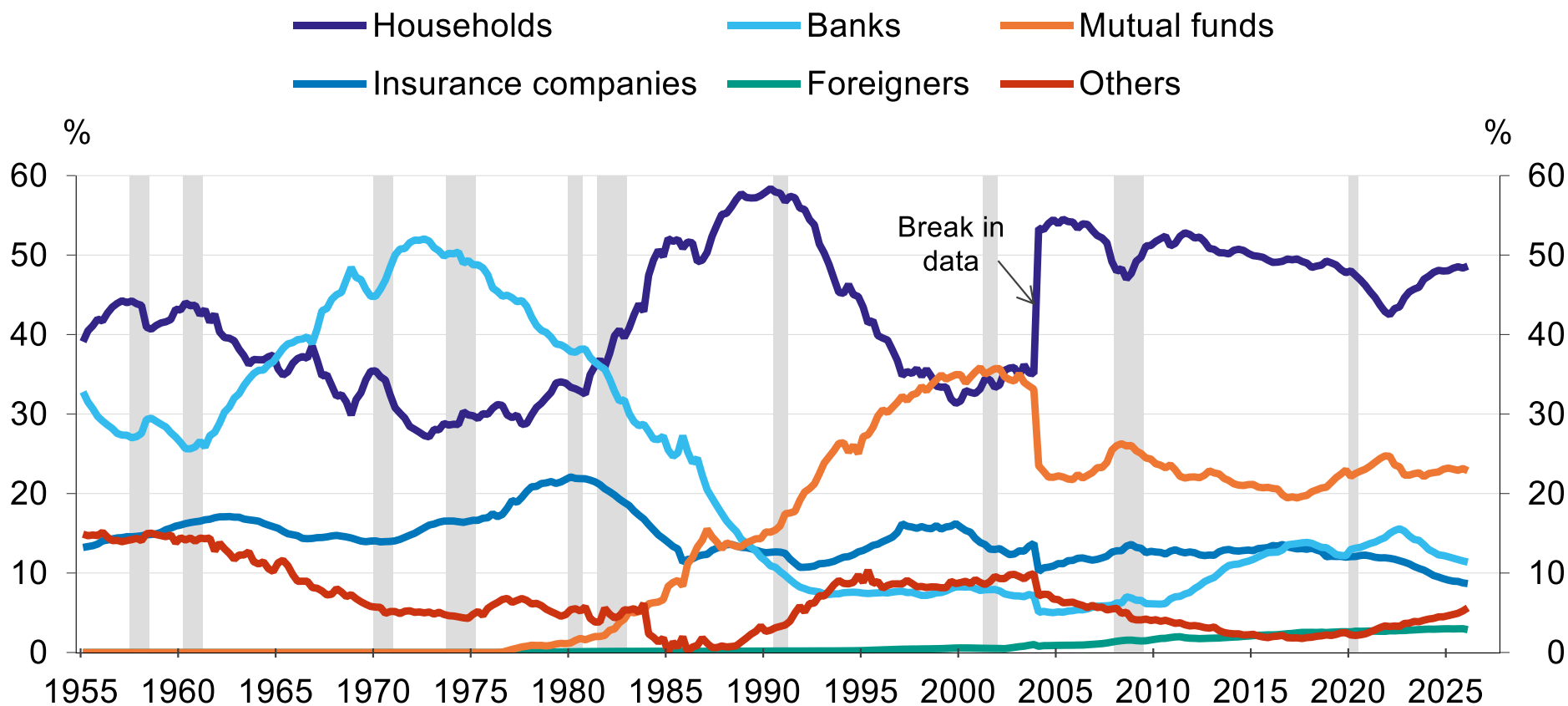
	2026 Q1	Q2 2019	Q2 2007
Households/hedge funds	48.6	48.8	52.2
Mutual funds and ETFs	29.1	25.2	25.9
Commercial banks	11.1	12.2	5.8
Insurance companies	8.7	12.1	11.9
Other	6.6	-4.3	2.0
Rest of world	2.9	2.6	1.1
Security brokers	0.4	0.4	1.6

Source: Fed, Macrobond, Deutsche Bank. Note “Others” mainly includes i) Non-financial corporate & non-corporate ii) S&L Government, iii. Credit Unions , iv) Government retirement funds v) GSEs and vi) Instrument Discrepancies. Total may not sum to total 100% due to rounding.

Key investor categories in municipal debt



Holdings of municipal securities as % of outstanding



Source: Fed, Macrobond, Deutsche Bank

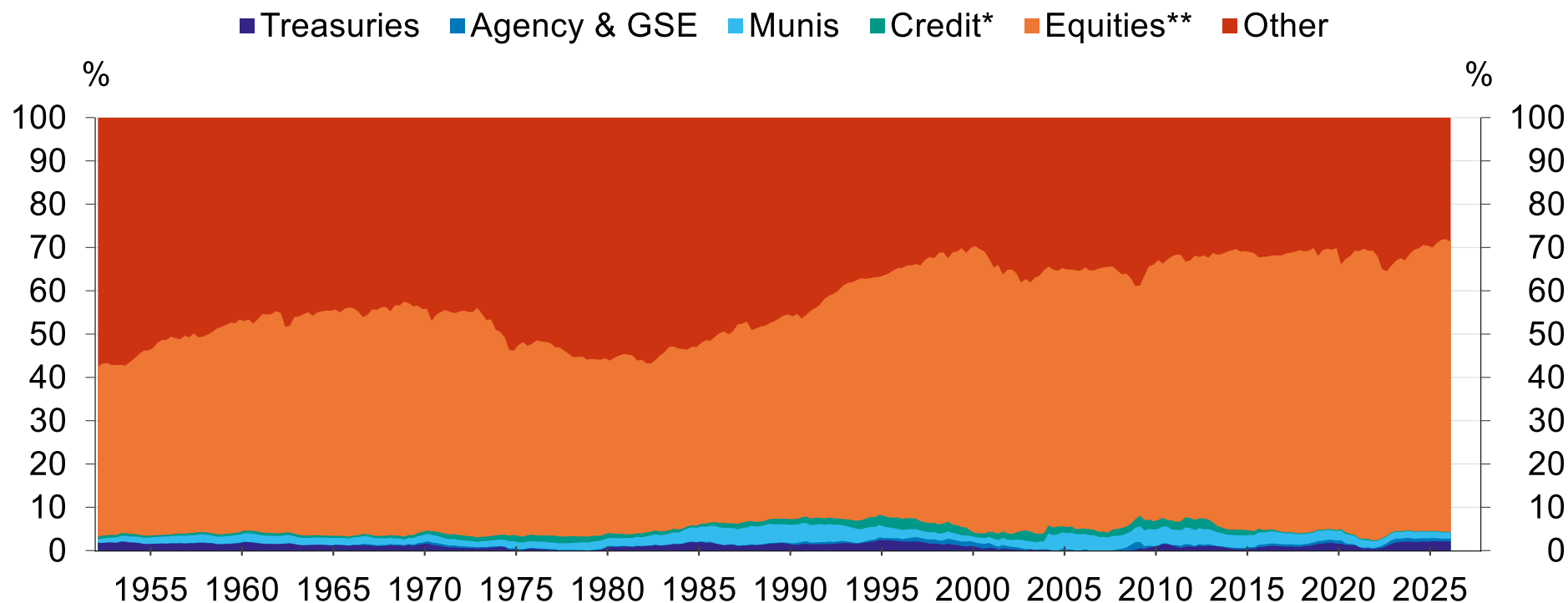


Asset allocations by investor type

US households (including hedge funds) are heavily overweight in equities relative to fixed income



Households' (incl. hedge fund) asset allocation



Source: Fed, Macrobond, Deutsche Bank.

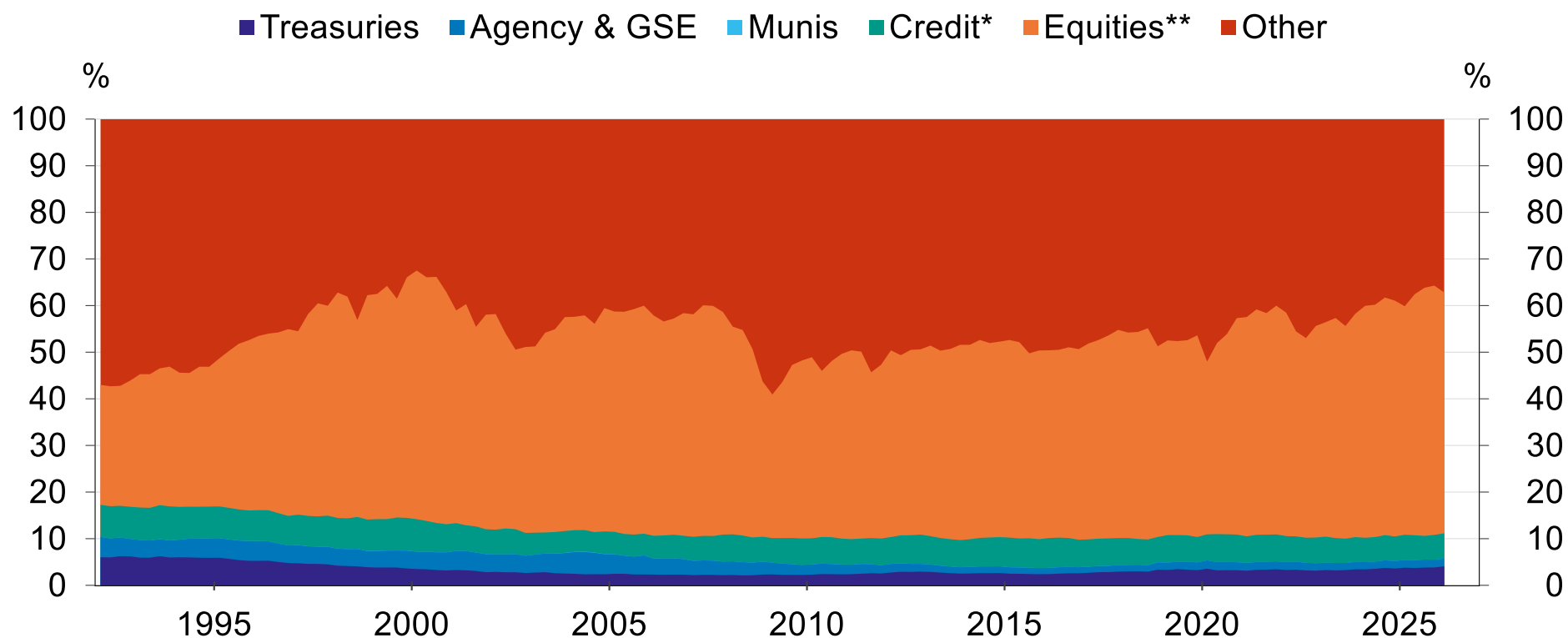
*Credit includes corporate & foreign bonds

**Equities includes corporate equities, mutual fund shares, life insurance reserves and pension funds reserves

Pension funds allocate about half of their assets in equities



Pension funds' asset allocation



Source: Fed, Macrobond, Deutsche Bank.

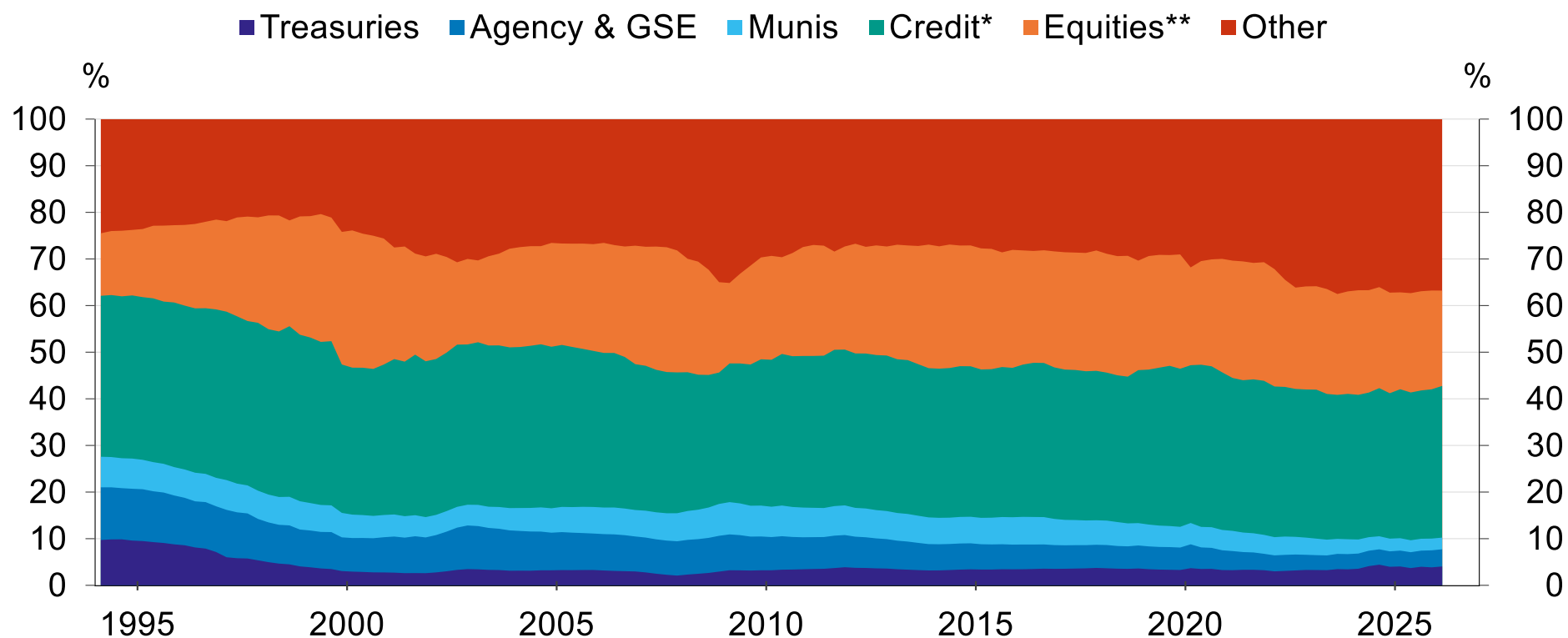
Pension funds incl. Private, State & local govt. retirement and Federal govt. retirement funds

*Credit includes corporate and foreign bonds **Equities includes corporate equities and mutual fund shares

Insurance companies are overweight credit



Insurance companies' asset allocation



Source: Fed, Macrobond, Deutsche Bank.

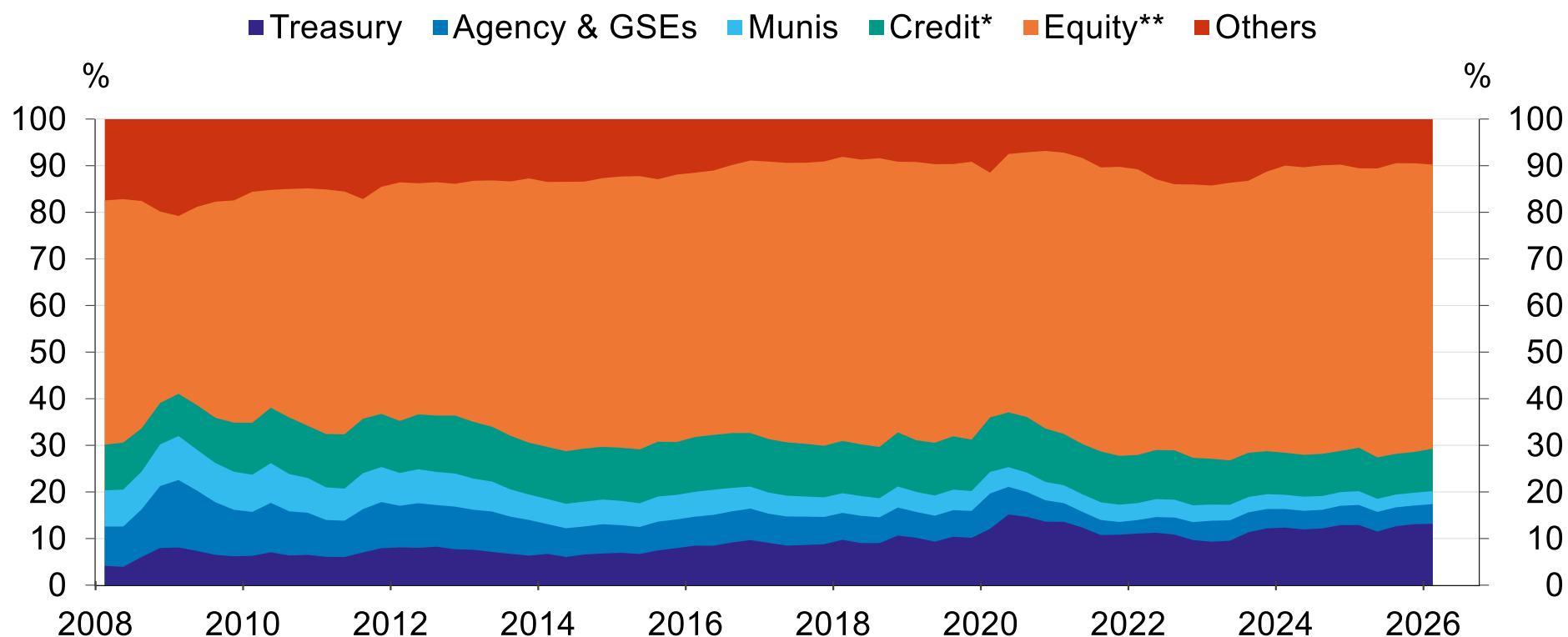
Insurance sector includes both life and non-life insurance

*Credit includes foreign and corporate bonds; **Equities include mutual fund shares and corporate equities

Mutual funds have over half of their allocation in equities



Mutual funds' asset allocation

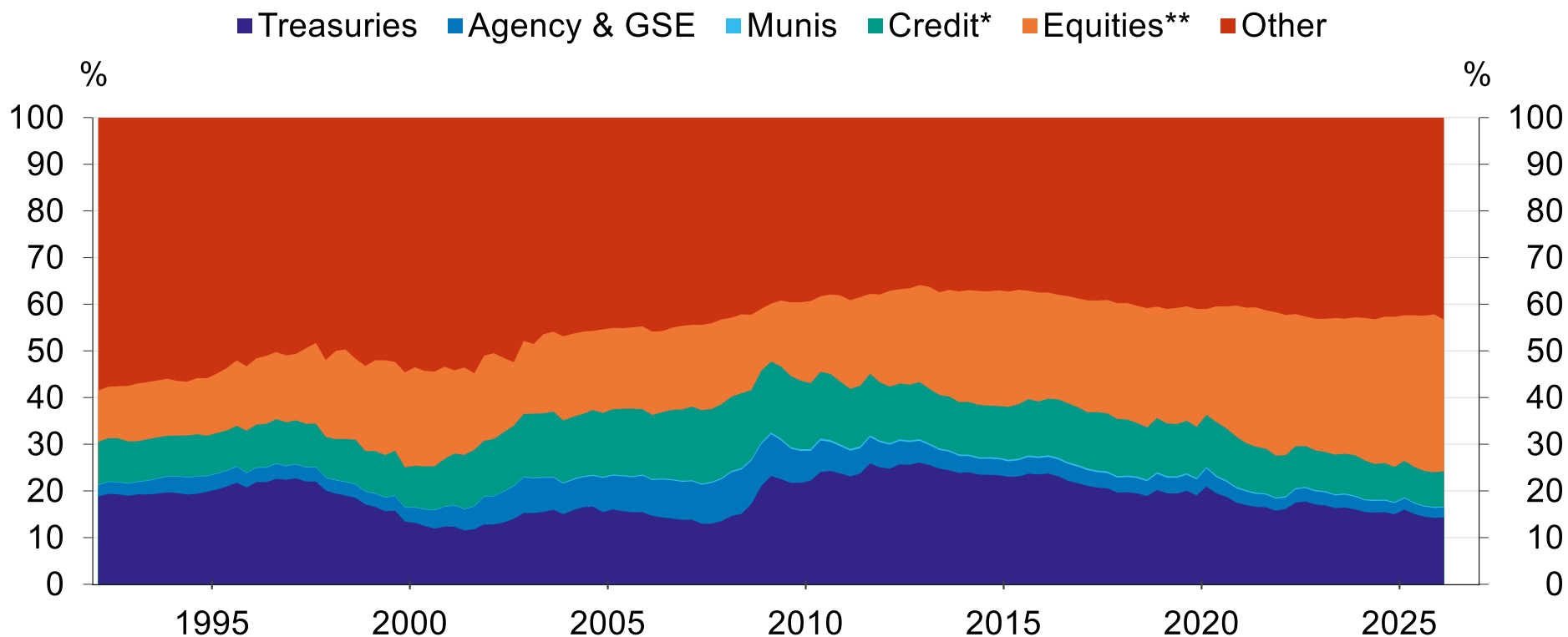


Source: Fed, Macrobond, Deutsche Bank. Mutual funds incl. traditional mutual funds, money mkt. MFs, exchange traded MFs & open ended MFs. *Credit includes corporate and foreign bonds; **Equity includes corporate equities and mutual fund shares.

Foreign investors increased allocations to US equities in the last decade



Asset allocation of foreigners in US assets



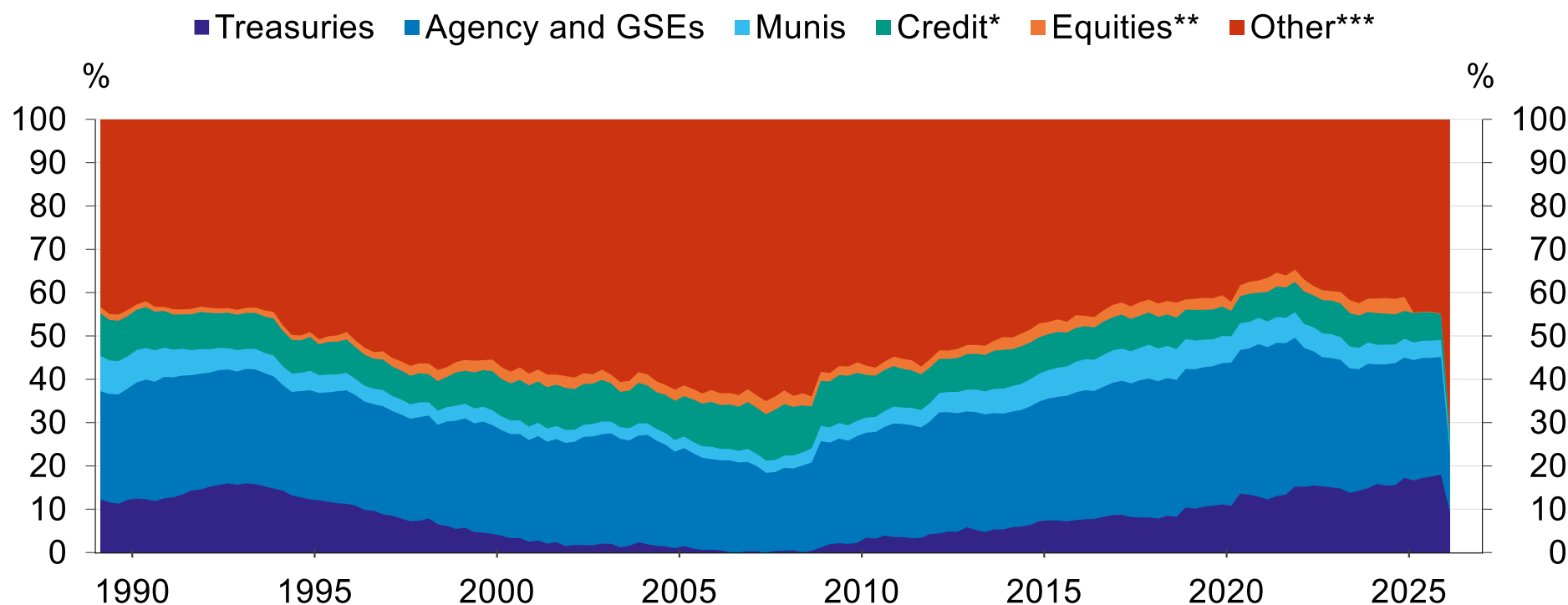
Source: Fed, Macrobond, Deutsche Bank.

*Credit is Corporate bonds held by Foreign sector; **Equities is calculated as sum of corporate equities and mutual fund shares

Commercial banks hold more Treasuries on their balance sheets after the Global Financial Crisis



Banks' asset allocation



Source: Fed, Macrobond, Deutsche Bank.

Banks incl. all commercial banks and brokers and dealers

*Credit includes corporate and foreign bonds; **Equities includes corporate equities and mutual fund shares,

***Excludes vault cash, reserves at Fed, fed funds and repos, loans, life ins. reserves and US direct investment abroad

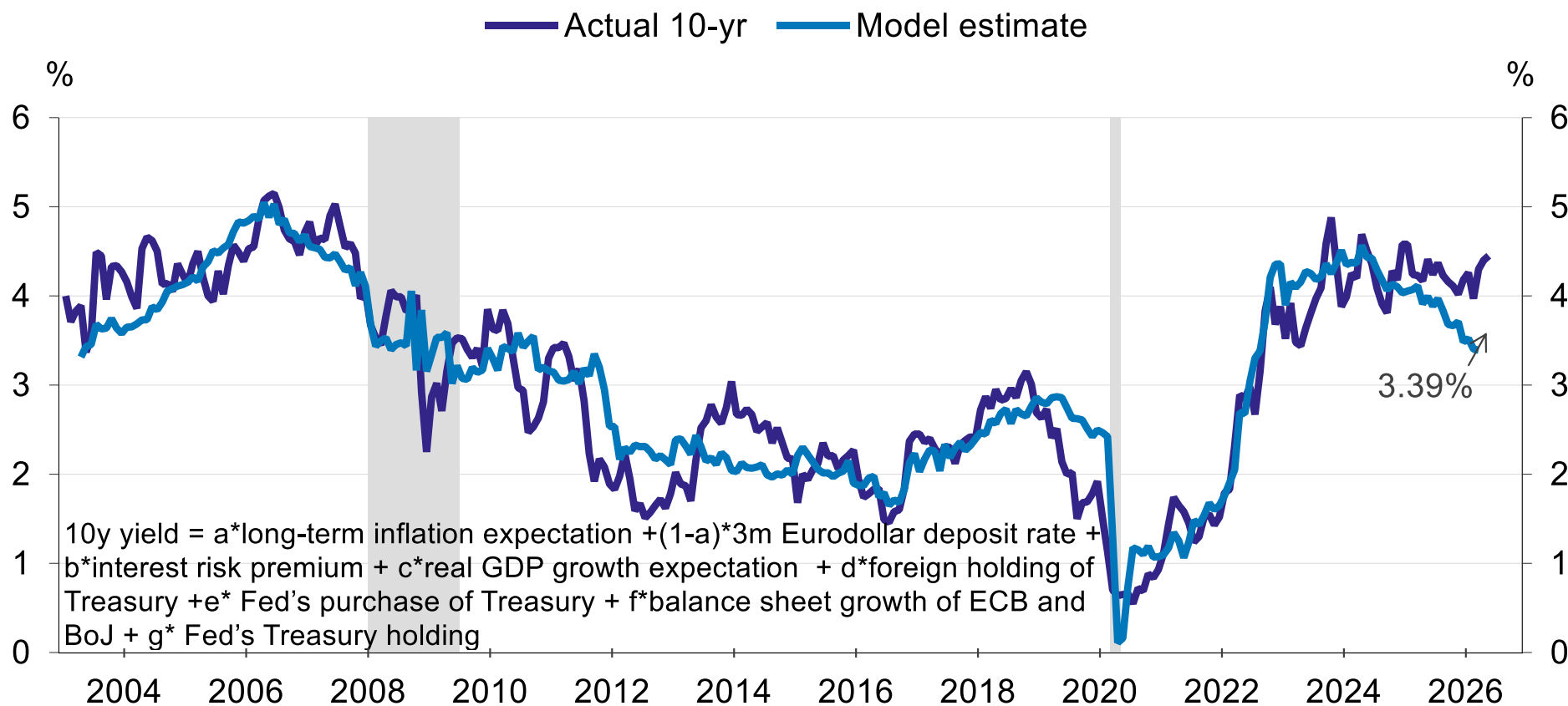


Models and outlook

10-year model accounting for central bank purchases suggests a fair value of ~3.4%



10-year Treasury yield model forecasts

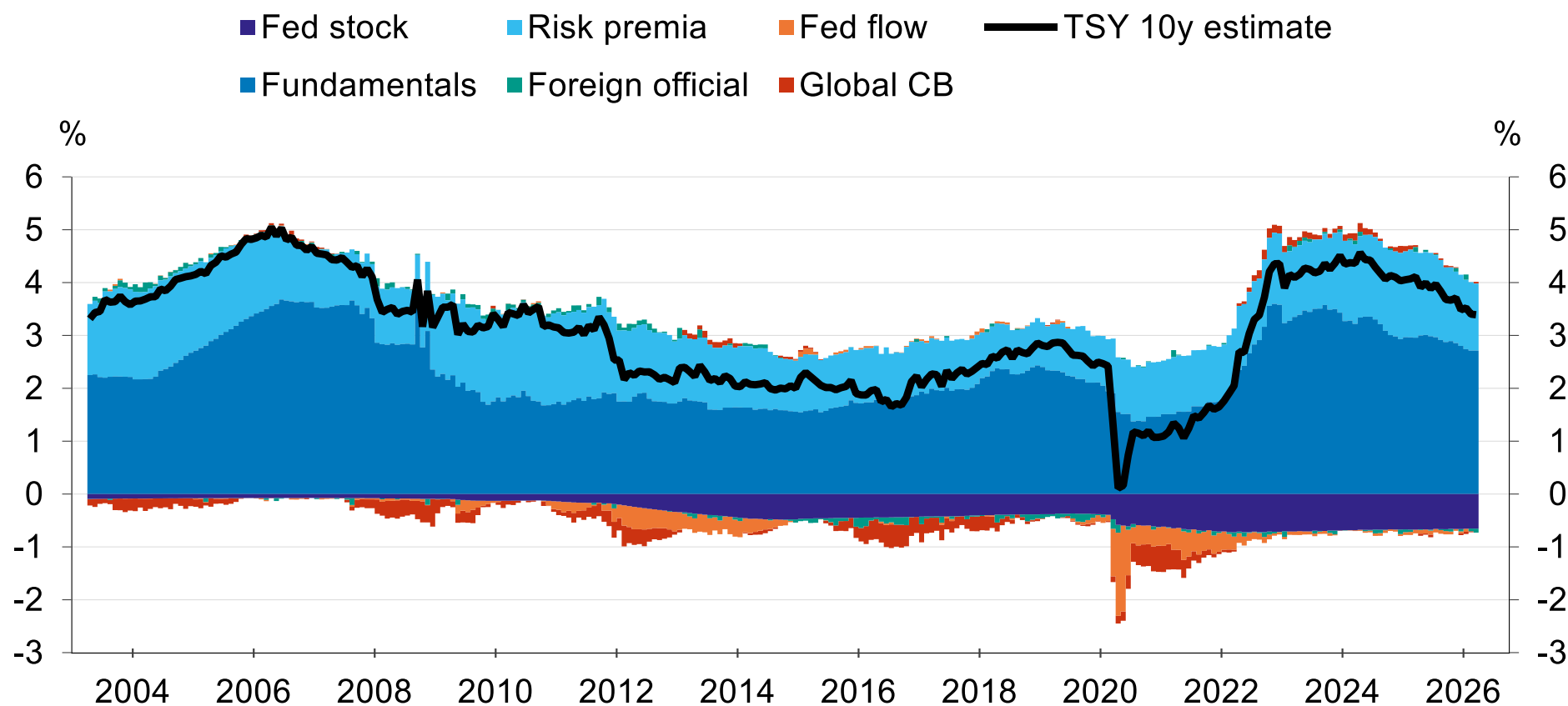


Source: U.S. Treasury, Philadelphia Fed, Riksbanken, BEA, Fed, ECB, BOJ, Macrobond, Deutsche Bank

Decomposition of 10y model estimates

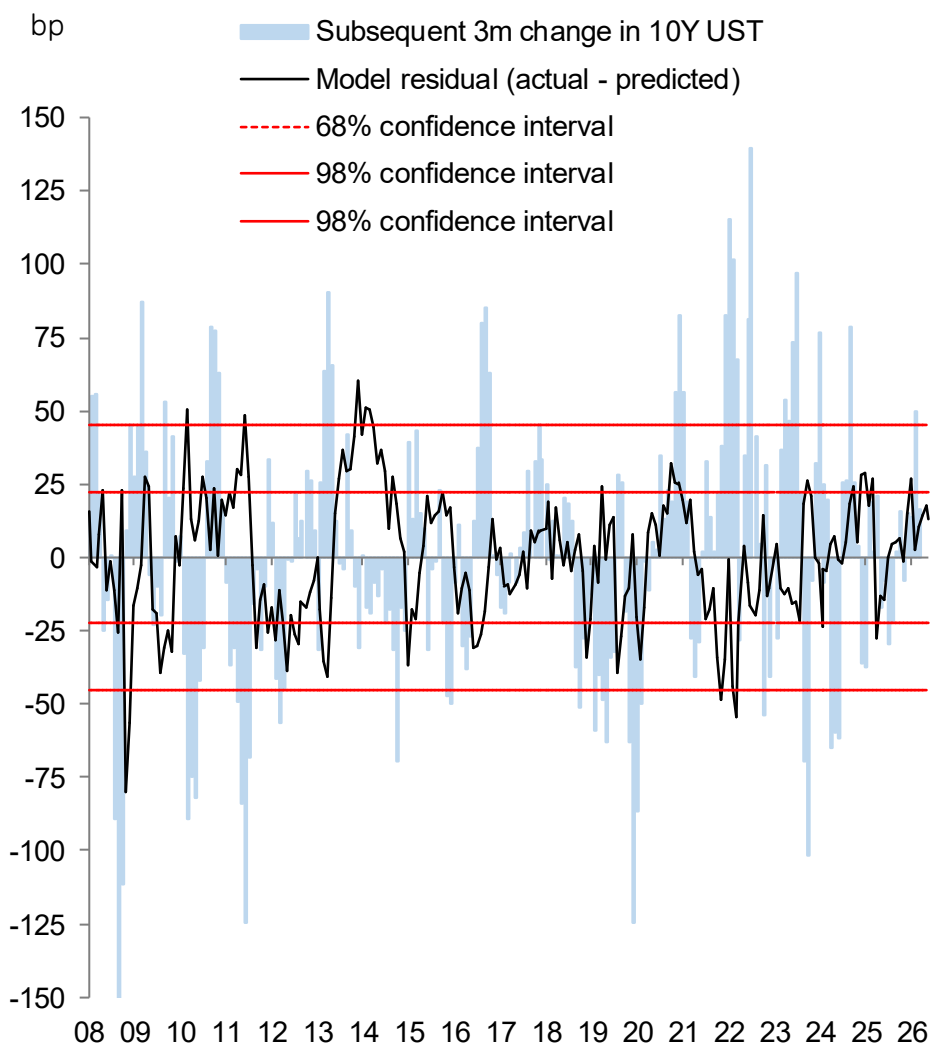
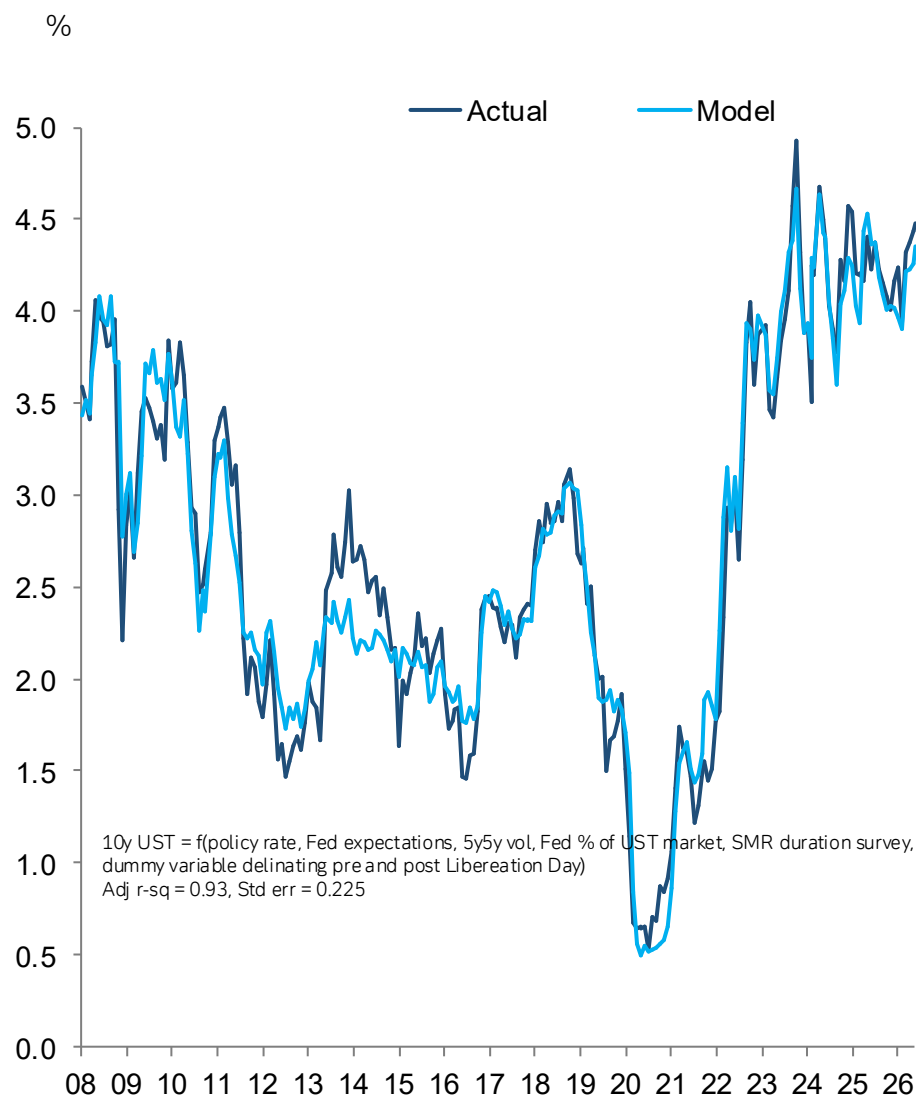


Model decomposition of 10-year Treasury yield



Source: U.S. Treasury, Philadelphia Fed, Riksbanken, BEA, Fed, ECB, BOJ, Macrobond, Deutsche Bank

10-year model using market explanatory variables suggests a fair value around 4.35% (as of June 12, 2026)

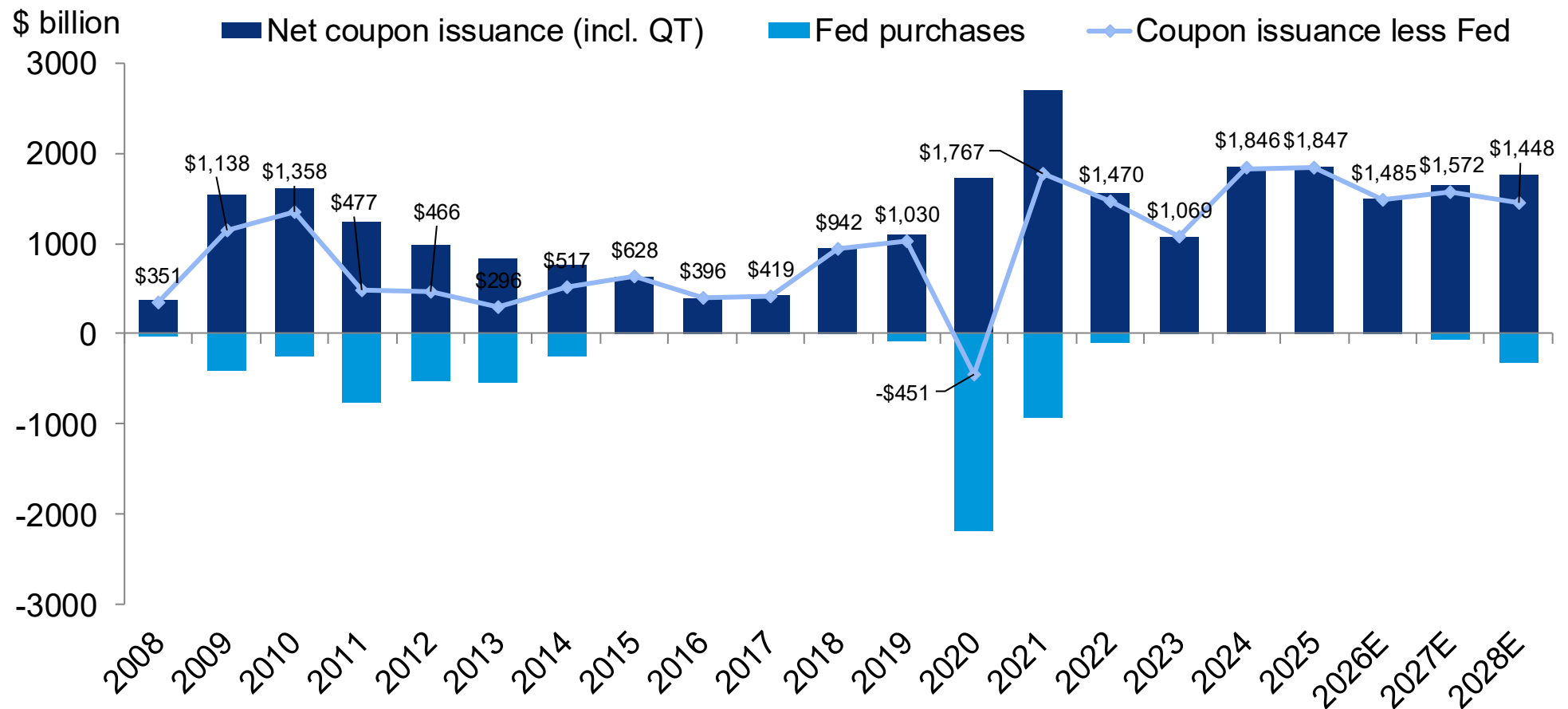


Source: FRB, BEA, Haver Analytics, Deutsche Bank

Treasury net coupon issuance set to fall by ~20% in 2026 versus last year



Treasury coupon issuance

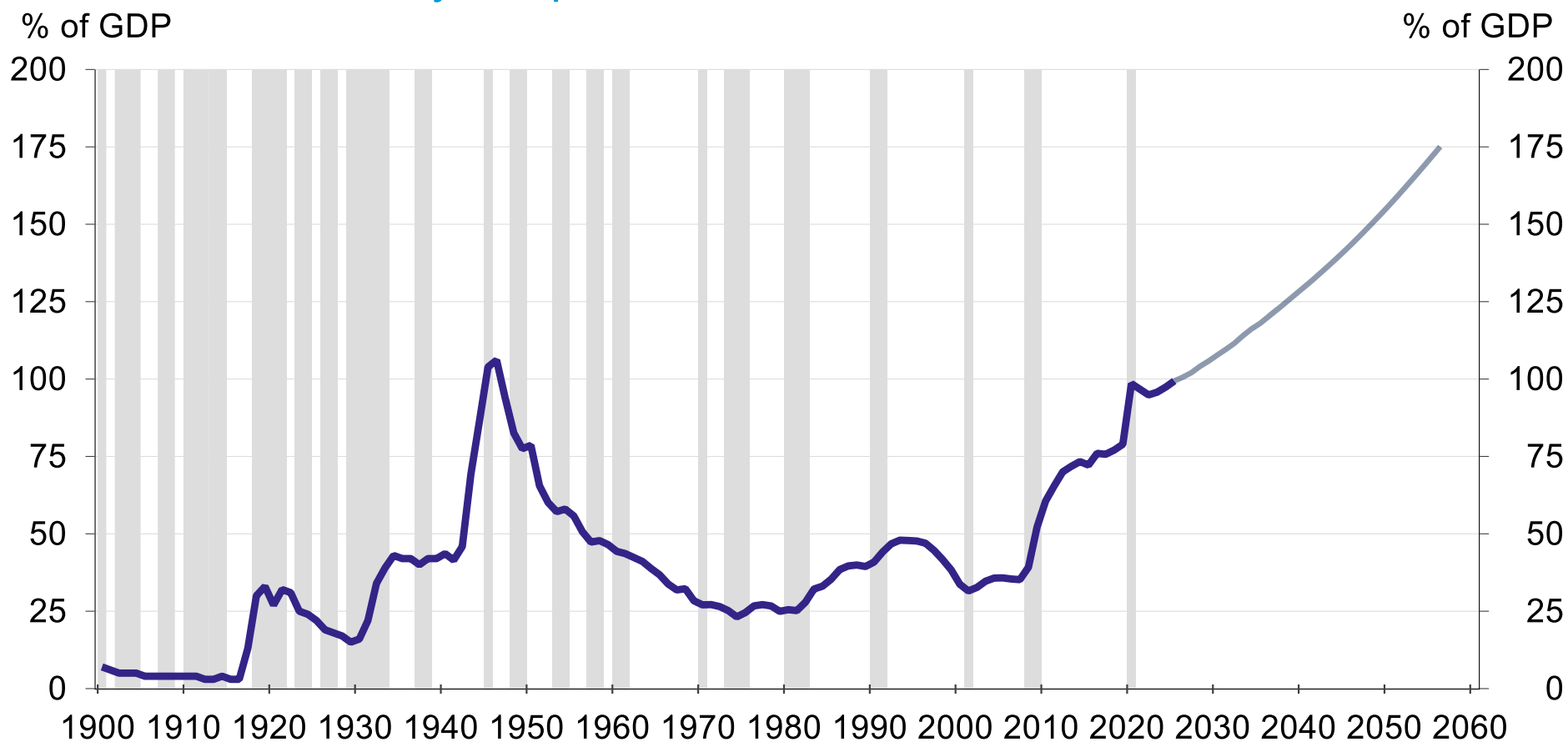


Source: FRB, BEA, Haver Analytics, Deutsche Bank

CBO's projection of US debt over the coming decades



Federal debt held by the public

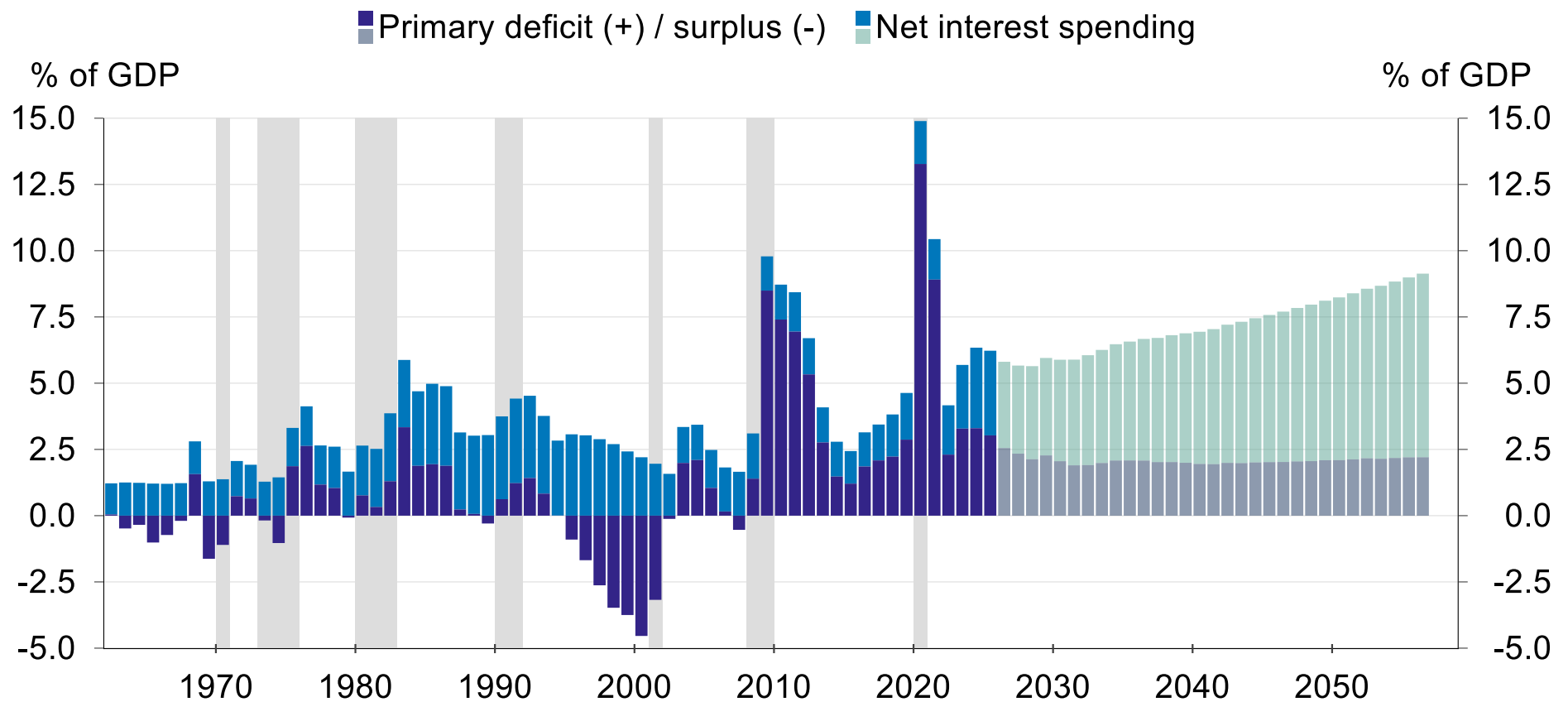


Source: CBO, Macrobond, Deutsche Bank

Interest expense projected to be a rising share of the US deficit due to high interest rates



Total Deficit, Net Interest Outlays, and Primary Deficit

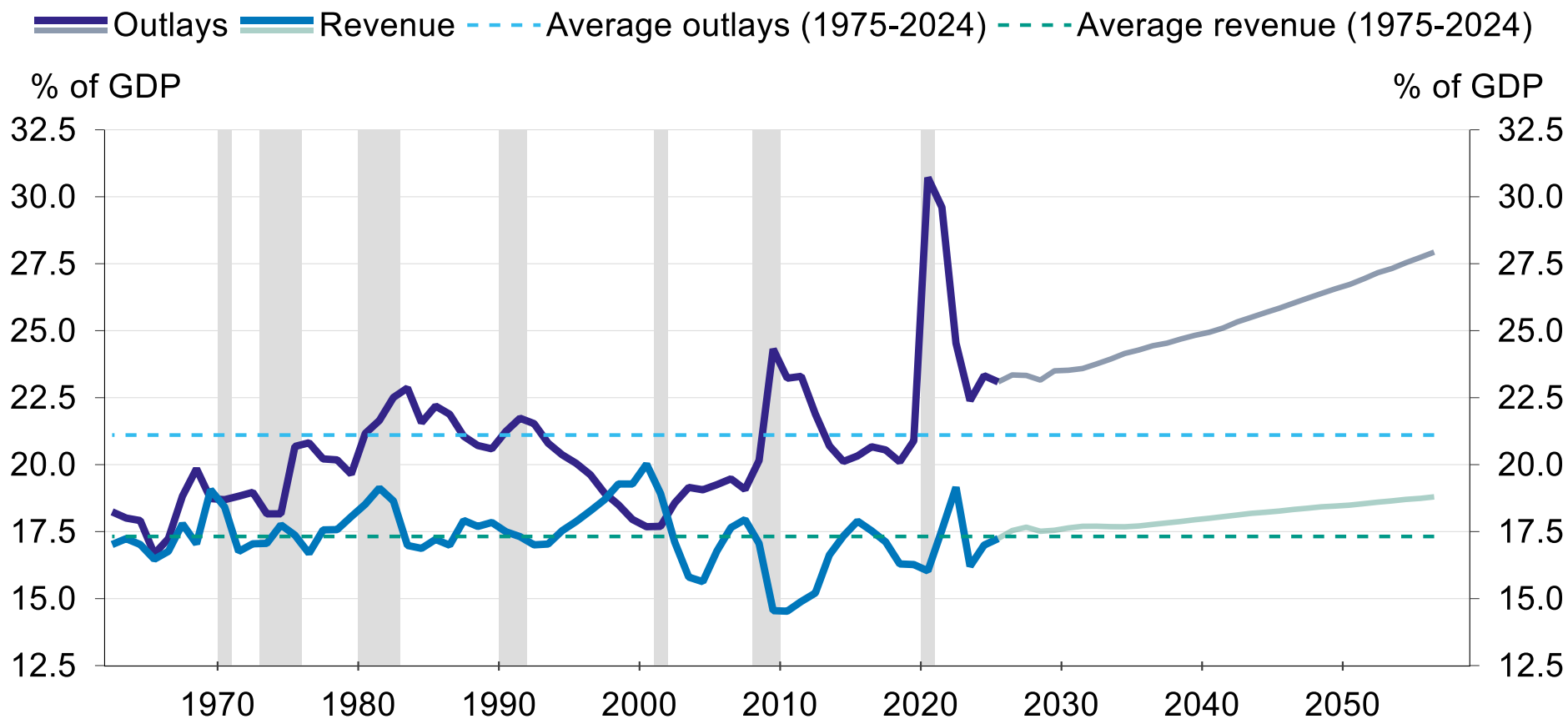


Source: CBO, Macrobond, Deutsche Bank

Rising outlays and a slower rise in revenue contributes to the primary deficit



Total Outlays and Revenues

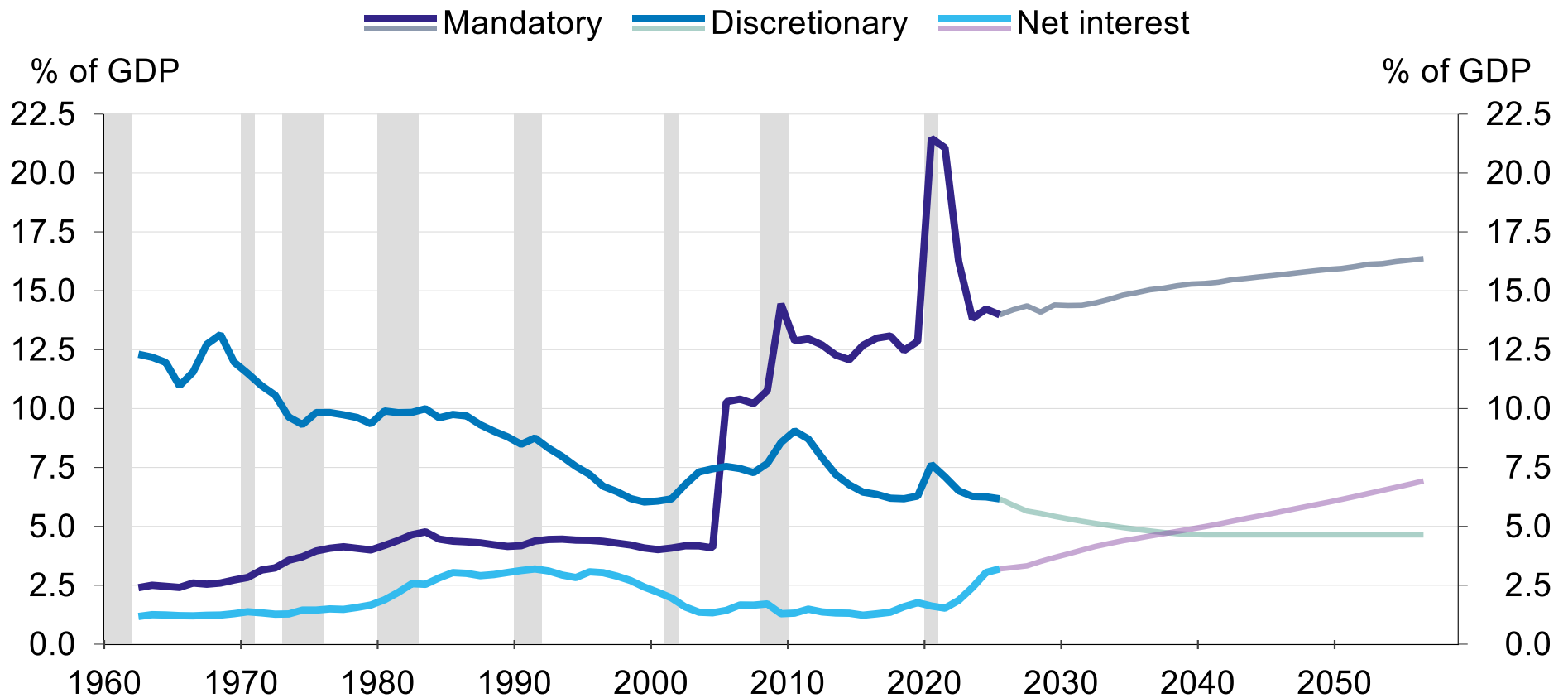


Source: CBO, Macrobond, Deutsche Bank

Mandatory spending expected to grow faster than discretionary as net interest rises too



Outlays, by Category

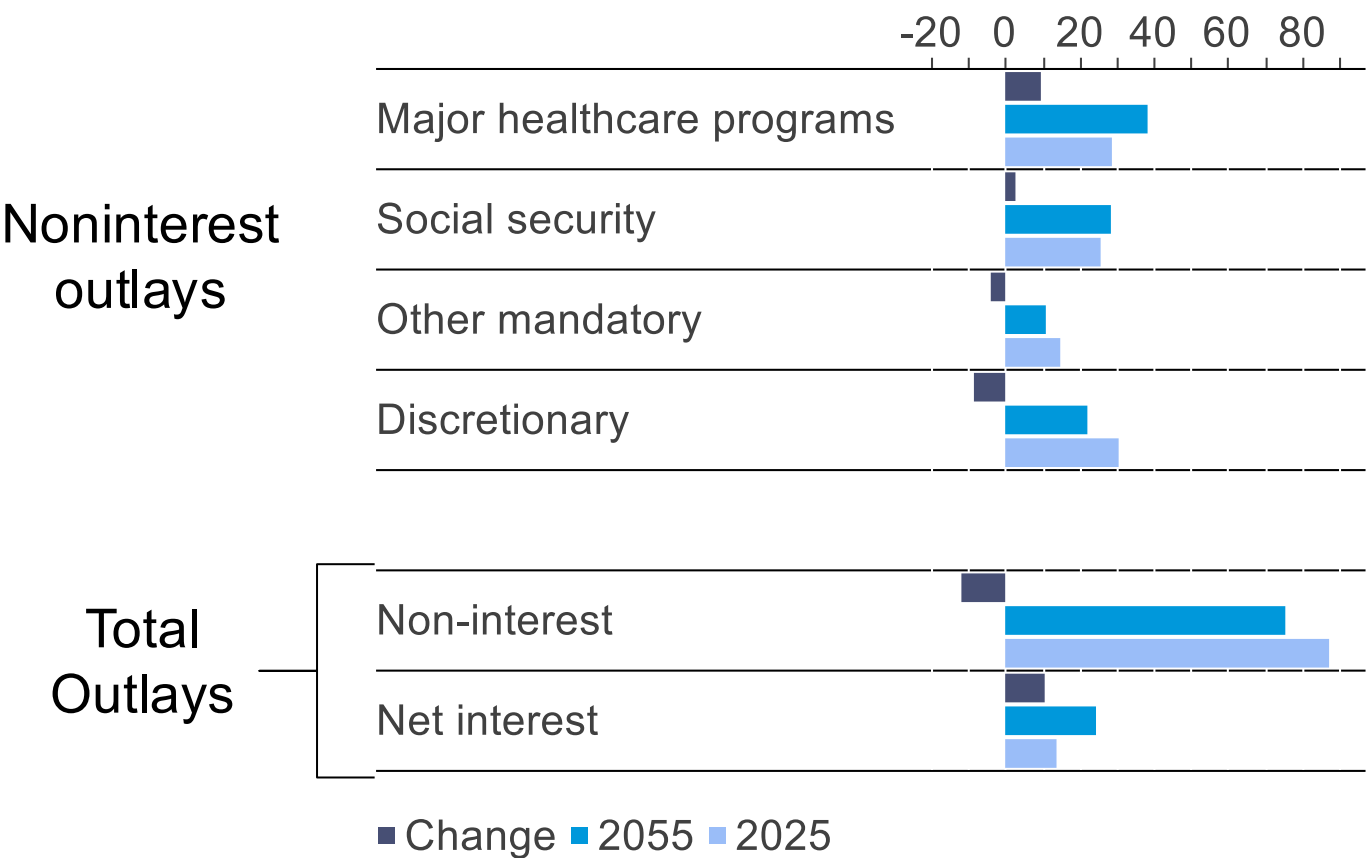


Source: CBO, Macrobond, Deutsche Bank

CBO projection of outlays from 2025 to 2055



Composition of Outlays 2025 & 2055 (%)

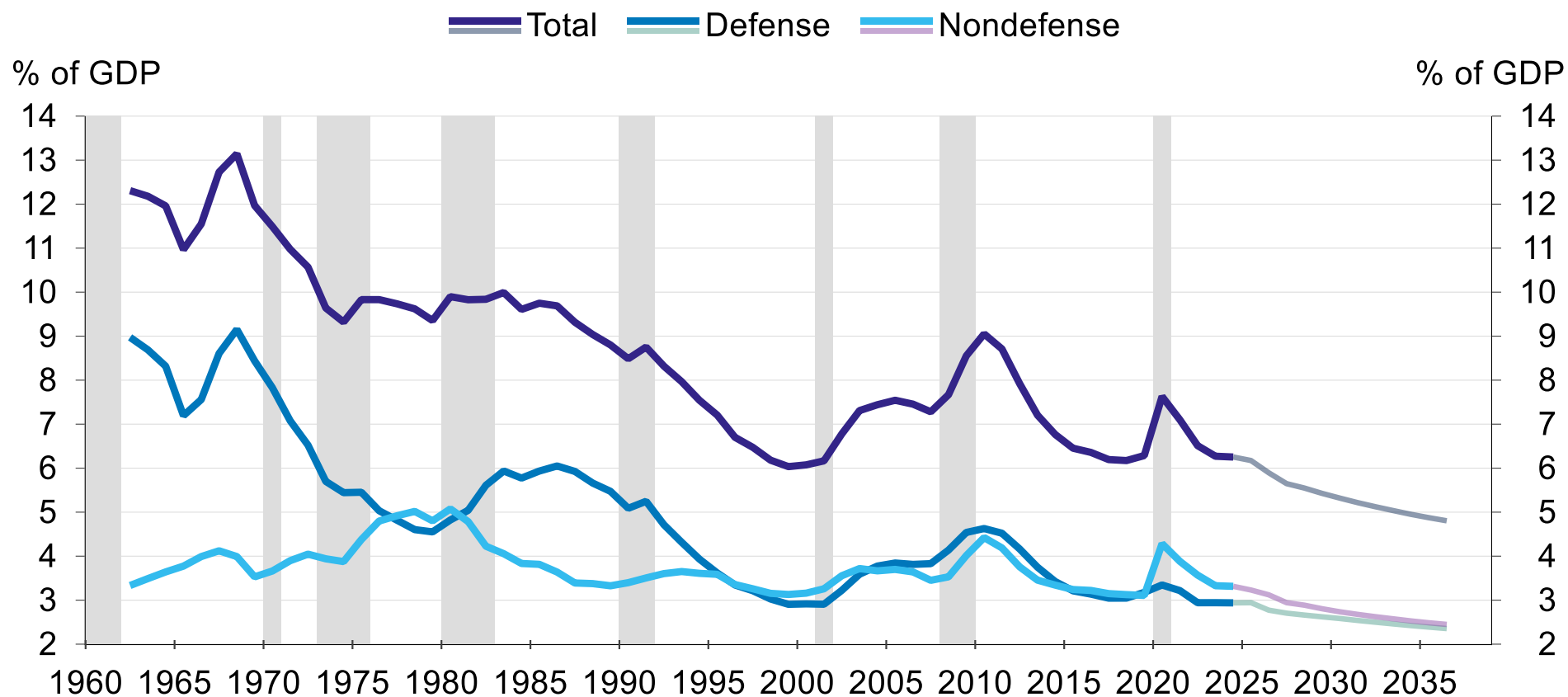


Source: CBO, Macrobond, Deutsche Bank

CBO projection of defense versus non-defense spending



Discretionary Outlays, by Category

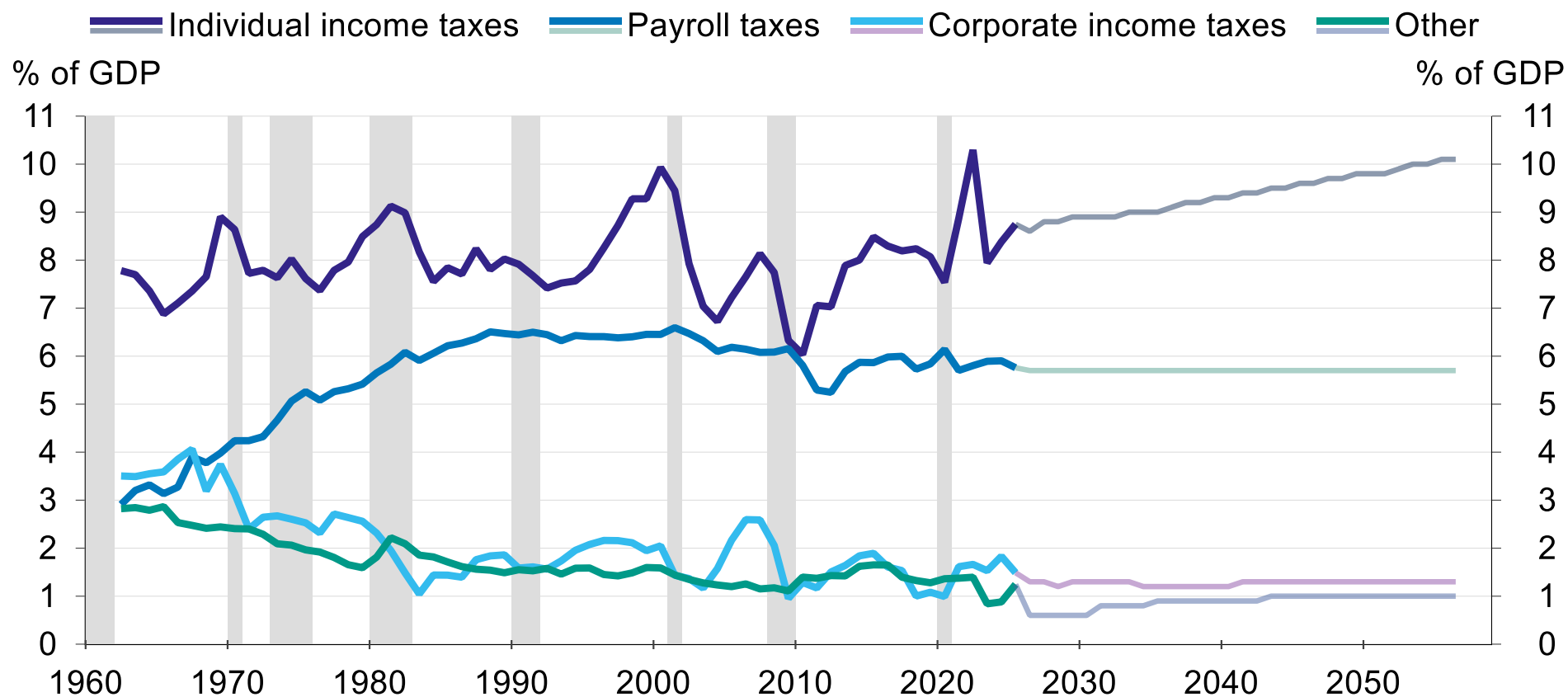


Source: CBO, Macrobond, Deutsche Bank

CBO projection of tax revenues by category



Revenues, by Category

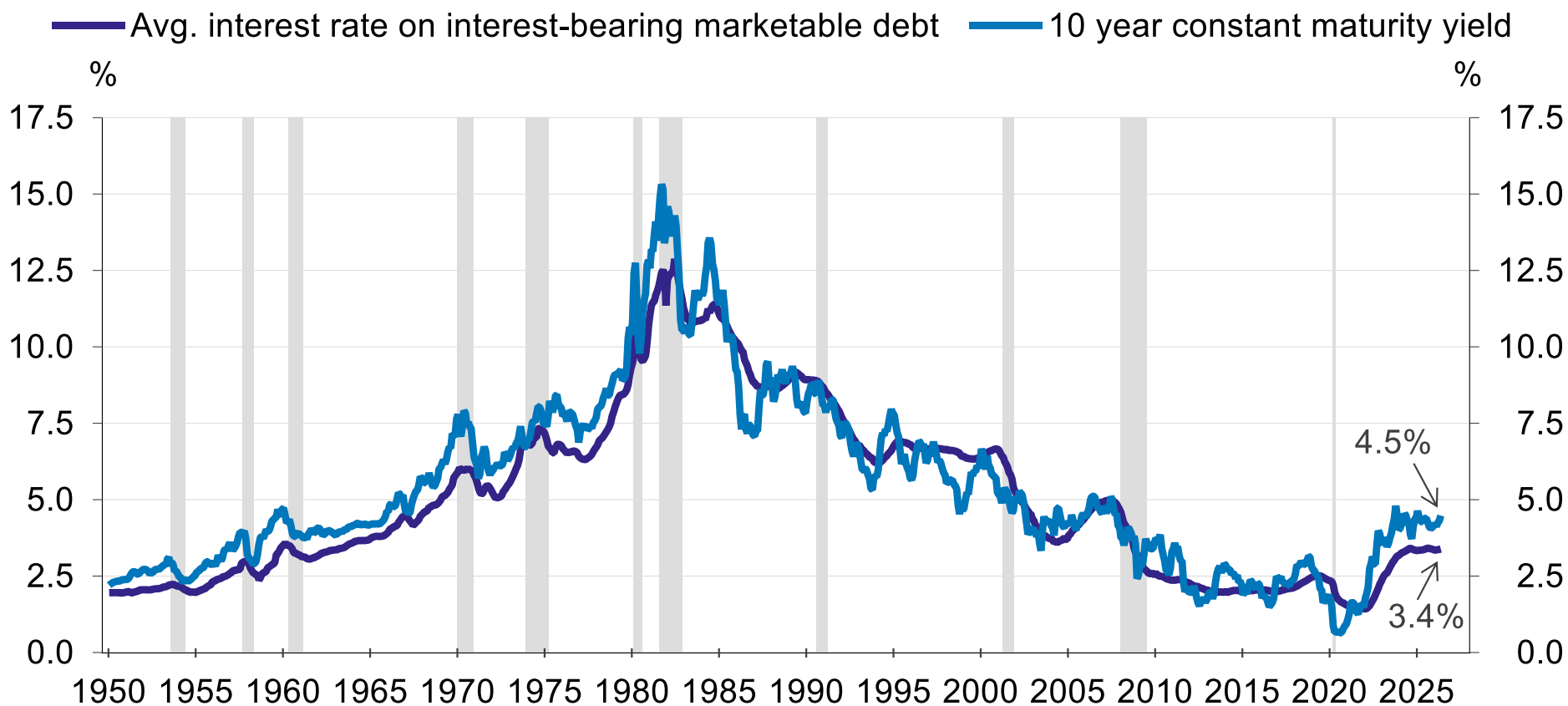


Source: CBO, Macrobond, Deutsche Bank

Average interest rate on public debt: 3.4%



10 yr Treasury and interest rate on government debt



Source: U.S. Treasury, Fed, Macrobond, Deutsche Bank



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Appendix 1

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